

DAVYDOV'S SOLITON

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NORTH-HOLLAND

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Received February 1992; editor: D.K. Campbell

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Abstract:

Theories and experiments related to the localization of Amide-I (or CO stretching) vibrational energy in protein are surveyed. Such localization arises through interaction of the Amide-I mode with lattice distortion. Generalizations to charge transport and biological applications are also considered.

Introduction

In 1973 a novel mechanism for the localization and transport of vibrational energy in protein was proposed by A.S. Davydov [Dav73]. Referring to the atomic structure of an alpha-helix region of protein shown in fig. 1, this mechanism can be described classically as follows. Vibrational energy of the CO stretching (or Amide-I) oscillators that is localized on the helix acts—through a phonon coupling effect—to distort the structure of the helix. The helical distortion reacts—again through phonon coupling—to trap the Amide-I oscillation energy and prevent its dispersion. This effect is called *self-localization* or *self-trapping*.

From the perspective of biochemistry it is an appealing mechanism because the energy involved is about right. To see this note that a fundamental means for the production of biological energy is the hydrolysis of adenosine triphosphate (ATP) into adenosine diphosphate (ADP) by the reaction



Under normal physiological conditions approximately 10 kcal/mole or 0.42 eV of free energy is

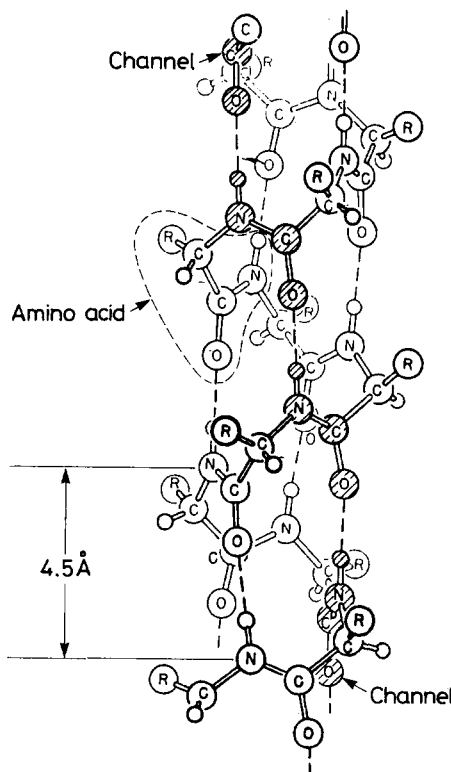


Fig. 1. An atomic model of the alpha-helix structure in protein. One "channel" is cross-hatched.

released by this reaction [Fox82] and this is about two quanta of Amide-I oscillator energy.

Over the past two decades this biological mechanism has been studied in detail by Davydov and his colleagues in many publications. To approach this work, the reader might begin with three books [Dav82a, Dav85, Dav91c], an extensive review [Dav82b], and several readily available English language papers [Dav74, Dav77, Dav79b, Dav79c] on what has become known as "Davydov's soliton".

The mathematical techniques that are used to analyze Davydov's soliton are similar to some that have been developed for the "polaron" effect suggested by Landau [Lan33] and studied by Pekar [Pek54], Fröhlich [Fro52], Holstein [Hol59] and many others [GL91]. It is not the purpose of this review to cover all aspects of polaron theory, but this work will often be used as a resource.

There is some confusion because the term polaron has been used over the past half-century to denote a wide variety of excitations (called *excitons*) that are self-localized through interactions with lattice phonons. In this context we shall see that Davydov's soliton corresponds to a polaron that is (i) *large* so the continuum limit approximation is justified, (ii) *acoustic* because the self-localization arises from interactions with acoustic modes of the lattice, and (iii) *weakly coupled* because the anharmonic energy is small compared with the phonon bandwidth.

More recently Davydov's theory has been critically examined by many scientists, and the following questions have been of particular concern:

- (i) What is the correct quantum mechanical description of Davydov's soliton at low temperature?
- (ii) How does the soliton get started on an alpha-helix?
- (iii) Is the soliton stable at normal physiological temperature (310 K)? If not, how long will it last?
- (iv) What is the experimental evidence for Davydov's soliton?
- (v) How may Davydov's theory be generalized to include charge transport and more general protein structures?
- (vi) What biological applications can be proposed for Davydov's soliton?

The structure of this review is organized around these questions. Thus attempts to answer question (i) are discussed in section 1, which begins with a brief but (it is hoped) understandable description of Davydov's theory as it has developed in his work over the past twenty years. This section then describes a simple model of the alpha-helix that many have found convenient for theoretical and numerical studies. In developing this model particular attention is given to establishing parameter values that are appropriate for the description of the alpha-helix. The last two subsections of section 1 consider ways in which Davydov's theory can be improved by introducing translational invariance and by a more accurate calculation of the soliton binding energy in the weak coupling limit. These considerations lead me to conclude that Davydov's theory is essentially classical although it has a clear quantum mechanical interpretation.

Section 2 describes theoretical and numerical results related to the initiation or "launching" of a soliton on the alpha-helix. Since this work is based on the standard theory of Davydov, the results obtained correspond to those of classical calculations. In section 3 various attempts to determine the effects of physiological temperature (310 K) are considered and evaluated. In section 4 some relevant experimental results are described. Of particular interest is acetanilide, a hydrogen-bonded molecular crystal that can be considered a "model protein". In the polaron jargon introduced above, self-localization in acetanilide is small, optical, and strongly coupled. Several generalizations of Davydov's theory are presented in section 5. These include the possibility of self-localization of electrons and also of protons. Finally some biological applications are sketched in section 6.

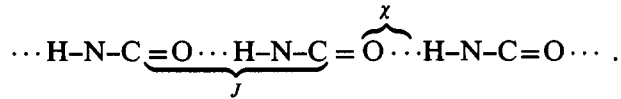
During the preparation of this review a draft copy was sent to about forty-five scientists who are actively interested in Davydov's soliton. Comments and criticisms were invited and those

received were of great assistance in putting the review into its final form. This process also led to disagreements, some of which could not be resolved. In those cases where my opinions differ from the views of others, I have so stated and have indicated the nature of the disagreement.

1. Low temperature theory

1.1. Davydov's theory for the alpha-helix

As shown in fig. 1, the alpha-helix region of a protein is a chain of amino acids held in helical shape by longitudinal hydrogen bonds. There are three "channels" (one is cross-hatched in fig. 1) running along the helix with the structure



Davydov and his colleagues have proposed the following quantum mechanical description of this system [DES78, Dav85, Dav91c]. The energy operator

$$\hat{H} = \hat{H}_{\text{ex}} + \hat{H}_{\text{ph}} + \hat{H}_{\text{int}} \quad (1.1)$$

is composed of three components. The first of these, called the "exciton" energy operator, is

$$\hat{H}_{\text{ex}} = \sum_{n,\alpha} \left[E_0 \hat{B}_{n\alpha}^\dagger \hat{B}_{n\alpha} - J \left(\hat{B}_{n\alpha}^\dagger \hat{B}_{n+1,\alpha} + \hat{B}_{n\alpha}^\dagger \hat{B}_{n-1,\alpha} \right) + L \left(\hat{B}_{n\alpha}^\dagger \hat{B}_{n,\alpha+1} + \hat{B}_{n\alpha}^\dagger \hat{B}_{n,\alpha-1} \right) \right]. \quad (1.2)$$

With reference to fig. 1, the subscript n ($= 1, 2, \dots, N$) counts molecules along a single channel and the subscript α ($= 1, 2, 3$) specifies a particular channel. Thus the index pair (n, α) selects an individual amino acid. The $\hat{B}_{n\alpha}^\dagger$ ($\hat{B}_{n\alpha}$) are boson creation (annihilation) operators for the Amide-I (CO stretching) oscillators, E_0 is the Amide-I site energy, $-J$ is the nearest neighbor dipole-dipole coupling energy along a channel, and $+L$ is the nearest neighbor dipole-dipole coupling energy between channels [NC76].

The phonon energy operator is

$$\hat{H}_{\text{ph}} = \frac{1}{2} \sum_{n,\alpha} \left[\hat{p}_{n\alpha}^2 / M + w (\hat{u}_{n+1,\alpha} - \hat{u}_{n\alpha})^2 \right], \quad (1.3)$$

where $\hat{p}_{n\alpha}$ and $\hat{u}_{n\alpha}$ are, respectively, the momentum and position operators for longitudinal displacement of an amino acid, M is the mass of an amino acid, and w is the spring constant of a hydrogen bond.

Finally the exciton-phonon interaction operator is

$$\hat{H}_{\text{int}} = \sum_{n,\alpha} \chi (\hat{u}_{n+1,\alpha} - \hat{u}_{n\alpha}) \hat{B}_{n\alpha}^\dagger \hat{B}_{n\alpha}, \quad (1.4)$$

where χ is the exciton-phonon coupling parameter, which determines the level of anharmonicity in the corresponding classical problem. From comparison with the first term on the right hand side of eq. (1.2) it is evident that

$$\chi = dE_0/dR, \quad (1.5)$$

where R is the length of the hydrogen bond directly adjacent to a particular Amide-I oscillator.

Equations (1.2)–(1.4) are rather unusual in the area of theoretical biophysics because each parameter can be independently determined.

Davydov's analysis of eq. (1.1) is based upon a product trial wave function [DE77, Dav85, Dav91c]

$$|\psi\rangle = |\Psi\rangle|\Phi\rangle \quad (1.6)$$

in which $|\Psi\rangle$ describes a single quantum excitation of the Amide-I system as

$$|\Psi\rangle = \sum_{n,\alpha} a_{n\alpha}(t) \hat{B}_{n\alpha}^\dagger |0\rangle_{\text{ex}}, \quad (1.7)$$

where $|0\rangle_{\text{ex}}$ is the vacuum state of the Amide-I oscillators, and $|\Phi\rangle$ is a coherent phonon state [Gla63a, Gla63b] for which

$$\langle\Phi|\hat{u}_{n\alpha}|\Phi\rangle = \beta_{n\alpha}(t), \quad (1.8a)$$

$$\langle\Phi|\hat{p}_{n\alpha}|\Phi\rangle = \pi_{n\alpha}(t). \quad (1.8b)$$

In eq. (1.7), $a_{n\alpha}(t)$ is a complex number representing the probability amplitude for finding a quantum of Amide-I energy in a particular amino acid, and in eqs. (1.8) $\beta_{n\alpha}(t)$ and $\pi_{n\alpha}(t)$ are, respectively, the average values of the longitudinal displacement and momentum of an amino acid.

Davydov minimizes the average value of $\hat{H}$ with respect to the product wave function by calculating $\langle\psi|\hat{H}|\psi\rangle = H(a_{n\alpha}, a_{n\alpha}^*, \beta_{n\alpha}, \pi_{n\alpha})$ and requiring that the corresponding Hamilton's equations

$$\begin{aligned} i\hbar\dot{a}_{n\alpha} = & [E_0 + W + \chi(\beta_{n+1,\alpha} - \beta_{n\alpha})]a_{n\alpha} \\ & - J(a_{n+1,\alpha} + a_{n-1,\alpha}) + L(a_{n,\alpha+1} + a_{n,\alpha-1}), \end{aligned} \quad (1.9a)$$

$$M\ddot{\beta}_{n\alpha} - w(\beta_{n+1,\alpha} - 2\beta_{n\alpha} + \beta_{n-1,\alpha}) = \chi(|a_{n\alpha}|^2 - |a_{n-1,\alpha}|^2), \quad (1.9b)$$

are satisfied. In eq. (1.9a)

$$W = \frac{1}{2} \sum_{n,\alpha} [M\dot{\beta}_{n\alpha}^2 + w(\beta_{n+1,\alpha} - \beta_{n\alpha})^2] \quad (1.10)$$

is the total phonon energy.

Under certain approximations, Davydov shows that eqs. (1.9) reduce to the nonlinear Schrödinger (NLS) equation that arises in the study of solitons [ZS71]. According to this theory, each soliton transports an amount of energy approximately equal to E_0 along the alpha-helix.

The solution of eq. (1.9b) can be divided into two components: (i) a particular solution that represents the “phonon dressing” of the soliton, and (ii) a solution of the homogeneous equation that can represent the effects of ambient temperature. At low temperature the second component can be neglected and this point of view is taken in the present section.

For a single quantum state, the normalization condition $\langle\psi|\psi\rangle = 1$ requires

$$\sum_{n,\alpha} |a_{n\alpha}|^2 = 1. \quad (1.11)$$

Equations (1.9) support a (symmetric) A-mode for which $a_{n1} = a_{n2} = a_{n3}$ and two E-modes with $[a_{n1}, a_{n2}, a_{n3}] \propto [1, \exp(\pm 2\pi i/3), \exp(\pm 4\pi i/3)]$. For the A-mode it is convenient to denote $a_n = \sqrt{3}a_{n\alpha}$ and $\beta_n = \beta_{n\alpha}$. Then eq. (1.11) becomes

$$\sum_n |a_n|^2 = 1 \quad (1.12)$$

and eqs. (1.9) and (1.10) reduce to

$$i\hbar \dot{a}_n = [E_0 + W + 2L + \chi(\beta_{n+1} - \beta_n)]a_n - J(a_{n+1} + a_{n-1}), \quad (1.13a)$$

$$3M\ddot{\beta}_n - 3w(\beta_{n+1} - 2\beta_n + \beta_{n-1}) = \chi(|a_n|^2 - |a_{n-1}|^2), \quad (1.13b)$$

$$W = \frac{1}{2} \sum_n [3M\dot{\beta}_n^2 + 3w(\beta_{n+1} - \beta_n)^2]. \quad (1.13c)$$

With the gauge transformation $a_n = \phi_n \exp[-it(E_0 + W + 2L - 2J)/\hbar]$, one obtains

$$i\hbar \dot{\phi}_n = \chi(\beta_{n+1} - \beta_n)\phi_n - J(\phi_{n+1} - 2\phi_n + \phi_{n-1}), \quad (1.14a)$$

$$3M\ddot{\beta}_n - 3w(\beta_{n+1} - 2\beta_n + \beta_{n-1}) = \chi(|\phi_n|^2 - |\phi_{n-1}|^2). \quad (1.14b)$$

Assuming a stationary solution ($\dot{\beta}_n = 0$) gives the discrete nonlinear Schrödinger equation

$$i(\hbar/J)\dot{\phi}_n + (\phi_{n+1} - 2\phi_n + \phi_{n-1}) + \sigma_0^{-1}|\phi_n|^2\phi_n = 0, \quad (1.15)$$

where

$$\sigma_0 = 3wJ/\chi^2 \quad (1.16)$$

and the corresponding lattice distortion is

$$\beta_{n+1} - \beta_n = -(\chi/3w)|\phi_n|^2. \quad (1.17)$$

For $\sigma_0 \gg 1$ and $N \gg 1$, an approximate stationary solution centered at $n = n_0$ has the squared amplitude

$$|\phi_n|^2 = \frac{1}{8\sigma_0} \operatorname{sech}^2\left(\frac{n - n_0}{4\sigma_0}\right), \quad (1.18)$$

thus σ_0 determines the size of a stationary soliton solution in units of the distance a ($= 4.5 \text{ \AA}$) between successive Amide-I oscillators (see fig. 1). More precisely, the distance between points where $|\phi_n|^2$ is equal to half of its maximum value is $7.05\sigma_0 a$.

The energy of the stationary soliton solution given in eq. (1.18) is

$$E_{\text{sol}}(0) = E_0 + 2L - 2J - E_B, \quad (1.19)$$

where

$$E_B = -\langle \psi | \hat{H}_{\text{int}} | \psi \rangle - W, \quad (1.20)$$

$$W = \frac{\chi^2}{6w} \sum_n |\phi_n|^4, \quad (1.21)$$

in the adiabatic approximation ($\dot{\beta}_n = 0$). For $N \gg 1$ eq. (1.15) also has a delocalized solution of small amplitude for which the phonon dressing is zero. This solution, called an *exciton*, has the energy

$$E_{\text{ex}}(0) = E_0 + 2L - 2J \quad (1.22)$$

when its group velocity is zero. Thus E_B , given in eq. (1.20), is the *binding energy* of a soliton with respect to an exciton.

The second term ($-W$) in eq. (1.20) is negative, but the first term is positive and of twice the magnitude. Thus

$$E_B = W, \quad (1.23)$$

where W is the energy of the phonon dressing given in eq. (1.21). From eqs. (1.18) and (1.21)

$$E_B = \frac{\chi^2}{6w} \frac{1}{(8\sigma_0)^2} \sum_n \text{sech}^4(n/4\sigma_0). \quad (1.24)$$

Approximating the sum by an integral one finds

$$E_B = \chi^4/216w^2J. \quad (1.25)$$

Equation (1.25) differs from previous results of Davydov and his colleagues because a somewhat different expression has been assumed for the interaction energy operator. Instead of eq. (1.4) Davydov et al. assume [DES78]

$$\hat{H}_{\text{int}} = \sum_{n,\alpha} \chi (\hat{u}_{n+1,\alpha} - \hat{u}_{n-1,\alpha}) \hat{B}_{n\alpha}^\dagger \hat{B}_{n\alpha}, \quad (1.26a)$$

while Davydov assumes [Dav85, Dav91c]

$$\hat{H}_{\text{int}} = \frac{1}{2} \chi \sum_n (\hat{u}_{n+1,\alpha} - \hat{u}_{n-1,\alpha}) \hat{B}_{n\alpha}^\dagger \hat{B}_{n\alpha}. \quad (1.26b)$$

Equations (1.26) both suppose that a particular Amide-I oscillator is equally influenced by stretching of the hydrogen bonds on either side of the peptide group (N-C=O). Equation (1.4) assumes, more correctly, that stretching of the hydrogen bond immediately adjacent to a C=O oscillator is of primary significance. See section 2 of ref. [CS90] for a detailed discussion of this point.

Within this model, the exciton number operator

$$\hat{Q} = \sum_{n,\alpha} \hat{B}_{n\alpha}^\dagger \hat{B}_{n\alpha} \quad (1.27)$$

commutes with the energy operator $\hat{H}$ defined in eq. (1.1). Thus the exciton number is conserved and exciton lifetime is infinitely long. In the real world, however, exciton lifetime is of the order of a picosecond [KCD84]. Davydov claims that the lifetime of a soliton should be longer than that of an exciton for the following reasons [Dav90]:

(i) The soliton does not interact with acoustic phonons because this interaction is accounted for in the phonon dressing.

(ii) Since the soliton velocity is less than that of longitudinal sound, it does not emit acoustic phonons.

(iii) The soliton binding energy (E_B) must be supplied for the soliton to decay into an exciton.

(iv) As the soliton passes a point in the helix, all molecules behind it remain displaced from their equilibrium positions by the distance $\chi/3w$. To destroy the soliton, these molecules must return to their equilibrium positions. Thus the Franck–Condon factor is small [Dav82a].

1.2. A simple model

Although the description of Amide-I oscillations on the alpha-helix proposed by Davydov and described above involve several approximations (mechanical interactions between oscillators are neglected [OSPN88] as are dipole–dipole interactions beyond nearest neighbors [Sco82a] and more complex phonon spectra [BE84, BE91]), it is often convenient to use an even simpler model for theoretical studies. Such a model can be motivated by the description of the symmetric mode in eqs. (1.13), for which

$$\hat{H} = \hat{H}_{\text{ex}} + \hat{H}_{\text{ph}} + \hat{H}_{\text{int}}, \quad (1.28)$$

where

$$\hat{H}_{\text{ex}} = \sum_n \left[E_0 \hat{B}_n^\dagger \hat{B}_n - J (\hat{B}_n^\dagger \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n-1}) \right], \quad (1.29a)$$

$$\hat{H}_{\text{ph}} = \frac{1}{2} \sum_n \left[\hat{p}_n^2 / \tilde{M} + \tilde{w} (\hat{u}_{n+1} - \hat{u}_n)^2 \right], \quad (1.29b)$$

$$\hat{H}_{\text{int}} = \chi \sum_n (\hat{u}_{n+1} - \hat{u}_n) \hat{B}_n^\dagger \hat{B}_n. \quad (1.29c)$$

In this model “bending” and “twisting” of the alpha-helix have been eliminated by assumption; only longitudinal compression waves are described. Thus it is important to use correct values for the parameters in the simple model. Comparison of eqs. (1.29) with eqs. (1.13) shows that

$$\tilde{w} = 3w, \quad \tilde{M} = 3M. \quad (1.30)$$

Several authors have extended Davydov's analysis to include an arbitrary number (Q) of Amide-I quanta. Zolotaryuk [Zol88] starts with a rather general trial wave function and considers two limiting cases: (i) a dilute gas of one-quantum solitons, and (ii) a single multi-quantum soliton. Case (ii) corresponds to that discussed by Kerr and Lomdahl [KL90] involving a product wave function as in eq. (1.6) but with eq. (1.7) changed to the Hartree approximate eigenstate that is used in the study of soliton propagation on an optical fiber [LH89, Wri90],

$$|\Psi\rangle = \frac{1}{\sqrt{Q!}} \left(\sum_n a_n(t) \hat{B}_n^\dagger \right)^Q |0\rangle_{\text{ex}}. \quad (1.31)$$

In this case the low temperature, gauge transformed system—corresponding to eqs. (1.14)—becomes

$$i\hbar \dot{\phi}_n = \chi (\beta_{n+1} - \beta_n) \phi_n - J (\phi_{n+1} - 2\phi_n + \phi_{n-1}), \quad (1.32a)$$

$$\tilde{M} \ddot{\beta}_n - \tilde{w} (\beta_{n+1} - 2\beta_n + \beta_{n-1}) = Q\chi (|\phi_n|^2 - |\phi_{n-1}|^2). \quad (1.32b)$$

Table 1
Parameter values for eqs. (1.29).

Parameter	Value	Units
χ	35–62	pN
$\tilde{w}$	39–58.5	N/m
$\tilde{M}$	5.7×10^{-25}	kg
J	1.55×10^{-22}	J

If this model is to be used to make quantitative statements about a real alpha-helix, it is necessary to know the value of the exciton–phonon coupling parameter χ . Ab initio calculations of this parameter have been conducted over the past decade [KK82a, KK82b, KK82c, LMC⁺86, Ost91] and recent values are:

$$\text{Kuprievich [Kup90]:} \quad -30 \text{ to } -60 \text{ pN}, \quad (1.33a)$$

$$\text{Pierce [Pie90]:} \quad +26 \text{ to } -7 \text{ pN}. \quad (1.33b)$$

These can be compared with estimates from experimental data:

$$\text{Careri et al. [CBG⁺84]:} \quad +62 \text{ pN}, \quad (1.34a)$$

$$\text{Scott [CS90]:} \quad +35 \text{ pN}. \quad (1.34b)$$

The ab initio calculations are particularly difficult for several reasons: (i) It is necessary to restrict the size of the molecule studied so the computations become economically feasible. At present all studies have been for formamide dimer, the simplest possible case. (ii) Even for the formamide dimer, it is necessary to restrict the size of the basis set, or level of truncation. (iii) Calculation of χ requires the evaluation of a third difference on the potential energy surface. (iv) As Careri has emphasized [Car90], the amide CO bond in a typical protein is *intermediate*, thus one expects its electron cloud to be a sensitive function of hydrogen bond length.

Østergård has recently completed a doctoral thesis that attempted to provide definitive ab initio calculations of χ [Ost91]. Although there is general agreement that the Born–Oppenheimer approximation (upon which such calculations are based) remains valid for studies of the intermediate CO bond [Pie91, Kup91, Ret91], he was unable to find convergence using currently available codes and computers. Thus the experimental values given in eqs. (1.34) are the most reliable estimates of χ at the present time.

Best currently available parameter values for the simple model are summarized in table 1. In this table χ is estimated to lie in the range of the experimental values given by eqs. (1.34). As indicated in eq. (1.30a), the longitudinal spring constant $\tilde{w}$ is three times that for a single hydrogen bond, which, in turn, is estimated experimentally to be 13 N/m [IS72] and from ab initio calculations as 19.5 N/m [KK82c]. The mass $\tilde{M}$ is three times the average mass of an amino acid in myosin [Sco82a]. Finally J has been calculated from electromagnetic theory [CN73, NC76, ELS84].

Although convenient for theoretical analysis and numerical computations, the model of eqs. (1.29) is an overly simplified representation of a real alpha-helix. Bending or twisting of the helix could lead to quite different values of χ and $\tilde{w}$. Mechanical coupling along the backbone might increase $\tilde{w}$ or alter J . More than nearest neighbor dipole–dipole interactions might be included. And so on. Quantitative results based on this model must therefore be considered provisional.

1.3. Introducing translational invariance

Although the quantitative description of Davydov's soliton (DS) presented above has a well defined quantum mechanical interpretation, the DS so described is essentially a classical object. Since its position and its momentum are well defined, the DS does not satisfy Heisenberg's uncertainty principle. This is to be expected because eqs. (1.9) (with W removed through a gauge transformation) are identical to the classical equations upon which the energy operator $\hat{H}$, defined in eq. (1.1), is based. If the alpha-helix is assumed to be of infinite length or of finite length with periodic boundary conditions, this deficiency can be stated in another way: Although the energy operator $\hat{H}$ commutes with the translation operator, the wavefunction defined in eqs. (1.6)–(1.10) is not an eigenfunction of the translation operator.

If the exciton phonon coupling (χ) is zero, the exact one-quantum wave functions for the simple model of eq. (1.29a) are

$$|\psi(k)\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N \exp(ikan) \hat{B}_n^\dagger |0\rangle_{\text{ex}}, \quad (1.35)$$

where N is the number of molecules in the simple model or one third the number of amino acids in the original alpha helix. With periodic boundary conditions (and N even) $k = 2\pi\nu/N$, where $\nu = 0, \pm 1, \pm 2, \dots, N/2$. The energy is

$$E(k) = E_0 - 2J \cos(ka). \quad (1.36)$$

Near the energy minimum

$$E(k) \doteq E_0 - 2J + \hbar^2 k^2 / 2m^*, \quad (1.37)$$

where

$$m^* = \hbar^2 / 2a^2 J. \quad (1.38)$$

Arbitrary initial conditions can be represented as a wave packet of these momentum eigenstates. With the particular initial conditions $a_0(0) = 1$ and $a_n(0) = 0$ for $n \neq 0$, the probability for finding a quantum of Amide-I oscillator energy at site n evolves with time as [KP84]

$$|a_n(t)|^2 = J_n^2(t/\tau), \quad (1.39)$$

where $\tau \equiv \hbar/2J$ and J_n is the Bessel function of the first kind of order n .

If the exciton phonon coupling (χ) is not zero, the quantum picture is more complicated. First of all there are one quantum *exciton* wavefunctions [Hol59]

$$|\psi(k)\rangle_{\text{ex}} = \frac{1}{\sqrt{N}} \sum_{n=1}^N \exp(ikan) \hat{B}_n^\dagger |0\rangle_{\text{ex}} + O(1/N), \quad (1.40)$$

which approach the exact $\chi = 0$ wavefunctions in the limit of large N .

In addition there are the *soliton* wavefunctions approximately described by Davydov as the product [DE77, Dav85, Dav91c]

$$|\psi_D(V)\rangle_{\text{sol}} = |\Psi\rangle |\Phi\rangle, \quad (1.41)$$

where

$$|\Psi\rangle = \sum_{n=1}^N a_n(t) \hat{B}_n^\dagger |0\rangle_{\text{ex}}, \quad (1.42a)$$

$$|\Phi\rangle = \exp\left(-\frac{i}{\hbar} \sum_{n=1}^N [\beta_n(t) \hat{p}_n - \pi_n(t) \hat{u}]\right) |0\rangle_{\text{ph}}, \quad (1.42b)$$

$\pi_n = \tilde{M} \dot{\beta}_n$ and $a_n(t)$ and $\beta_n(t)$ are determined by the system

$$i\hbar \dot{a}_n = [E_0 + W + \chi(\beta_{n+1} - \beta_n)] a_n - J(a_{n+1} + a_{n-1}), \quad (1.43a)$$

$$\tilde{M} \ddot{\beta}_n - \tilde{w}(\beta_{n+1} - 2\beta_n + \beta_{n-1}) = \chi(|a_n|^2 - |a_{n-1}|^2), \quad (1.43b)$$

$$W = \frac{1}{2} \sum_{n=1}^N [\tilde{M} \dot{\beta}_n^2 + \tilde{w}(\beta_{n+1} - \beta_n)^2]. \quad (1.43c)$$

Since the size parameter $7\tilde{w}J/\chi^2$ is in the range 11–52 for the parameters listed in table 1, an approximate solution of eqs. (1.43) is

$$a_n(t) \doteq \frac{1}{\sqrt{8\sigma_0(1-v^2/V_0^2)}} \text{sech}\left(\frac{a(n-n_0)-vt}{4a\sigma_0(1-v^2/V_0^2)}\right) \\ \times \exp\{(i/\hbar)[m^*va(n-n_0)-Et]\}, \quad (1.44a)$$

$$\beta_{n+1} - \beta_n \doteq -\frac{\chi}{\tilde{w}} \frac{|a_n|^2}{(1-v^2/V_0^2)}, \quad (1.44b)$$

where m^* is the effective mass of the exciton defined in eq. (1.38), and $V_0 = a(\tilde{w}/\tilde{M})^{1/2}$ is the sound speed,

$$\sigma_0 = \tilde{w}J/\chi^2, \quad (1.45)$$

$$E = E_0 - 2J - E_{\text{BD}} + \frac{1}{2}M_{\text{sol}}v^2 + O(v^4). \quad (1.46)$$

In eq. (1.46)

$$E_{\text{BD}} = \chi^4/24\tilde{w}^2J \quad (1.47)$$

is the *soliton binding energy*, and

$$M_{\text{sol}} = m^* \left(1 + \frac{1}{3}\tilde{M}\chi^4/\tilde{w}^3\hbar^2\right) \quad (1.48)$$

is its *effective mass*.

Substituting eqs. (1.44) into (1.42) gives an approximate expression for Davydov's product wavefunction for a soliton traveling with velocity v and centered at molecule $n = n_0$ at time $t = 0$. Let us denote this wavefunction as $|\psi_D(n_0, v)\rangle$. These results also arise in polaron theory.

By extending Pekar's theory [Pek54] to a continuum approximation of our simple model of eqs. (1.29), Young et al. [YSW79] obtain results similar to eqs. (1.41)–(1.48). Young et al. refer to this as the “moving Pekar theory”.

The problem of introducing translational invariance in polaron theory has been considered by Pekar [Pek54] and discussed in detail by Höhler [Hoh61]. Applying Höhler's approach to the moving Pekar theory, Young et al. construct soliton wave functions with translational symmetry as

$$|\tilde{\psi}(k)\rangle = C \sum_{n_0=1}^N \exp(ikan_0) |\psi_D(n_0, v)\rangle, \quad (1.49)$$

where

$$\hbar k = M_{\text{sol}} v \quad (1.50)$$

and C is a normalizing constant to be chosen so $\langle \tilde{\psi}(k) | \tilde{\psi}(k) \rangle = 1$. More recently Čápek and Krausová and (in greater detail) Wagner and Köngeter have shown that for any localized state there is an extended state with lower energy [CK87, WK89]. A time dependent wave packet of extended eigenstates then has the form

$$|\psi(t)\rangle = \sum_k F(k) |\tilde{\psi}(k)\rangle \quad (1.51)$$

and Heisenberg's uncertainty principle arises from the dispersion of this wave packet.

The reader should be aware that several scientists disagree with the main ideas presented in this section. In particular: (i) Although the importance of introducing translational invariance has been emphasized by Bolterauer, he claims that the lifetime of the soliton is the time for dispersion of the wave packet in eq. (1.51) [Bol90, BO91]. (ii) Davydov argues that the product wave function of eq. (1.41) with the parameters as in eqs. (1.44a,b) is an eigenfunction of the translation operator because the position n_0 can be located arbitrarily along the chain [Dav91a, Dav91b]. (iii) Finally Brown believes that I overemphasize the classical nature of the soliton described by eqs. (1.41) and (1.44a,b) [Bro91].

1.4. Accuracy of Davydov's wave function

Up to this point the aim has been to present the basic features of Davydov's theory in a simple and understandable way. Over the past few years a number of scientists have discussed the accuracy of this theory [BLW86, BWL86, KL87, CK87, SKS88, ZRRS88, Bro88, WK89] and some of these concerns are considered in the present section. Brown, Lindenberg and West, for example, showed that the product form of the trial wave function assumed by Davydov in eq. (1.41) is not exact [BLW86] and that Hamilton's equations derived from the averaged Hamiltonian

$$H(a_n, a_n^*, \beta_n, \pi_n) = \langle \psi | \hat{H} | \psi \rangle \quad (1.52)$$

do not produce a wave function that satisfies Schrödinger's equation exactly [BWL86]. Škrinjar et al. [SKS88] and Zhang et al. [ZRRS88] have suggested using the time dependent variational principle [LEK72] to optimize a more complicated trial wave function that was originally introduced by Davydov in connection with the question of thermal stability [Dav80]—see section 3.1. As Brown, Lindenberg and Wang have emphasized [BLW90], many of these questions have their roots in

polaron theory and the basic task is to determine what theoretical perspective is appropriate for a particular range of parameters.

Before taking up this task, it is necessary to sort out two details. First note that for the values presented in table 1 the soliton size parameter

$$\sigma_0 \equiv \tilde{w}J/\chi^2 \quad (1.53)$$

lies between 1.6 and 7.4. Since the half-width of $|a_n|^2$, defined in eq. (1.44a), is about seven times this value, a continuum approximation is justified. Second is the question of how to represent the phonon interaction operator. In eq. (1.29c) it is written

$$\hat{H}_{\text{int}} = \chi \sum_n (\hat{u}_{n+1} - \hat{u}_n) \hat{B}_n^\dagger \hat{B}_n, \quad (1.54)$$

where the sum is over the N molecules of the chain. In ref. [DES78] Davydov et al. use the form

$$\hat{H}_{\text{int}} = \tilde{\chi} \sum_n (\hat{u}_{n+1} - \hat{u}_{n-1}) \hat{B}_n^\dagger \hat{B}_n, \quad (1.55)$$

while in his book [Dav85] Davydov uses

$$\hat{H}_{\text{int}} = \frac{\chi}{2} \sum_n (\hat{u}_{n+1} - \hat{u}_{n-1}) \hat{B}_n^\dagger \hat{B}_n. \quad (1.56)$$

There are reasons for these choices. As has been noted above, eq. (1.54) should be used to describe phonon coupling of the Amide-I (C=O stretching) mode in alpha-helix because this mode interacts primarily with the adjacent hydrogen bond. Equation (1.55), on the other hand, is for excitons that have no favored orientation such as the electrons considered in polaron theory. Finally eq. (1.56) represents an attempt to approximate the effect of a unilateral interaction, as in eq. (1.54), by a bilateral interaction. Equation (1.55) is the form of the interaction that is commonly used in recent polaron studies [VF84, Tak84, Tak85, Nag87, BLW87, KL87, CK87, MS88, WK89, WBL89c, WBL89a, WBL89b, BC89, BI89, RN90]. It should be clear that $\tilde{\chi}$ in eq. (1.55) must be taken as half of the value given in table 1 when results from polaron studies are applied to Davydov's soliton.

In comparing various corrections to Davydov's theory, it is helpful to consider them as functions of a composite coupling parameter that, in the notation of Young et al. [YSW79], can be written as

$$4\pi\alpha \equiv \tilde{\chi}^2/2\tilde{w}\hbar\omega_D, \quad (1.57)$$

where $\omega_D = (\tilde{w}/\tilde{M})^{1/2}$ is the band edge for acoustic phonons (Debye frequency). If $4\pi\alpha \gg 1$, the coupling is said to be *strong*, and if $4\pi\alpha \ll 1$, it is said to be *weak*.

For the interaction energy as in eq. (1.56), Davydov calculates the soliton binding energy to be [Dav85]

$$E_{\text{BD}} = \tilde{\chi}^4/3\tilde{w}^2J. \quad (1.58)$$

Introducing translational symmetry as is indicated in section 1.3, Young et al. find for strong coupling that the binding energy is increased to

$$E_B = E_{\text{BD}} (1 + 3.439/4\pi\alpha + \dots). \quad (1.59)$$

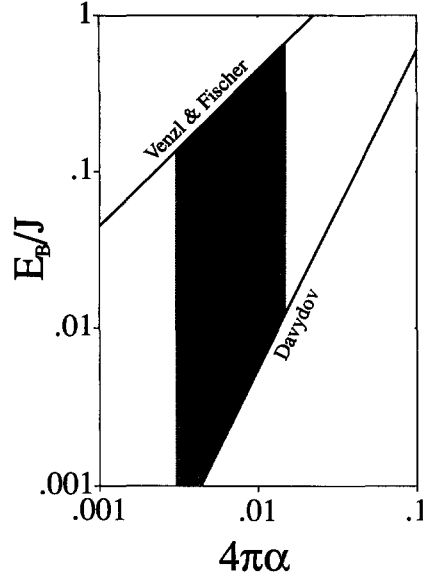


Fig. 2. Calculations of binding energy (E_B) in units of the dipole interaction energy (J) as a function of the coupling constant ($4\pi\alpha$), which is defined in eq. (1.57). $V = 0.08$ and the continuum approximation is valid. The curve labeled Davydov is calculated from eq. (1.61) and the curve labeled Venzl & Fischer is from eq. (1.62). The coupling constant for the alpha-helix lies in the range 0.003 to 0.015, which is indicated by the gray area.

Using the notation of Venzl and Fischer [VF84], of Nagy [Nag87], and of Wagner and Köngeter [WK89] it is convenient to define another composite parameter

$$V \equiv J/2\hbar\omega_D. \quad (1.60)$$

In terms of these two composite parameters ($4\pi\alpha$ and V) Davydov's expression for the soliton binding energy can be written as

$$E_B/J = \frac{1}{3} (4\pi\alpha/V)^2. \quad (1.61)$$

From the parameter values in table 1, $V = 0.08$. Using this value eq. (1.61) is plotted in fig. 2.

Equation (1.61) indicates that the binding energy rises quadratically with the coupling constant ($4\pi\alpha$) for weak coupling. This differs with weak coupling calculations from polaron theory, which indicate a linear dependence [LLP53]. The book by Haken (see chapter V of [Hak76]) presents a particularly clear discussion of this point. Thus Venzl and Fischer [VF84] focus their attention on a critical value of the coupling constant where the linear dependence from weak coupling polaron theory intersects with the quadratic dependence from soliton theory. Below this critical value, they suggest, the binding energy should be computed from weak coupling polaron theory and above it by Davydov's soliton theory or, equivalently, the strong coupling polaron theory of Pekar [Pek54] and Holstein [Hol59, Hol81]. Weak coupling calculations according to Venzl and Fischer lead to the relation

$$E_B/J = 3.541 \cdot 4\pi\alpha/V, \quad (1.62)$$

which is also plotted in fig. 2 for $V = 0.08$.

There have been two recent attempts to improve upon these calculations. The first by Nagy [Nag87] generalizes a variational method introduced by Huybrechts [Huy77] and obtains a ground state energy that approaches that for a weakly coupled polaron as $(4\pi\alpha) \rightarrow 0$. For strong coupling this analysis indicates a binding energy that is slightly larger than that obtained by Davydov. A second attempt by Wagner and Köngeter [WK89] uses the Fulton–Gouterman transcription [FG61] to construct the ground state. Their analysis indicates that the binding energy approaches the weakly coupled polaron result for weak coupling and the Pekar–Davydov result for strong coupling. Furthermore they obtain groundstate energies that are lower than those of Venzl and Fischer [VF84] and of Nagy [Nag87].

Recalling that

$$\tilde{\chi} = \chi/2, \quad (1.63)$$

where χ is the value from table 1, the coupling constant for the simple model of the alpha-helix that was described in section 1.2 lies in the range

$$4\pi\alpha = 0.003 \text{ to } 0.015. \quad (1.64)$$

Thus the simple model of the alpha-helix is weakly coupled, and from fig. 2, Davydov's theory is expected to underestimate the binding energy by about two orders of magnitude for a typical alpha-helix.

There is a physical reason why one expects the weak coupling limit to be appropriate for Davydov's soliton and not necessarily for a polaron. To see this, note from eqs. (1.53) and (1.57) that

$$\sigma_0(4\pi\alpha) = J/8\hbar\omega_D. \quad (1.65)$$

For Davydov's soliton, J is a dipole–dipole interaction energy of about 10 cm^{-1} while $\hbar\omega_D$ (the Debye energy) is about 100 cm^{-1} . Thus if $\sigma_0 > 1$, so the continuum approximation holds, then $4\pi\alpha \ll 1$ and the coupling is weak. For a polaron, however, J is the overlap energy between the tails of adjacent electronic states. This can be larger than $\hbar\omega_D$, so the strong coupling approximation is often appropriate.

In the weak coupling limit ($4\pi\alpha \ll 1$) Davydov's analysis—see eq. (1.48)—gives

$$M_{\text{sol}} = m^* \left[1 + \frac{2}{3}(4\pi\alpha)^2 + \dots \right]. \quad (1.66)$$

Thus from eq. (1.64) we see that the effective mass of Davydov's soliton is

$$M_{\text{sol}} \doteq m^* \quad (1.67)$$

to an accuracy that is better than 0.1%. This means that the wave packet describing the uncertainties in position and momentum of Davydov's soliton—see eq. (1.51)—will disperse almost exactly as the wave packet for a weakly coupled polaron.

From eq. (1.38), the value of J from table 1 and the fact (see fig. 1) that the longitudinal spacing between Amide-I oscillators in an alpha-helix $a = 4.5 \text{ \AA}$,

$$m^* = 1.76 \times 10^{-28} \text{ kg}, \quad (1.68)$$

which is 193 times the mass of an electron or about 10% of the mass of a proton. Heisenberg's uncertainty principle for linear motion is

$$\Delta x \Delta p \geq \hbar/2, \quad (1.69)$$

where Δx and Δp are, respectively, the rms uncertainties in position and momentum. Using eq. (1.68) this becomes

$$\Delta x \Delta v \geq 3 \times 10^{-7} \text{ m}^2/\text{s}, \quad (1.70)$$

where Δv is the rms uncertainty in velocity. If we take eq. (1.51) to represent a wave packet for which Δx is (rather arbitrarily) chosen to localize the soliton to five turns of the helix, then $\Delta v = 133 \text{ m/s}$ or about 3% of the speed of longitudinal sound waves. Thus the quantum corrections to Davydov's soliton that are required by Heisenberg's uncertainty relation are important.

Throughout the foregoing discussion it has been assumed that the continuum approximation was justified. If this is not the case, different analytical techniques are required. In section 4.1 some experimental results related to the self-trapping of Amide-I energy in crystalline acetanilide are discussed for which the continuum approximation is not appropriate. In this case we find that Davydov's method of analysis gives reasonable agreement with a more detailed theory.

Brown and Ivić have recently constructed a theory that interpolates between weak and strong coupling [BI89]. Although their theory captures the transition between linear and quadratic dependence of binding energy on the coupling parameter ($4\pi\alpha$), their comments concerning the application of this theory to the alpha-helix are, in my view, misleading because they underestimate the size of the classical soliton. This is my reasoning:

(i) The value chosen for the longitudinal spring constant was that for a single hydrogen bond and therefore misses the factor of three in eq. (1.30a).

(ii) The interaction energy operator used was as in eq. (1.55) rather than eq. (1.56); thus the factor of two in eq. (1.63) was neglected.

(iii) With reference to table 1, the lowest value (13 N/m) for the hydrogen bond spring constant (w) was selected and the largest value (62 pN) was chosen for χ .

Putting these items together, the mean value of the soliton size parameter ($\sigma_0 \approx 4.5$) is underestimated by a factor of about twenty-four. Brown states: " σ_0 plays a subordinate role in the Brown-Ivić theory and the essential conclusions of the latter are insensitive to the above changes in parameters" [Bro91].

The quantum Monte Carlo (QMC) technique has been used by Wang et al. to numerically calculate eigenstates of the simple Hamiltonian given in eq. (1.29) that represent a single quantum of self-trapped Amide-I energy [WBL89a, WBL90]. Since these QMC calculations do not require analytic approximations, the accuracy of the results is limited only by grid-size effects and statistical errors due to limited run times. Wang et al. estimate their numerical accuracy to be a few per cent, and they find evidence that at low temperatures the excitation and the lattice participate cooperatively in a coherent structure. Clearly such calculations are useful in settling various theoretical questions. Since they were carried out for the same interaction Hamiltonian and parameter values used by Brown and Ivić, however, they are not the best characterization of the equilibrium properties of the alpha-helix that could be achieved by this method.

Recently Bernstein has been studying the dimer (two coupled molecules) in an attempt to determine the accuracy of Davydov's wavefunction [Ber91a, Ber91b]. It turns out that this quantum mechanical problem can be solved exactly, thus providing another context for comparing the various analytical approaches.

2. Launching the soliton

2.1. General considerations

In sections 1.3 and 1.4 we have become aware of two important limitations of the standard theory of Davydov's soliton that is presented in sections 1.1 and 1.2: (i) The standard Davydov soliton is a classical object. It does not satisfy Heisenberg's uncertainty relation unless (or until) translational invariance is imposed on the wavefunction. (ii) For weak coupling the standard Davydov theory leads to a binding energy that is much less than that obtained from polaron theory. [See fig. 2 and compare eqs. (1.61) and (1.62).]

Nonetheless a substantial amount of scientific work has appeared that is based upon the standard theory. How should this work be viewed? Since the basic dynamical equations of Davydov's theory—eqs. (1.43), for example—are identical (under a gauge transformation that eliminates W) to the classical equations underlying the quantum mechanical energy operator, this work can always be considered as a classical analysis of the problem. In the intermediate and strong coupling range ($4\pi\alpha \geq 1$), the translationally invariant version of the Pekar–Holstein–Davydov theory developed by Höhler [Hoh61] and by Young et al. [YSW79] is in reasonable agreement with more detailed studies by Venzl and Fischer [VF84], Nagy [Nag87], and Wagner and Köngeter [WK89].

It is in this spirit that the results presented in the present section should be considered.

2.2. Inverse scattering calculations

Davydov's product wave function for the simple model of the alpha-helix described in section 1.2 is presented in eqs. (1.42). Assuming $\beta_n = 0$ (the adiabatic approximation) in eqs. (1.43) and introducing the gauge transformation

$$a_n = \phi_n \exp[-it(E_0 + W - 2J)/\hbar] \quad (2.1)$$

leads to the discrete nonlinear Schrödinger equation

$$i\dot{\phi}_n + (\phi_{n+1} - 2\phi_n + \phi_{n-1}) + \sigma_0^{-1}|\phi_n|^2\phi_n = 0, \quad (2.2)$$

where time is measured in units of $\hbar/J$ and $\sigma_0 = \tilde{w}J/\chi^2$ determines the size of a soliton in units of the lattice spacing a ($= 4.5 \text{ \AA}$). For $\sigma_0 > 1$, it is appropriate to use the continuum approximation

$$i\frac{\partial\phi}{\partial t} + \frac{\partial^2\phi}{\partial x^2} + \frac{1}{\sigma_0}|\phi|^2\phi = 0, \quad (2.3)$$

where distance (x) is measured in units of the lattice spacing. Equation (2.3) is “exactly integrable” using the inverse scattering transform (ISTM) method devised by Zakharov and Shabat [ZS71] and this property can be used to calculate the probability of a soliton emerging from specified initial conditions [HMS81, Sco82a, BD83, Sco84, BGVV88, Bri90]. For a semi-infinite system with $n = 1, 2, 3, \dots, \infty$ and initial conditions

$$a_1(0) = 1, \quad a_n(0) = 0 \quad \text{for } n > 1, \quad (2.4)$$

soliton analysis of eq. (2.3) indicates a threshold value of χ for soliton formation. This value is

$$\chi_{t1} = \frac{1}{4}\pi\sqrt{\tilde{w}J}. \quad (2.5)$$

At this point two comments are in order: (i) The initial conditions indicated in eqs. (2.4) are particularly convenient for numerical studies, and (ii) the appearance of a threshold is an artifact of these initial conditions. From eq. (1.44a) it is evident that for the initial conditions

$$a_n(0) = \frac{1}{\sqrt{8\sigma_0(1-v^2/V_0^2)}} \operatorname{sech}\left(\frac{a(n-n_0)}{4a_0(1-v^2/V_0^2)}\right) \exp[(i/\hbar)m^*va(n-n_0)], \quad (2.6)$$

where $n_0 \gg \sigma_0$, a soliton (traveling with velocity v) would be launched at any value of χ .

2.3. Numerical studies

The primary motivation for early numerical studies of Davydov's equations (1.13) was to determine whether the magnitude of the exciton-phonon coupling constant (χ) in a real alpha-helix is large enough to support soliton formation. To proceed conservatively, therefore, two assumptions were made that tend to inhibit soliton formation.

(i) Initial Amide-I energy was localized in a single turn of the helix. A typical initial condition to launch a symmetric (A-mode) soliton was

$$a_{11}(0) = a_{12}(0) = a_{13}(0) = \sqrt{2/3}, \quad a_{n\alpha}(0) = 0 \quad \text{for } n > 1. \quad (2.7)$$

This puts two Amide-I quanta into the first turn at one end of the helix.

(ii) All important dipole-dipole interactions were considered. Thus, in addition to the nearest interactions (J and L) in eqs. (1.13), eight longer range interaction terms were included.

A numerical code that embodied these two assumptions was developed over a period of several years in the early 1980s [HMS81, Sco82a, MS84]. This code represented a helix of 200 turns (as noted in section 6.3, this is about the length of the alpha-helix in myosin) with free end boundary conditions. With the initial conditions indicated in eqs. (2.7), a threshold for soliton formation was observed at an exciton-phonon coupling constant

$$\chi_{t1} \simeq 50 \text{ pN}. \quad (2.8)$$

This compares favorably with the range of experimental values indicated in eq. (1.34). It should be emphasized that assumption (i) listed above is conservative. If one assumes that the initial transition is directly into a soliton state, then the threshold for soliton formation is zero (no threshold).

It is also interesting to note that if the two quanta of the initial conditions (2.7) are accounted for by the wave function in eq. (1.31) suggested by Kerr and Lomdahl [KL90], then the exciton-phonon coupling threshold for the simple model becomes

$$\chi_{t1} = \frac{1}{4}\pi\sqrt{\tilde{w}J/Q}, \quad (2.9)$$

where Q is the number of Amide-I quanta carried by the soliton. With $Q = 2$ and parameter values from table 1, eq. (2.9) indicates $\chi_{t1} = 43$ to 53 pN, which is in reasonable agreement with eq. (2.8).

The conclusion that the exciton-phonon coupling constant for a real alpha-helix is large enough to ensure soliton formation has been challenged recently by Mechtly and Shaw [MS88]. They construct a wave function for the simple Hamiltonian operator of eqs. (1.29) that is based upon a time dependent unitary transformation acting on the phonon Fock space. This wave function is

more accurate than Davydov's product wave function shown in eqs. (1.41) to (1.44), because it is an eigenfunction of the translation operator over the range

$$0 \leq 4\pi\alpha \leq 1, \quad (2.10)$$

where the coupling parameter $4\pi\alpha$ is defined in eq. (1.57). The ordinary differential equations that describe time evolution of the wave function parameters in this formulation differ significantly from those in Davydov's formulation, eq. (1.43). Since numerical studies of their equations with the initial conditions indicated in eqs. (2.4) show soliton formation only for

$$4\pi\alpha > 1, \quad (2.11)$$

Mechtly and Shaw conclude that the previous studies [Sco82a, MS84] underestimate the threshold value of χ for soliton formation by a factor of at least four.

From the point of view presented in section 1.3, however, the numerical results of Mechtly and Shaw are to be expected. Since their wave function must be a packet of momentum eigenstates over the parameter range indicated in eq. (2.10), it is only by moving outside this range—as indicated in eq. (2.11)—that soliton formation can be observed in their formulation. Equation (1.66) for the effective mass of a soliton shows why an exact wave function should become more classical as the coupling parameter $4\pi\alpha$ is increased above unity: The effective mass rises quadratically and Heisenberg's uncertainty relation in the form

$$\Delta x \Delta v \geq \hbar/2M_{\text{sol}} \quad (2.12)$$

becomes closer to that of a classical object.

In a related study, Rhodes and Nicholls [RN90] also introduce a wave function that is based on a time dependent unitary transformation. Again numerical computations show delocalization of the wave packet as is required by the uncertainty relation of eq. (2.12).

The salient points of this subsection are summarized as follows:

(i) Numerical studies of Davydov's (classical) equations show a threshold level of the exciton-phonon coupling parameter (χ_{tl}) above which Amide-I energy, that is initially localized on a single turn of the alpha-helix, is converted into a soliton.

(ii) This threshold level ($\chi_{\text{tl}} \simeq 50$ pN) is in good agreement with independent experimental measurements ($\chi = 35$ to 62 pN).

(iii) Numerical studies of an exact time dependent wave function, as in eq. (1.51), cannot show localization of Amide-I energy because this wave packet must obey the Heisenberg uncertainty relation given in eq. (2.12).

(iv) Numerical studies using improved wave functions (i.e., improved over Davydov's product wave function) that suggest higher threshold levels of the exciton-phonon coupling constant are misleading. The higher threshold is an artifact of the wave packet nature of the improved wave function.

(v) These considerations are conservative. If it is assumed that the initial state has the shape of a soliton—see eq. (2.6)—then the threshold value of the exciton-phonon coupling parameter is reduced to zero.

2.4. The effects of disorder

The effects of random variations in the parameter values have been discussed in detail by Förner and his colleagues [MFL89, FL90, For91b, For91e]. To this end the simple model of eqs. (1.29)

has been generalized to

$$\hat{H}_{\text{ex}} = \sum_n \left[(E_0 + E_n) \hat{B}_n^\dagger \hat{B}_n - J_n (\hat{B}_n^\dagger \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n-1}) \right], \quad (2.13a)$$

$$\hat{H}_{\text{ph}} = \frac{1}{2} \sum_n \left[\hat{p}_n^2 / \tilde{M}_n + \tilde{w}_n (\hat{u}_{n+1} - \hat{u}_n) \right], \quad (2.13b)$$

$$\hat{H}_{\text{int}} = \sum_n \chi_n (\hat{u}_{n+1} - \hat{u}_n) \hat{B}_n^\dagger \hat{B}_n, \quad (2.13c)$$

where E_n , J_n , $\tilde{M}_n$, $\tilde{w}_n$, and χ_n are permitted to vary randomly with n . Using Davydov's product wave function defined in eqs. (1.41) and (1.42), a gauge transformed version of eqs. (1.43) becomes

$$i\hbar \dot{a}_n = [E_n + \chi_n (\beta_{n+1} - \beta_n)] a_n - (J_n a_{n+1} + J_{n-1} a_{n-1}), \quad (2.14a)$$

$$\tilde{M}_n \ddot{\beta}_n + \tilde{w}_{n+1} (\beta_{n+1} - \beta_n) - \tilde{w}_n (\beta_n - \beta_{n-1}) = \chi_n |a_n|^2 - \chi_{n-1} |a_{n-1}|^2. \quad (2.14b)$$

With $n = 1, 2, \dots, 200$ and the initial conditions of eqs. (2.4), the details of soliton formation were studied under the following conditions.

(i) *Mass disorder.* Disorder in the masses was found to destroy the soliton only for very large variations. For random values over the range $0.66M \leq \tilde{M}_n \leq 1.79M$ (where $M = 114 m_p$ or one-third of the value in table 1) virtually no change in soliton dynamics was found. This range corresponds to the mass variation of natural amino acids.

(ii) *Spring constant disorder.* Keeping the mass disorder as in (i) and introducing a random variation in the spring constant over the range $0.8w \leq \tilde{w}_n \leq 1.2w$ (where $w = 13 \text{ N/m}$ or one-third of the lowest value in table 1) introduced no change in the soliton dynamics.

(iii) *Dipole coupling disorder.* For variations in J_n alone or together with the mass disorder in (i) the soliton is stable over the range $0.95J \leq J_n \leq 1.05J$ (where J is the value in table 1).

(iv) *Exciton-phonon coupling disorder.* For variations in χ_n alone or together with the mass disorder in (i) the soliton is stable over the range $0.82\chi \leq \chi_n \leq 1.2\chi$ (where χ is the maximum value in table 1).

(v) *Combined disorder.* With the mass disorder as in (i), the maximum variations in $\tilde{w}$, J , and χ that permit soliton formation from the initial conditions of eqs. (2.4) are $\pm 0.2w$, $\pm 0.025J$, and $\pm 0.1\chi$, respectively.

(vi) *Diagonal disorder.* A random variation greater than $\pm 8 \text{ cm}^{-1}$ in E_n prevented soliton formation.

Thus disorders in the dipole-dipole interaction energies (J_n) and in the site energies (E_n) appear to be of greatest importance. These results are conservative, however, because they apply to the emergence of a soliton from the initial conditions of eqs. (2.4). With the initial conditions indicated in eq. (2.6), larger ranges of disorder are expected to be tolerated.

2.5. Biochemical considerations

It has been emphasized by Knox et al. [KMW90] that the mechanism for injecting energy into (and extracting it from) a Davydov soliton has not yet received adequate theoretical study. Phonon coupling of the sort described in section 1 is not sufficient because this mechanism conserves Q

and therefore permits the soliton only to give its kinetic energy to the lattice, not the rest energy E_0 [Sco82a]. Other mechanisms must be supposed to transfer the main portion of the soliton energy, and one possibility is related to McClare's "excimer" [McC72].

The basic idea can be appreciated by considering the classical system

$$\left\{ i \frac{d}{dt} - \begin{bmatrix} \omega_1 & -J(x) \\ -J(x) & \omega_2 \end{bmatrix} \right\} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \bar{0}, \quad (2.15)$$

where a_1 represents the complex amplitude of a soliton and a_2 is the complex amplitude of some external oscillator that is either feeding energy into or accepting it from the soliton. Then $J(x)$ is the dipole-dipole coupling between the soliton and the external oscillator, and this coupling depends upon the distance, x , between the two. The soliton is assumed to have an unperturbed frequency ω_1 and the external oscillator has frequency ω_2 . Thus a significant parameter is the detuning

$$\Delta \equiv \omega_2 - \omega_1. \quad (2.16)$$

As the mechanical coordinate is decreased from x_1 to x_2 , the energy transmitted will be [Sco90]

$$U = \sqrt{J^2(x_2) + \Delta^2/4} - \sqrt{J^2(x_1) + \Delta^2/4}. \quad (2.17)$$

Under the most favorable circumstances, $x_1 = \infty$ and $\Delta = 0$ (perfect tuning). Then

$$U = J(x_2) = \mu_1 \mu_2 / 4\pi \epsilon_0 x_2^3, \quad (2.18)$$

where μ_1 and μ_2 are, respectively, the transition dipole moments of the soliton and the external oscillator and ϵ_0 ($= 10^{-9}/36\pi$ F/m) is the dielectric permittivity of free space. If both the soliton and the external oscillator have the dipole moment of an Amide-I oscillator (i.e., $\mu_1 = \mu_2 = 0.3$ D $= 1.3 \times 10^{-30}$ C m) and $x_2 = 2$ Å then eq. (2.18) indicates that only about 6% of the Amide-I vibrational energy can be converted to mechanical work. This is an order of magnitude less than measured values of muscular efficiency [ABL+83].

To increase this efficiency, larger dipole moments are needed. One might, for example, suppose that the soliton interacts with a low-lying electronic state so $\mu_2 \gg \mu_1$.

3. The effects of physiological temperature

3.1. Soliton stability

The effects of temperature on an alpha-helix soliton were first discussed by Davydov in a paper that appeared in 1980 [Dav80]. Starting with the simple model of eqs. (1.29), his analysis proceeds by expressing the longitudinal displacement operator ($\hat{u}$) in terms of phonon creation and annihilation operators ($\hat{b}_q^\dagger$ and $\hat{b}_q$) as

$$\hat{u} = \left(\frac{a\hbar}{2\tilde{M}NV_0} \right)^{1/2} \sum_q \frac{\exp(inaq)}{\sqrt{|aq|}} (\hat{b}_q + \hat{b}_q^\dagger), \quad (3.1)$$

where q is the phonon wave number. The trial wave function is assumed to be the sum of products

$$|\psi\rangle = \sum_n |\Psi(n)\rangle |\Phi(n, q)\rangle, \quad (3.2)$$

with

$$|\Psi(n)\rangle = a_n(t) \hat{B}_n^\dagger |0\rangle_{\text{ex}}. \quad (3.3)$$

The phonon factor, however, is taken as

$$|\Phi(n, q)\rangle = \prod_q \hat{S}_{nq} |\nu\rangle, \quad (3.4)$$

where $|\nu\rangle$ is a thermally populated phonon state,

$$\hat{S}_{nq}(t) \equiv \exp[\beta_{qn}^*(t) \hat{b}_q - \beta_{qn}(t) \hat{b}_q^\dagger] \quad (3.5)$$

is a unitary displacement operator, and the β_{qn} represent modulated plane waves,

$$\beta_{qn}(t) = \tilde{\beta}_{qn}(t) \exp(-inaq). \quad (3.6)$$

The actions of the unitary operator, $\hat{S}_{qn}$, upon $\hat{b}_q$ and $\hat{b}_q^*$ lead to their displacement by β_{qn} and β_{qn}^* because

$$\hat{\tilde{b}}_q \equiv \hat{S}_{nq}^\dagger \hat{b}_q \hat{S}_{nq} \quad (3.7a)$$

$$= \hat{b}_q - \beta_{qn}. \quad (3.7b)$$

Thus exciton-phonon coupling leads to vibrations about new equilibrium positions, the β_{qn} . These vibrations are characterized by new creation and annihilation operators $\hat{\tilde{b}}_q^\dagger$ and $\hat{\tilde{b}}_q$.

Averaging the Hamiltonian operator defined in eqs. (1.29) with respect to $|\psi\rangle$ leads to [CHC+88]

$$\begin{aligned} \langle \psi | \hat{H} | \psi \rangle = & \sum_n \left[E_0 |a_n|^2 - J (a_n^* a_{n-1} e^{-W_{n,n-1}} + a_n^* a_{n+1} e^{-W_{n,n+1}}) \right] \\ & - |a_n|^2 \sum_q F_q e^{iqna} (\beta_{qn} + \beta_{-qn}^*) + |a_n|^2 \sum_q \hbar \omega_q (\bar{\nu}_q + |\beta_{qn}|^2) + \frac{1}{2} \sum_q \hbar \omega_q \end{aligned} \quad (3.8)$$

where

$$W_{n,n\pm 1} = \sum_q \left[(\bar{\nu}_q + \frac{1}{2}) |\beta_{qn} - \beta_{qn\pm 1}|^2 + \frac{1}{2} (\beta_{qn} \beta_{qn\pm 1}^* - \beta_{qn}^* \beta_{qn\pm 1}) \right] \quad (3.9)$$

is a Debye-Waller exponent, ω_q is the frequency of a phonon state, $\bar{\nu}_q = [\exp(\hbar \omega_q / kT) - 1]^{-1}$ is the thermal equilibrium number of phonons in the state with wavenumber q , and

$$F_q \equiv \left(\frac{\hbar}{2\tilde{M}N\omega_q} \right)^{1/2} 2i\chi \sin |qa|. \quad (3.10)$$

Davydov proceeds by treating $\langle \psi | \hat{H} | \psi \rangle$ as a classical Hamiltonian that determines the dynamics of the $a_n(t)$ and the $\beta_{qn}(t)$. Cruzeiro et al. [CHC+88] have studied these Hamiltonian equations numerically, but Davydov introduces a gauge transformation from a_n to ϕ_n and makes a continuum

approximation. He finds that the dynamical equations derived from eq. (3.8) can be approximated as the nonlinear Schrödinger equation

$$i\hbar \frac{\partial \phi}{\partial t} + \frac{\hbar^2}{2m^*} \frac{\partial^2 \phi}{\partial x^2} + G|\phi|^2 \phi \doteq 0, \quad (3.11)$$

where

$$m^* \equiv (\hbar^2/2a^2J) \exp(W) \quad (3.12)$$

and W is an average Debye–Waller exponent. Above the Debye temperature, $kT > \hbar(\tilde{\omega}/\tilde{M})^{1/2}$, he obtains the approximate expression

$$G \doteq (\chi^2/\tilde{\omega})(1 - T/T_0), \quad (3.13)$$

where

$$T_0 = \frac{1}{2}\pi(\hbar/k)\sqrt{\tilde{\omega}/\tilde{M}}. \quad (3.14)$$

Thus at temperature T_0 the anharmonic term in eq. (3.11) is reduced to zero and the localized (soliton) state no longer exists. For the parameter values in table 1, T_0 lies between 100 K and 120 K.

The approximations made in Davydov's thermal analysis have been criticized by several authors [SKS88, ZRRS88, SIZ90]. In particular it has been suggested that his treatment of $\langle \psi | \hat{H} | \psi \rangle$ as a classical Hamiltonian leads to unnecessary errors. More precise is the time dependent variation principle [LEK72]

$$\delta \int_{t_1}^{t_2} dt \langle \psi(t) | i\hbar d/dt - \hat{H} | \psi(t) \rangle = 0, \quad (3.15)$$

which leads to correction terms in the dynamical equations for the $a_n(t)$ and the $\beta_{qn}(t)$. The improvements in accuracy of this approach for realistic model parameters have not yet been estimated.

Another means to calculate the equilibrium properties of soliton states is the quantum Monte Carlo (QMC) technique of Wang et al. [WBL89a, WBL90]. These authors find that the soliton state is no longer an equilibrium solution above a temperature of just a few degrees kelvin, which is an order of magnitude less than the value obtained from eq. (3.14). As was noted in section 1.4, Wang et al. use parameters for the simple Hamiltonian of eqs. (1.29) that greatly underestimate the size of the soliton. It would be interesting to have these QMC calculations repeated for the more realistic parameters of table 1.

3.2. Numerical studies of thermal lifetime

Davydov's thermal analysis assumes that the number of solitons is conserved. Reference to eqs. (1.13), however, suggests that this assumption may not be correct. As temperature is increased from zero degrees kelvin, the homogeneous part of the solution to eq. (1.13b) will become increasingly important. This will contribute a random component to the factor $(\beta_{n+1} - \beta_n)$ in eq. (1.13a), which, in turn, may induce scattering from soliton (localized) states to exciton (extended) states.

Since the exciton states have short lifetimes (ca. 10^{-12} s) [KCD84], this mechanism could lead to rapid loss of Amide-I vibrational energy.

As was noted at the end of section 1.1, Davydov argues that the soliton state is stable because (among other reasons) its destruction requires a longitudinal displacement of the alpha-helix by a distance equal to the contraction introduced by the soliton. From eq. (1.44b) this contraction is equal to $\chi/\tilde{w}(1 - v^2/V_0^2)$ for the simple model of eqs. (1.29). To quantify this effect, one can calculate the Franck-Condon factor [Sco87] for a stationary soliton ($v = 0$) to make a transition to an exciton state. This factor is the overlap integral between the phonon wavefunctions before and after the transition. For the simple model of eqs. (1.29) with $n = 0, 1, 2, \dots, N$ at zero degrees kelvin, the Franck-Condon factor is

$$F(0) = \exp \left[-\frac{4}{\pi} \frac{\chi^2}{\hbar \omega_D \tilde{w}} n_0 \left(1 - \frac{n_0}{N} \right) \right], \quad (3.16)$$

where n_0 is the location of the center of the soliton and $\omega_D = (\tilde{w}/\tilde{M})^{1/2}$. At higher temperatures

$$F(T) < F(0). \quad (3.17)$$

The value of $F(0)$ given in eq. (3.16) equals unity for $n_0 = 0$ or N (i.e. at the ends of the molecule) and attains a minimum value of

$$F_{\min}(0) = \exp \left(-\frac{N}{\pi} \frac{\chi^2}{\hbar \omega_D \tilde{w}} \right) \quad (3.18)$$

at the center. With $N = 200$ and other parameters as in table 1, $F_{\min}(0) \doteq 0.042$.

This stabilizing quantum effect is not apparent from a direct integration of eqs. (1.32) because these are identical (under a gauge transformation) to the classical equations underlying the energy operator defined in eqs. (1.29). Nonetheless several groups have studied thermal lifetime of the soliton state by direct integration of eqs. (1.32) with thermal noise introduced into eq. (1.32b) [LK85, LMC⁺86, LK90]. Such studies are logically sound if one concludes that the lifetime is long enough to be biologically interesting (a positive conclusion) for then the Franck-Condon factor provides a margin of safety for that conclusion. On the other hand, if one decides from a numerical study of eqs. (1.32) that the soliton lifetime is too short to be biologically useful (a negative conclusion), then it is necessary to consider whether the Franck-Condon factor (a quantum effect) could render this negative conclusion invalid.

By far the most ambitious numerical study of soliton lifetime is being carried out by Förner et al. [MFL89, FL90, For91a, For91c, For91d, For91e]. They assume $N = 200$ and introduce a random energy of NkT onto eq. (1.32b) 120 ps (typically) prior to the start of the numerical experiment. As in section 2.2, the soliton is introduced by the initial conditions

$$\phi_1(0) = 1, \quad \phi_n(0) = 0 \quad \text{for } n > 1, \quad (3.19)$$

thus the number of Amide-I quanta Q is equal to unity in these calculations.

Typical results are shown in fig. 3a for $T = 0$ K and in fig. 3b for $T = 300$ K and a rather wide range of the parameters χ and $\tilde{w}$. If a soliton was observed to form and propagate across the entire length of the helix (200 turns), that point in parameter space is marked by a solid black circle ($\bullet$). If the Amide-I vibrational energy is observed to be self-trapped but pinned to the lattice, that point is indicated by a cross ($\times$). At other (unmarked) points in the parameter space solitons did not develop or soliton lifetime was not long enough to permit transit of the entire molecule.

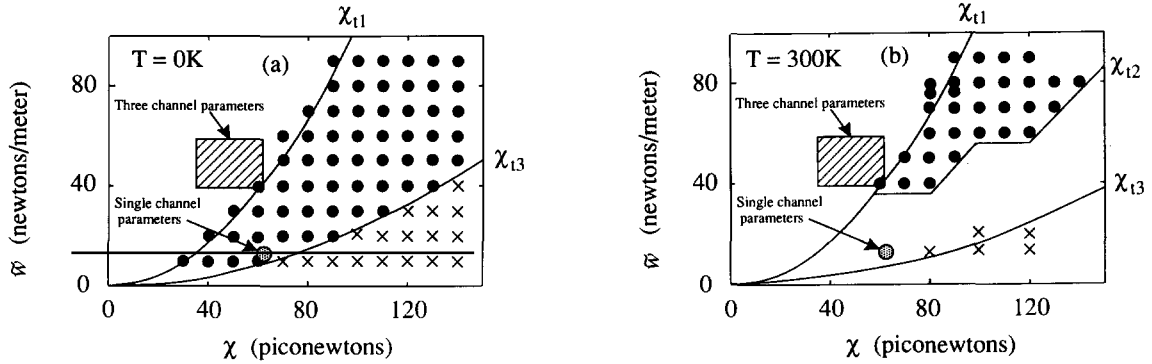


Fig. 3. Numerical results obtained by Förner for the propagation of solitons on the system of eqs. (1.32) with $\tilde{M} = 1.9 \times 10^{-25}$ kg, $N = 200$ and the initial conditions of eqs. (3.19) [FL90,FL91a]. A soliton that forms and propagates across the entire molecule is indicated by (•) and a pinned soliton by (×). (a) $T = 0$ K, (b) $T = 300$ K.

Consider first the calculations for $T = 0$ K that are shown in fig. 3a. There are two critical levels of the exciton–phonon coupling parameter: χ_{t1} and χ_{t3} . The larger of these, χ_{t3} , is readily understood from eq. (1.45) as the condition for the soliton size to be approximately equal to the lattice spacing. Larger values of χ lead to pinning. The smaller threshold, χ_{t1} , indicates the level of anharmonicity for which the soliton will form with the initial conditions of eqs. (3.19). This is a question that was discussed in detail in sections 2.1 and 2.2 and the smooth curve labeled “ χ_{t1} ” in fig. 3a is plotted directly from eq. (2.9) with $Q = 1$. The dependence of this agreement upon the dipole coupling parameter, J , has been verified in recent numerical studies by Förner [For91a].

Next consider the calculations for $T = 300$ K that are shown in fig. 3b. The threshold level for soliton formation (χ_{t1}) and for pinning (χ_{t3}) remain unchanged but a new threshold level (χ_{t2}) appears that is clearly dependent on temperature. With reference to eqs. (1.32), this threshold indicates the value of χ above which the random (thermal) part of the term $\chi(\beta_{n+1} - \beta_n)\phi_n$ becomes large enough to induce significant scattering from soliton to exciton states.

Finally let us consider where a “real alpha-helix” lies in this parameter space. Remember that these results are based upon the simple model of eqs. (1.29), which is a poor representation of the alpha-helix shown in fig. 1. Nevertheless the parameter values given in table 1 provide the best available representation of an alpha-helix in this model and are indicated as the “three channel parameters” in fig. 3. These parameters lie well to the left of χ_{t2} in fig. 3b and one must conclude that the soliton does not experience significant thermal scattering into exciton states at 300 K. This conclusion is reinforced by the fact that the Franck–Condon factor, indicated in eq. (3.16), has not been included in this numerical study.

Since the region of the “three channel parameters” lies to the left of χ_{t1} in figs. 3a and 3b, one might be tempted to conclude that the exciton–phonon coupling is too weak for the soliton to form with the initial conditions of eqs. (3.19). This question, however, has been studied in much greater detail and the results are summarized above in section 2. In particular, the agreement between the value of χ necessary for soliton formation, given in eq. (2.8), compares favorably with the range of experimental values given in eqs. (1.34). Thus soliton formation is not unexpected on a real alpha-helix.

Also indicated in fig. 3 are the “single channel parameters” that Lomdahl and Kerr used to characterize a real alpha-helix in the context of the simple model of eqs. (1.29) [LK85, LK90].

They assume that a real alpha-helix can be represented by using the parameters of a single channel (shown cross-hatched in fig. 1) with the interaction Hamiltonian as in eq. (1.55). As was noted in section 1.4, this underestimates the size of the classical soliton by a factor of about twenty-four. Although these "single channel parameters" are often referred to as the "standard" or "widely accepted" alpha-helix parameters, such terms should be used with caution. At the present time it is best to take the parameters from table 1 and use the interaction Hamiltonian of eq. (1.54).

3.3. Analytic studies of thermal lifetime

An analytic approach to the problem of the thermal lifetime of the Davydov soliton has recently been presented by Schweitzer and Cottingham [CS89, SC90]. Using a representation developed by Eremko et al. [EGV85] they write a continuum approximation to the simple Hamiltonian operator in eqs. (1.29) and are able to obtain an analytical expression for the average thermal lifetime of the Davydov soliton within lowest order perturbation theory. The basis states for this representation of the Hamiltonian, transformed to a frame of reference moving with the soliton velocity, consist of the Davydov soliton state and delocalized exciton states. The Hamiltonian is of the form $\hat{H}_0 + \hat{V}$, where the exact eigenstates of $\hat{H}_0$ are the Davydov soliton with phonons relative to the distorted lattice and extended exciton states with phonons relative to the undistorted lattice. The soliton state is the single bound state for the potential well associated with the lattice distortion described by the coherent phonon state, and the exciton states are the continuum of unbounded states for this potential. The binding energy is just the Davydov soliton binding energy and the perturbation $\hat{V}$ couples these two sets of states.

Schweitzer and Cottingham calculate in lowest order perturbation theory the rate of transition from an initial state corresponding to the Davydov soliton plus phonons relative to the distorted lattice. The effect of temperature is included by averaging over a thermal equilibrium ensemble of initial phonon states. For the "single channel" parameters of fig. 3 they found a soliton lifetime of 0.32 ps at 300 K [SC90]. Furthermore, at the upper left hand corner of the "three channel" box in fig. 3, a soliton lifetime of 3.5 ps was calculated. Thus they conclude that "the Davydov soliton is not nearly stable enough to be a practical mechanism for energy transfer for any reasonable choice of parameters" [Sch91]. However, they are planning to make lifetime calculations over a larger range of parameter values in the near future.

There are several reasons for suggesting caution in accepting this conclusion:

(i) The calculation only considers the standard Davydov soliton described in eqs. (1.41)–(1.44) of section 1.3. Other localized states could have very different lifetimes.

(ii) Recently Förner has begun to find approximate agreement between his numerical calculations and the analytic results of Schweitzer and Cottingham [For91a]. Notice from fig. 3 that the parameter values chosen to date by Schweitzer and Cottingham lie in regions of the parameter space where Förner also finds that the soliton has a short lifetime.

(iii) Davydov points out that the division of the Hamiltonian into $\hat{H}_0 + \hat{V}$ is not unique [Dav91a]. For another division into (say) $\hat{H}'_0 + \hat{V}'$ Schweitzer and Cottingham would necessarily compute different lifetimes.

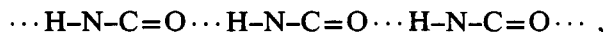
(iv) The analysis of Schweitzer and Cottingham does not include the extra binding energy (see fig. 2) that arises from the weak coupling calculations of Venzl and Fischer, which were described in section 1.4.

4. Experimental observations

4.1. The acetanilide story

In the early 1970s, Giorgio Careri began a novel experimental investigation of protein dynamics. His idea was to circumvent many of the problems arising in the study of real protein by looking at hydrogen bonded crystals that might be regarded as "model proteins". The basic idea can be appreciated by comparing the molecular structure of acetanilide ($\text{CH}_3\text{CONHC}_6\text{H}_5$) or ACN and the structure of a typical polypeptide chain in natural protein as shown in fig. 4. The similarity of bond lengths and angles for the peptide group (HNCO) suggests that the dynamical behavior of ACN might provide clues to the corresponding behavior of natural protein.

The comparison becomes more striking if we compare the structure of crystalline ACN shown in fig. 5 with that of the alpha helix shown in fig. 1. In both cases one finds hydrogen bonded peptide channels with the atomic structure



which was the starting point for the theories reviewed in section 1.

Careri's intuition was rewarded by the discovery of an anomalous resonance at 1650 cm^{-1} in crystalline ACN [Car73], as shown in fig. 6. A decade of effort by Careri and his co-workers at the University of Rome established that this 1650 cm^{-1} peak was an Amide-I (CO stretching)

	ACETANILIDE	POLYPEPTIDE CHAIN
$\text{R}[\text{\AA}]$	2.97	2.79 ± 0.12
$\text{CO}[\text{\AA}]$	1.22	1.24
$\text{CN}[\text{\AA}]$	1.35	1.32
$\widehat{\text{OCN}}[^\circ]$	123	124

Fig. 4. A comparison of the peptide geometry in acetanilide (ACN) and in natural protein.

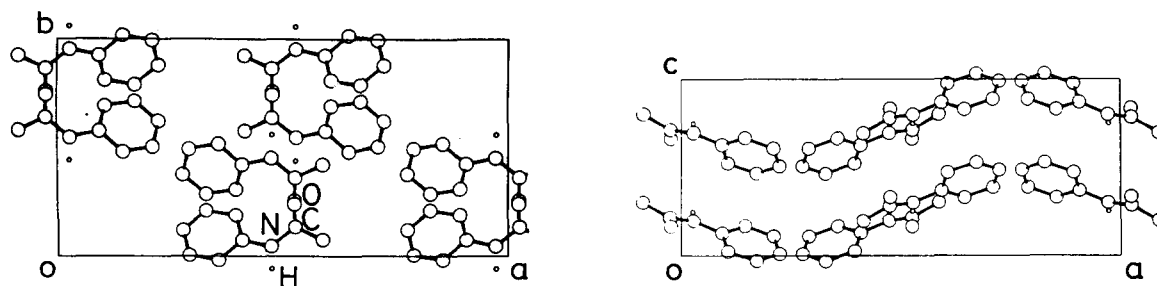


Fig. 5. A unit cell of crystalline acetanilide (ACN).

resonance that could not be conventionally assigned to a Fermi resonance, Davydov splitting, or a phase transition [CBG+84]. It is important to note that Careri's discovery and the subsequent experimental studies were made without any knowledge of Davydov's theory of self-trapped molecular vibrations. It was only in 1982, after he had exhausted all possibilities for conventional assignment of the 1650 cm^{-1} band, that dynamical self-trapping was considered.

The original idea [CBG+84] was that the 1650 cm^{-1} band is evidence of a self-trapped Amide-

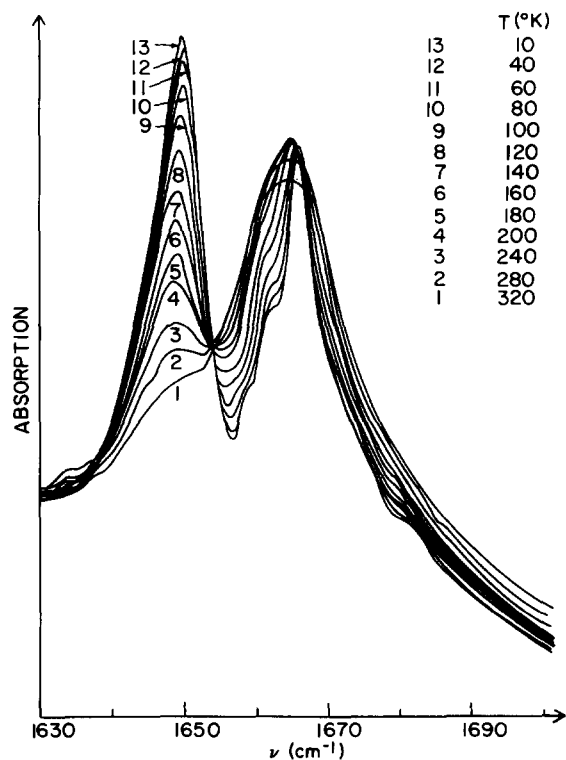


Fig. 6. An infrared absorption spectrum of crystalline acetanilide in the Amide-I region. Notice the anomalous peak at 1650 cm^{-1} , which indicates a self-trapped state.

I state arising through interaction with an intramolecular or optical phonon mode. An optical mode was considered more likely than an acoustic mode as the source of self-trapping because it avoids the small Franck–Condon factor indicated in eq. (3.16), and a small Franck–Condon factor would, in turn, preclude the strong optical absorption that is evident in the data of fig. 6. The theoretical model, therefore, is the linear chain with an Einstein oscillator of frequency ω_0 coupled to each Amide-I mode through a phonon coupling parameter χ_0 . In the same spirit as section 1, an appropriate energy operator is

$$\hat{H} = \hat{H}_{\text{ex}} + \hat{H}_{\text{ph}} + \hat{H}_{\text{int}}, \quad (4.1)$$

where

$$\hat{H}_{\text{ex}} = \sum_n \left[E_0 \hat{B}_n^\dagger \hat{B}_n - J (\hat{B}_n^\dagger \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n-1}) \right], \quad (4.2a)$$

$$\hat{H}_{\text{ph}} = \frac{1}{2} \sum_n \left(\hat{p}_n^2/m + K \hat{q}_n^2 \right), \quad (4.2b)$$

$$\hat{H}_{\text{int}} = \chi_0 \sum_n \hat{q}_n \hat{B}_n^\dagger \hat{B}_n, \quad (4.2c)$$

and n ($= 1, 2, \dots, N$) counts the ACN molecules in the linear chain. Equation (4.2a) is identical to eq. (1.29a) in the simple model of section 1. Equation (4.2b) represents the energy of the Einstein oscillator; thus

$$\omega_0 = \sqrt{K/m}. \quad (4.3)$$

Finally eq. (4.2c) represents the coupling of the Einstein oscillator to the Amide-I mode through the exciton–phonon coupling parameter χ_0 .

At this point it is convenient to transform from position and momentum operators ($\hat{q}_n$ and $\hat{p}_n$) describing the Einstein oscillator to boson creation and annihilation operators ($\hat{b}_n^\dagger$ and $\hat{b}_n$). Thus

$$\hat{b}_n = (2m\hbar\omega_0)^{-1/2} (m\omega_0 \hat{q}_n + i\hat{p}_n), \quad (4.4a)$$

$$\hat{b}_n^\dagger = (2m\hbar\omega_0)^{-1/2} (m\omega_0 \hat{q}_n - i\hat{p}_n), \quad (4.4b)$$

whereupon eqs. (4.2b) and (4.2c) become

$$\hat{H}_{\text{ph}} = \sum_n \hbar\omega_0 (\hat{b}_n^\dagger \hat{b}_n + \frac{1}{2}), \quad (4.5a)$$

$$\hat{H}_{\text{int}} = \lambda_0 \sum_n (\hat{b}_n + \hat{b}_n^\dagger) \hat{B}_n^\dagger \hat{B}_n, \quad (4.5b)$$

where

$$\lambda_0 \equiv \sqrt{\hbar/2m\omega_0} \chi_0. \quad (4.6)$$

Let us assume a single Amide-I quantum ($Q = 1$) and a temperature of zero degrees kelvin and use the method of Davydov (outlined in section 1) to study the system. He would assume a product trial wave function

$$|\psi\rangle = |\Psi\rangle|\Phi\rangle, \quad (4.7)$$

where

$$|\Psi\rangle = \sum_n a_n(t) \hat{B}_n^\dagger |0\rangle_{\text{ex}} \quad (4.8)$$

and $|\Phi\rangle$ is a product of a coherent states for the Einstein oscillators,

$$|\Phi\rangle = \sum_n \exp [\beta_n(t) \hat{b}_n^\dagger - \beta_n^*(t) \hat{b}_n] |0\rangle_{\text{ph}}. \quad (4.9)$$

Then treating $\langle \psi | \hat{H} | \psi \rangle$ as a classical Hamiltonian leads to the dynamic equations

$$i\hbar \dot{a}_n = [E_0 + W + \lambda_0(\beta_n + \beta_n^*)] a_n - J(a_{n+1} + a_{n-1}), \quad (4.10a)$$

$$i\hbar \dot{\beta}_n = \hbar\omega_0 \beta_n + \lambda_0 |a_n|^2, \quad (4.10b)$$

where $W = N\hbar\omega_0/2$ is the total ground state energy of the Einstein oscillators. Assuming $\dot{\beta}_n = 0$ (the adiabatic assumption) and introducing the gauge transformation $a_n = \phi_n \exp[-it(E_0 + W)/\hbar]$ leads to the discrete nonlinear Schrödinger equation

$$i\hbar \dot{\phi}_n + \frac{2\lambda_0^2}{\hbar\omega_0} |\phi_n|^2 \phi_n + J(\phi_{n+1} + \phi_{n-1}) = 0, \quad (4.11)$$

where from eqs. (4.3) and (4.6)

$$2\lambda_0^2/\hbar\omega_0 = \chi_0^2/K. \quad (4.12)$$

In section 1 we considered this discrete nonlinear Schrödinger equation for the ratio of the dispersive parameter (J) to the anharmonic parameter (χ_0^2/K) greater than unity. This ratio was between 1.6 and 7.4 for the alpha-helix parameter from table 1, which justifies the continuum approximation. In ACN, on the other hand, it is evident that [CBG⁺84, SGSC85]

$$J \ll \chi_0^2/K. \quad (4.13)$$

To see this note first that, from detailed electromagnetic calculations [ELS84], $J = 4 \text{ cm}^{-1}$. Second, if the inequality is satisfied then the binding energy of a self-trapped state (soliton) with respect to a ($k = 0$) exciton is

$$E_{\text{BD}} = (\chi_0^2/2K) - 2J. \quad (4.14)$$

Finally, in fig. 6, the 1650 cm^{-1} peak is assigned to a soliton and the peak at 1665 cm^{-1} to an exciton. This implies that $E_{\text{BD}} = 15 \text{ cm}^{-1}$ so that $(\chi_0^2/2K) = 23 \text{ cm}^{-1}$ and confirms that eq. (4.13) is satisfied.

Davydov's theory for crystalline acetanilide, modeled as in eqs. (4.2), indicates the soliton binding energy is given by eq. (4.14). In the context of fig. 3a, ACN lies well within the region of $\times$'s and the soliton is not mobile, but "pinned" to one molecule in the lattice [ELS84]; thus

$$M_{\text{sol}} = \infty. \quad (4.15)$$

Since J is small compared with the anharmonic parameter (χ_0^2/K) it is possible to solve the problem by using a perturbation expansion in J . This has two advantages: (i) It provides

quantitative estimates of the accuracy of Davydov's analysis and the conditions under which it breaks down, and (ii) it gives solutions when the number of Amide-I quanta Q is greater than unity. To proceed, it is convenient to write eq. (4.1) in the form

$$\hat{H} = \sum_n \hat{h}_n - J\hat{V}, \quad (4.16)$$

where

$$\hat{h}_n = [E_0 + \lambda_0(\hat{b}_n + \hat{b}_n^\dagger)] \hat{B}_n^\dagger \hat{B}_n + \hbar\omega_0 \hat{b}_n^\dagger \hat{b}_n, \quad (4.17)$$

$$\hat{V} = \sum_n (\hat{B}_n^\dagger \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n-1}), \quad (4.18)$$

and the ground state phonon energy ($N\hbar\omega_0/2$) has been ignored.

Equation (4.17) is the energy operator for the "displaced harmonic oscillator" (see section 6 of [Hak76]), for which exact eigenfunctions are known. Dropping the molecule index n ($= 1, 2, \dots, N$) for typographical convenience, these eigenfunctions can be written in the form [SBJ89]

$$\hat{h}|u\rangle = E|u\rangle, \quad (4.19a)$$

$$|u\rangle = |Q\rangle|\phi(Q, \tilde{n})\rangle, \quad (4.19b)$$

$$|\phi(Q, \tilde{n})\rangle = \sqrt{\tilde{n}!} \exp[-\frac{1}{2}(Q\lambda_0/\hbar\omega_0)^2] \sum_{m=0}^{\infty} \frac{(-Q\lambda_0/\hbar\omega_0)^{m-\tilde{n}}}{\sqrt{m!}} L_{\tilde{n}}^{m-\tilde{n}}[(Q\lambda_0/\hbar\omega_0)^2]|m\rangle, \quad (4.19c)$$

$$E = QE_0 + \tilde{n}\hbar\omega_0 - Q^2\lambda_0^2/\hbar\omega_0. \quad (4.19d)$$

In these equations $\hat{B}^\dagger \hat{B}|Q\rangle = Q|Q\rangle$ and $\hat{b}^\dagger \hat{b}|m\rangle = m|m\rangle$. Also $L_n^m[\cdot]$ is an associated Laguerre polynomial using the normalization in Gradshteyn and Ryzhik [GR80] and not (for example) that in Morse and Feshbach [MF53]. As $Q\lambda_0 \rightarrow 0$, $|\phi(Q, \tilde{n})\rangle$ reduces to the phonon number state $|\tilde{n}\rangle$, but in general $|\phi(Q, \tilde{n})\rangle$ is a sum over all phonon number states.

Returning to eq. (4.16) and assuming periodic boundary conditions, a zero order estimate of the wave function is

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N \exp(ikn)|u_n\rangle, \quad (4.20)$$

where $k = 2\pi\nu/N$ and $\nu = 0, \pm 1, \dots, N/2$ [or $\pm(N-1)/2$] for N even (or odd). As in eq. (1.49), this wave function is an eigenfunction of the translational symmetry operator. With $Q = 1$, a first order estimate of the energy is

$$E(k) \doteq E_0 - \frac{\lambda_0^2}{\hbar\omega_0} - 2J \exp[-(\lambda_0/\hbar\omega_0)^2] \cos k, \quad (4.21)$$

where " $\doteq$ " means that terms of order

$$\hbar\omega_0 J^2/\lambda_0^2 = 2KJ^2/\chi_0^2 \quad (4.22)$$

are neglected. From our previous quantitative considerations related to eqs. (4.13) and (4.14), the numerical value of eq. (4.22) is 0.7 cm^{-1} . This is commensurate with experimental precision [CBG⁺84].

Since the wavelength of infrared light is large compared with a lattice spacing, only $k = 0$ exciton states will be observed experimentally. The energy of such a state is

$$E_{\text{ex}}(k = 0) = E_0 - 2J. \quad (4.23)$$

If a wave packet of the eigenfunctions in eq. (4.20) is constructed to localize a quantum of Amide-I energy on a single molecule, then all values of k must be equally represented. From eq. (4.21) the average energy of the wavepacket will be

$$E_{\text{sol}} = E_0 - \lambda_0^2/\hbar\omega_0. \quad (4.24)$$

Assigning $E_{\text{ex}}(k = 0)$ to the 1665 cm^{-1} peak in fig. 6 and E_{sol} to the 1650 cm^{-1} peak leads to the same estimate of the anharmonic parameter $\lambda_0^2/\hbar\omega_0$ ($= \chi_0^2/2K = 23 \text{ cm}^{-1}$) as Davydov's analysis, but eq. (4.21) shows an additional contribution to the binding energy of a $k = 0$ soliton. Thus

$$E_{\text{B}} = \frac{\chi_0^2}{2K} - 2J\{1 - \exp[-(\lambda_0/\hbar\omega_0)^2]\}. \quad (4.25)$$

Also the effective mass of a $k = 0$ soliton will be

$$M_{\text{sol}} = m^* \exp[(\lambda_0/\hbar\omega_0)^2], \quad (4.26)$$

where $m^* = \hbar^2/2a^2J$ is the effective mass of a bare exciton.

Comparison of eqs. (4.25) and (4.26) with eqs. (4.14) and (4.15) provides quantitative estimates for some of the errors arising from Davydov's analysis. Evidently as $(\lambda_0/\hbar\omega_0)$ becomes greater than unity, these errors become smaller.

In addition to translational invariance, the wave function of eq. (4.20) has two properties that are not shared by Davydov's wave function of eqs. (4.7) through (4.10):

(i) *Higher levels of Amide-I excitation* ($Q > 1$). From eq. (4.19d) one expects to observe an overtone series at the frequencies $\nu(Q)$, where $\hbar\nu(Q) = E(Q) - E(0)$; thus in cm^{-1}

$$\nu(Q) = 1673Q - 23Q^2, \quad (4.27)$$

where the coefficient of Q has been chosen so $\nu(1) = 1650 \text{ cm}^{-1}$ as is shown in fig. 6. Measured overtone frequencies are given in table 2. If the coefficients in eq. (4.27) are adjusted to

$$\nu(Q) = 1674Q - 24.7Q^2, \quad (4.28)$$

the measured values are predicted to an rms difference of 0.6 cm^{-1} . Thus a better value for the parameter $\lambda_0^2/\hbar\omega_0 = \chi_0^2/2K$ is

$$\lambda_0^2/\hbar\omega_0 = 24.7 \text{ cm}^{-1}. \quad (4.29)$$

(ii) *Higher levels of phonon excitation* ($\tilde{n} > 0$). With $Q = 1$ and $\tilde{n} = 0$, eq. (4.19c) reduces to a coherent state [Gla63a, Gla63b]. With $\hat{p}_n = \lambda_0|a_n|^2/\hbar\omega_0$ and with $|a_n|^2 = 1$ (Amide-I energy localized on a single molecule), eq. (4.9) reduces to the same coherent state.

Table 2
A comparison of measured overtone frequencies from [SGSC85] with theoretical values calculated from eq. (4.27). All values are in cm^{-1} .

Q	$\nu(Q)$ measured	$\nu(Q)$ eq.(4.27)
1	1650.0 ± 0.5	1650
2	3250 ± 1	3254
3	4803 ± 3	4812
4	6304 ± 5	6324

What about $\tilde{n} > 0$? For $Q = 0$ eq. (4.19c) reduces to

$$|\phi(0, \tilde{n})\rangle = |\tilde{n}\rangle, \quad (4.30)$$

a phonon number state for which $b^\dagger b |\tilde{n}\rangle = \tilde{n} |\tilde{n}\rangle$. These ground states will be thermally populated when $\hbar\omega_0 \leq k_B T$. Taking this effect into consideration leads to a quantitative understanding of the temperature dependence of the 1650 cm^{-1} band, which is shown in fig. 6.

As has been pointed out by Alexander and Krumhansl [AK86], a correct calculation of the temperature dependent intensity of the 1650 cm^{-1} band follows that for a “zero phonon” band of a color center [Fit68]. This point is made with particular clarity by Krumhansl in ref. [Kru87]. Thus one seeks the sum of all transitions from the ground states $|\tilde{n}\rangle$ to first excited ($Q = 1$) states with $m = \tilde{n}$. Since the ground states are thermally populated with probabilities

$$P(\tilde{n}) = \left[1 - \exp\left(-\frac{\hbar\omega_0}{k_B T}\right) \right] \exp\left(-\tilde{n} \frac{\hbar\omega_0}{k_B T}\right), \quad (4.31)$$

the temperature dependence of the 1650 cm^{-1} band should be given by

$$W(T) = \sum_{\tilde{n}=0}^{\infty} P(\tilde{n}) |\langle \tilde{n} | \phi(1, m) \rangle|^2. \quad (4.32)$$

Using identity number (8.976) from Gradshteyn and Ryzhik [GR80], this sum is readily computed to be

$$W(T) = \exp\left[-\left(\frac{\lambda_0}{\hbar\omega_0}\right)^2 \coth\left(\frac{\hbar\omega_0}{2k_B T}\right)\right] I_0\left[\left(\frac{\lambda_0}{\hbar\omega_0}\right)^2 \text{csch}\left(\frac{\hbar\omega_0}{2k_B T}\right)\right], \quad (4.33)$$

where $I_0[\cdot]$ is the modified Bessel function of the first kind, of order zero.

Experimental values for $W(T)$ have been obtained by Bigio and Johnston [SBJ89] using Raman scattering rather than infrared absorption (as in fig. 6) because the temperature in the focal volume of the laser beam could be determined from the ratio of anti-Stokes to Stokes intensities. These results are presented in table 3.

Using the value for $\lambda_0^2/\hbar\omega_0$ given in eq. (4.29), the only free parameter in eq. (4.33) is ω_0 . Fitting calculations from eq. (4.33) to the data in table 3 requires $\hbar\omega_0$ to be 75 cm^{-1} . The data in table 3 can also be fitted to within experimental uncertainty by assuming coupling to several optical modes [SBJ89]. Thus the underlying theoretical model is altered to the linear chain with

Table 3
Normalized integrated Raman intensity values $W(T)/W(0)$ of the 1650 cm^{-1} band in single crystal acetanilide. Temperature uncertainties are due primarily to the method for estimating the heating caused by absorption in the focal volume of the laser beam [SBJ89].
The σ 's give rms errors.

Temperature (K)	σ	Relative intensity	σ
21	4	1.00	0.016
21	4	0.996	0.016
53	4	0.892	0.019
53	4	0.895	0.019
100	10	0.710	0.022
100	10	0.726	0.022
149	6	0.496	0.025
149	6	0.507	0.025
227	12	0.356	0.028
227	12	0.346	0.028
305	8	0.170	0.032
305	8	0.169	0.032

Amide-I modes coupled to Einstein oscillators with frequencies $\omega_1, \omega_2, \dots, \omega_M$ through the phonon coupling parameters $\chi_1, \chi_2, \dots, \chi_M$, whereupon eq. (4.29) changes to

$$\sum_{j=1}^M \frac{\lambda_j^2}{\hbar\omega_j} = 24.7\text{ cm}^{-1}. \quad (4.34)$$

Assuming that all M of the optical modes have the same frequency (ω_1), the relative intensity becomes

$$\frac{W(T)}{W(0)} = \left(\frac{\exp[-(24.7/M\omega_1) \coth(0.7174\omega_1/T)] I_0[(24.7/M\omega_1) \operatorname{csch}(0.7174\omega_1/T)]}{\exp(-24.7/M\omega_1)} \right)^M, \quad (4.35)$$

where ω_1 is in cm^{-1} and T is in degrees kelvin. With $\omega_1 = 71\text{ cm}^{-1}$ and $M \geq 5$, eq. (4.35) is in excellent agreement with the data from table 3.

The above discussion appears to present a convincing case for the fundamental conclusion of Careri et al. that self-trapping by optical phonon modes (along the lines introduced by Davydov) can explain the 1650 cm^{-1} band in crystalline acetanilide [CBG⁺84]. Furthermore, there is evidence for a similar red shifted, temperature sensitive, Amide-I band in parachloroacetanilide [CBG⁺84], methyl-three-fluorated-acetanilide [Bar91] and, perhaps, in N-methyl acetamide [HCSN86, ASNI91]. Much experimental work needs to be done to understand these examples thoroughly and to explore other possibilities. It is fair to state, however, that the fundamental conclusion of Careri et al. has not been passively accepted by the scientific community. Some of the questions that have been raised are listed below.

(i) *Which phonon modes are involved?* The theoretical considerations presented above indicate that a cluster of about five or more optical modes around 70 cm^{-1} are sufficient to explain the temperature dependence of the 1650 cm^{-1} band intensity. Careri et al. originally supposed that a single optical mode would be involved because—as Davydov has always insisted—acoustic phonon

modes are expected to lead to the small Franck–Condon factor indicated in eq. (3.18). Alexander originally argued that only self-trapping by acoustic modes could account for the temperature dependence of the intensity of the 1650 cm^{-1} band [Ale85]. This conclusion was based upon a misapprehension of some of the numerical parameters involved and was later amended by Alexander and Krumhansl to a suggestion that self-trapping in ACN could be explained by assuming acoustic modes plus a single optical mode at 70 cm^{-1} [AK86]. Although recent incoherent neutron scattering studies of ACN by Barthes et al. indicate that acoustic modes are not involved in the self-trapping mechanism [BAS⁺91], the work of Alexander and Krumhansl is of great value to me because it showed me how to correctly calculate the temperature dependence of the 1650 cm^{-1} band intensity [Kru87].

(ii) *Fermi resonance.* Although Careri had ruled out Fermi resonance as an explanation for the 1650 cm^{-1} band because the strong temperature dependence should imply a correspondingly strong displacement of frequency with temperature [CBG⁺84], Johnston and Swanson concluded that “the unusual behavior of the optically active Amide-I modes is most likely a result of Fermi coupling and not Davydov-like solitons” [JS85]. They proposed a set of interacting bands for this resonance, but subsequent spectral measurements as a function of pressure rendered this explanation invalid [Joh85]. Further evidence against Fermi resonance has been provided by a recent Raman study of deuterated ACN ($\text{CD}_3\text{CONHC}_6\text{D}_5$) [SDA⁺91]. The Johnston and Swanson publication is still interesting, however, because a comparison of their figs. 1a and 1b (at $T = 21\text{ K}$) clearly shows a red shift of 23 cm^{-1} for both the 1650 cm^{-1} band and the 1667 cm^{-1} band as ^{13}C is substituted for the amide carbon. Added to the ^{15}N data presented by Careri et al., this provides strong evidence that the 1650 cm^{-1} band is purely Amide-I.

(iii) *Topological defect state.* Careri's first suggestion for an explanation of the 1650 cm^{-1} band was to suppose regions of different crystalline phase within the ACN [Car73] and this possibility was carefully discussed and eliminated by Careri et al. [CBG⁺84]. Nonetheless Blanchet and Fincher proposed a “topological defect state” to explain the band. The lack of any observable quasi-elastic broadening of incoherent neutron scattering spectra reported recently by Barthes rules out such an explanation [Bar90].

(iv) *Molecular dynamics studies.* In the course of some molecular dynamics studies of crystalline ACN, Tenenbaum et al. proposed that the 1650 cm^{-1} band could be explained as “due to the coupling of the low frequency motion in the chain of molecules with the motion of the hydrogen bonded protons, at variance with current suggestions” [TGC87]. This proposal overlooks both the ^{15}N data [CBG⁺84] and the ^{13}C data [JS85], which show the 1650 cm^{-1} band to be Amide-I (C–O stretching) in character. Further evidence against this explanation is provided by Raman studies of deuterated ACN [SAMB89, SDA⁺91].

(v) *Fine structure of the 1665 cm^{-1} band.* The fine structure of the 1665 cm^{-1} band in crystalline acetanilide is clearly evident in the infrared absorption data of fig. 6, which show three components at 1665 , 1662 and 1659 cm^{-1} in order of decreasing intensity. The space group of the unit cell (shown in fig. 5) is $D_{2h}^{15}(\text{P}_{\text{bca}})$ and the factor group is D_{2h} ; thus there are three IR-active crystal modes. Following Chirgadze and Nevskaya [CN73], Careri et al. assigned the three components to B_{2u} , B_{1u} , and B_{3u} , respectively [CBG⁺84]. Exploratory infrared polarization studies performed on thin crystals have indicated some uncertainty in these assignments [CGS88]. It is an important point because calculations of dipole–dipole interaction energies (the parameter J in table 1, for example) follow directly from such assignments. Polarized Raman measurements should be made on carefully prepared single crystals in order to check the transition dipole moment calculated by Chirgadze and Nevskaya [CN73, NC76].

(vi) *Free electron laser experiments.* Transient bleaching–recovery measurements on the 1650

cm^{-1} band of crystalline ACN using a free electron laser (FEL) as a source have recently been published by Fann et al. [FRR⁺90]. According to these authors, their experiments “indicate that the anomalous 1650 cm^{-1} band which appears on cooling of acetanilide crystals persists for at least several microseconds following rapid pulse heating.” This observation is at variance with unpublished experiments carried out at Los Alamos using a pulsed CO laser source [Big89]. These experiments indicated that the 1650 cm^{-1} band changes its intensity on a microsecond time scale in a manner that is consistent with laser heating and the observations of Careri et al. [CBG⁺84]. Both measurements need to be carefully repeated and compared.

(vii) *Deuterated measurements.* In an important series of papers Barthes and her colleagues are pursuing spectral studies of various deuterations of crystalline ACN under infrared absorption, Raman scattering, coherent inelastic neutron scattering, and incoherent neutron scattering [BAS⁺88, SAMB89, Bar90, SDA⁺91, BAS⁺91]. In addition to sorting out many of the theoretical questions (as noted above), these measurements have uncovered a new effect in $\text{CD}_3\text{CONHC}_6\text{D}_5$, which is deuterated everywhere except at the hydrogen bonding proton. In this sample the 1650 cm^{-1} band is not observed at any temperature, but (surprisingly) a previously unobserved anomalous band appears at 1480 cm^{-1} in the region of the Amide-II or NH in-plane bending mode. This band has also been observed in N-methylacetamide [ASNI91]. These observations show that deuteration of the methyl and phenyl groups has a strong effect on the amide modes, which is not well understood.

Clearly the “acetanilide story” is not yet finished.

4.2. Local modes

The crux of Davydov's original idea [Dav73, DK73] was that vibrational energy can become localized in a molecular structure as a result of anharmonic effects; thus it is related to “local mode theory”, which arose at about the same time and is now an established concept in physical chemistry [Hen76].

Local mode effects are particularly pronounced in X–H stretching oscillations (as in H_2O , CCl_2H_2 , NH_3 , CH_4 , C_6H_6 , etc.) because the restoring force on a proton is soft. It has been known since 1929, for example, that the CH stretching oscillation in liquid benzene (C_6H_6) forms an overtone series [Ell29]

$$\nu(Q) = 3083Q - 57.5Q^2 \text{ cm}^{-1}, \quad (4.36)$$

where $Q = 1, 2, \dots, 9$. That these overtones ($Q > 1$) correspond to localized vibrational states is established beyond doubt by the experimental observation that exactly the same series occurs in pentadeuterated benzene ($\text{C}_6\text{D}_5\text{H}$) [RHB82].

Equation (4.36) reminds one of the overtone series generated by the 1650 cm^{-1} band in crystalline acetanilide and displayed in eq. (4.27) or (4.28). What is the difference? It arises from the fundamental character of the anharmonicity. To make a clear distinction it is convenient to classify atomic anharmonicity into two types: *intrinsic* and *extrinsic*.

(i) *Intrinsic anharmonicity.* Intrinsic anharmonicity arises when the restoring force of an atomic bond deviates from Hooke's law. A simple classical model that describes this effect in benzene is [SLE85]

$$(i \, d/dt - \omega_0) \mathbf{A} + \gamma \text{diag}(|A_1|^2, \dots, |A_6|^2) \mathbf{A} + \varepsilon \mathbf{M} \mathbf{A} = \mathbf{0}, \quad (4.37)$$

where the components of $\mathbf{A} = \text{col}(A_1, \dots, A_6)$ represent the complex mode amplitudes of the six CH stretching oscillators. The softness of an individual CH bond is represented by γ and the

dipole-dipole interactions between the CH oscillators are given by the 6×6 , real, symmetric matrix εM . Equation (4.37) has the technical advantage that it can be exactly quantized [SE86]. Doing so and choosing the five parameters

$$\begin{aligned}\omega_0 &= 3156.73 \text{ cm}^{-1}, & \gamma &= 117.45 \text{ cm}^{-1}, & \varepsilon &= -4.0 \text{ cm}^{-1}, \\ a &= 0.25, & b &= -0.25,\end{aligned}$$

where

$$M = \begin{bmatrix} 0 & 1 & a & b & a & 1 \\ 1 & 0 & 1 & a & b & a \\ a & 1 & 0 & 1 & a & b \\ b & a & 1 & 0 & 1 & a \\ a & b & a & 1 & 0 & 1 \\ 1 & a & b & a & 1 & 0 \end{bmatrix},$$

gives an excellent representation of the four components (A, E_1, E_2, B) of the fundamental ($Q = 1$) spectrum and ten overtones, a total of fourteen spectral lines.

In developing this quantum theory in detail, Bernstein has shown that at each level of excitation (value of Q) there are four lowest levels (or six lowest states because two are doubly degenerate) arranged over an energy band ΔE , where [BES90]

$$\frac{\Delta E}{\varepsilon} \sim \frac{Q}{(Q-1)!} \left(\frac{\varepsilon}{\gamma} \right)^{Q-1}, \quad (4.38)$$

as Q becomes large compares with unity. A local mode is formed as a wavepacket of these six lowest states. Since the dispersive lifetime of such a wavepacket is $O(\hbar/\Delta E)$, this lifetime grows factorially with Q . In this manner the fundamentally nonlinear character of eq. (4.37) is recovered in the correspondence limit ($\hbar \rightarrow 0, Q \rightarrow \infty$). Finally we note:

(a) Equation (4.38) is not limited to benzene; it holds for any number of degrees of freedom in a linear chain with periodic boundary conditions [BES90].

(b) For the fundamental level ($Q = 1$), the anharmonic term in eq. (4.37) has no qualitative effect on the spectrum; it merely displaces the ground state energy and shifts ω_0 .

(c) In real benzene, the overtone linewidths are much broader than is indicated by eq. (4.38). This is probably because the overtone at level Q is unstable to decay into two bending oscillations at level $(Q-1)$ [Hen87].

(ii) *Extrinsic anharmonicity.* The anharmonic effect proposed by Davydov is called extrinsic because it arises through interaction with a low frequency phonon field. As we saw in sections 1 and 4.1, extrinsic anharmonicity introduces a qualitatively new state (the $Q = 1$ soliton) at the fundamental level. Furthermore, the quantum theory is much more challenging. Nonetheless, in the spirit of eq. (4.38), a corresponding result is obtained from eq. (4.21) of the previous section. Taking $\Delta E = E(\pi) - E(0)$ one finds

$$\frac{\Delta E}{J} = 4 \exp[-(\lambda_0/\hbar\omega_0)^2] \quad (4.39)$$

for $Q = 1$.

4.3. Raman scattering from living cells

In 1981 I was engaged in a numerical investigation of eqs. (1.9) to better understand Davydov's theory of soliton propagation along a three dimensional alpha-helix. In the course of this work it became evident that, under certain initial conditions, energy would vary periodically between the three spines of the alpha-helix with a phase shift of 120° , much like the voltage on a standard three-phase power line [Sco82a]. The frequency of alternation, ν , is related to the transverse dipole-dipole coupling energy, L , by

$$\nu = L/h = 3.85 \times 10^{11} \text{ Hz}, \quad (4.40)$$

when L is measured in J [Sco82b]. Considering overtones and interactions of the moving soliton with a discrete lattice, a set of internal resonances $\{\nu_i(\text{th})\}$ was suggested. At about this time I became aware of some striking observations of Raman scattering from living cells (*E. coli*) that appeared only when the cell population was metabolically active (eating, growing, dividing, etc.) by Sydney Webb. Among these observations were a set of spectral lines—let us call them $\{\nu_i(\text{exp})\}$ —that indicated internal resonances in the far infrared range, between 10^{11} and 10^{12} Hz [WS77]. Webb's observations were summarized in a review article [Web80] and confirmed by similar observations in the Soviet Union [BBG⁺80, ABV⁺88], and it was interesting to notice that

$$\{\nu_i(\text{th})\} \doteq \{\nu_i(\text{exp})\}. \quad (4.41)$$

This agreement seemed especially significant because the $\{\nu_i(\text{exp})\}$ were measured with no knowledge of the existence of Davydov's theory, and the theoretical model was constructed with no knowledge of the existence of the experimental data. These theoretical calculations were first reported in 1981 [Sco81] and later rechecked and compared with the best available experimental data [LMS⁺82].

Shortly thereafter it was suggested that Mie scattering from cell density fluctuations (clumps of cells) could provide an explanation for Webb's observations [OS81, CA83]. In order to settle the matter an experimental program was organized at the Los Alamos National Laboratory (under the aegis of the Center for Nonlinear Studies) to remeasure laser Raman scattering from metabolically active cells using an optical multichannel analyzer (OMA) to record the spectra. The point of using an OMA is that it records the entire spectrum at the same instant of time, thus eliminating artifacts arising from variations of the preparations during the scanning time of a conventional instrument. These experiments did not show Webb's $\{\nu_i(\text{exp})\}$, but they did show an increase in fluorescence while the cells were synchronously dividing, which could lead to spurious lines in a scanning experiment [LBSL85, PFV91]. At this point Webb was invited to spend several weeks at Los Alamos to check the experimental procedure and make suggestions for improvements. He proposed changing the cells from *E. coli* to *B. megaterium* and using a different technique to synchronize division of the cell population. This was tried, but again the results were negative: no $\{\nu_i(\text{exp})\}$ were observed [LB86].

From all this I must conclude that OMA Raman scattering experiments on synchronized, metabolically active cell populations do not provide evidence for the existence of Davydov's soliton in living organisms. Sydney Webb does not agree with this; he says the experiment is tricky and was not done correctly at Los Alamos [Web85]. The remarkable agreement indicated in eq. (4.41) remains unexplained.

4.4. Pump-probe experiments

As Knox et al. have pointed out [KMW90], a decisive experiment to demonstrate the existence of Davydov's soliton would involve the transmission of vibrational energy from point A to point B along an alpha-helix. The time-of-flight and efficiency of transmission could then be measured as functions of distance between A and B, temperature, humidity, pH, and other biochemically relevant parameters.

To this end, Knox et al. have described the preliminary stages of an experiment designed to search for long range transfer vibrational energy. The basic idea is a pump-probe measurement that can be described in the context of fig. 7. A segment of alpha-helix of known length (30 to 100 Å) has a donor chromophore (A) attached to the input end and an acceptor chromophore (B) attached to the output end. Pumping A with a short red pulse (λ_A) induces a soliton of energy E_S on the alpha-helix. From eqs. (1.23) and (1.25)

$$E_S(V) = E_S(0) + \frac{1}{2}M_{\text{sol}}v^2, \quad (4.42a)$$

where

$$E_S(0) = E_0 + 2L - 2J - E_B. \quad (4.42b)$$

This energy is transported along the alpha-helix and eventually raises chromophore B above its ground state. A blue probe pulse (λ_B) then induces fluorescence (λ_B). The time delay (t_2) between pump and probe pulses is of central importance. The total fluorescence caused by a probe pulse at time t_2 is related to the signal $S(t_2)$ arriving at the probe by

$$I_f(t_2) = I_f(\infty)[1 + KS(t_2)], \quad (4.43)$$

where $I_f(\infty)$ is the total fluorescence generated with no pump input and K is a constant that depends upon various coupling efficiencies. (Many more details will be found in ref. [KMW90].)

As was discussed in section 1, the shape of the pulse, $S(t_2)$, will be that of a wavepacket dispersing like an (undressed) exciton. This is because

$$M_{\text{sol}} \doteq m^*, \quad (4.44)$$

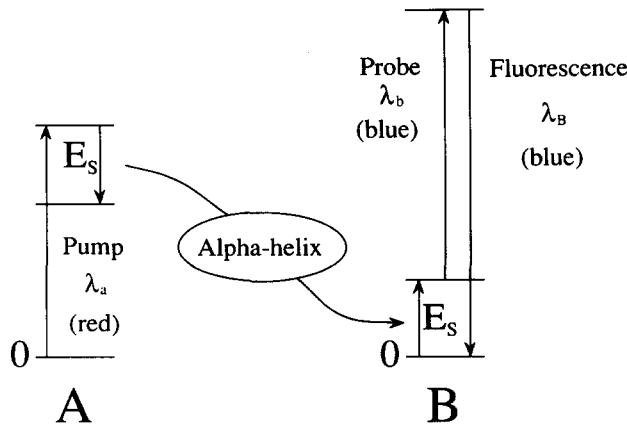


Fig. 7. Sketch of the pump-probe experiment proposed by Knox. E_S is the energy of a soliton that is transported from a donor chromophore (A) to an acceptor chromophore (B).

where m^* is the effective mass of an exciton. The velocity of the wavepacket (v) will depend upon the initial conditions that chromophore A imposes on the alpha-helix. This does not contradict the discussions in section 2 because the inverse scattering method of soliton theory predicts that the soliton will be launched at a speed equal to the group velocity of the dispersive linear wavepacket [HMS81].

How, then, will this experiment distinguish between a soliton and an exciton? Because the soliton is expected to have a longer lifetime for the reasons listed at the end of section 1.1.

5. Generalizations of Davydov's theory

5.1. The vibron soliton

An important property of the Hamiltonian operators studied by Davydov is that they commute with a number operator, $\hat{Q}$. Thus with $\hat{H}$ as in eqs. (1.29) and

$$\hat{Q} = \sum_n \hat{B}_n^\dagger \hat{B}_n, \quad (5.1)$$

$[\hat{Q}, \hat{H}] = 0$, which, in turn, implies $\hat{Q}|\psi\rangle = Q|\psi\rangle$ with

$$\dot{Q} = 0. \quad (5.2)$$

Thus Q (the eigenvalue of $\hat{Q}$) is a constant of the motion that counts the number of Amide-I quanta represented by the wave function. Usually we have chosen $Q = 1$ although in eqs. (1.31) and (1.32) the Kerr-Lomdahl wave function for $Q \neq 1$ was indicated [KL90] and in section 4.1 we considered wave functions with $Q \geq 1$ in order to calculate the overtone spectrum of crystalline acetanilide.

As Takeno has pointed out, it is not strictly correct to assume a Hamiltonian operator that commutes with $\hat{Q}$ [Tak84, Tak85, Tak90a, Tak90b]. Equation (1.29a) should be written

$$\hat{H}_{\text{osc}} = \sum_n [E_0 \hat{B}_n^\dagger \hat{B}_n - J (\hat{B}_n^\dagger \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n-1} + \hat{B}_n \hat{B}_{n+1} + \hat{B}_n \hat{B}_{n-1})]. \quad (5.3)$$

This, in turn, implies that $\dot{Q} \neq 0$. Takeno refers to the excitation related to eq. (5.3) as a “vibron” rather than an “exciton” and a self-trapped vibron is called a “vibron soliton”. (Note that this usage differs from that of Rashba [Ras68], who used the term to denote an electronic-vibrational excited state.)

It is straightforward to estimate the quantitative consequences of including the last terms of eq. (5.3) on Davydov's results. Since $i\hbar\dot{Q} = [\hat{Q}, \hat{H}]$,

$$\dot{Q} = \frac{J}{i\hbar} \sum_n (\hat{B}_n^\dagger \hat{B}_{n+1}^\dagger - \hat{B}_n \hat{B}_{n+1}), \quad (5.4)$$

and using Davydov's wave function from eq. (1.44a) one finds

$$\dot{Q} = (2J/\hbar) \sin(2E_0 t/\hbar). \quad (5.5)$$

Thus Davydov's analysis amounts to approximating $\dot{Q}$ in eq. (5.5) by its time average, which, in turn, implies the neglect of variations in Q of $\pm J/E_0$ or about $\pm 0.5\%$ for $Q = 1$. Although small, this effect could be important because it might open channels of communication (so to speak) between hitherto isolated sectors of constant Q .

Including the $\hat{B}_n \hat{B}_{n+1}$ and $\hat{B}_n^\dagger \hat{B}_{n+1}^\dagger$ terms in eq. (5.3) and allowing Q to be an arbitrary parameter leads Takeno to a nonlinear Klein–Gordon equation [Tak85, Sco69] that reduces to the nonlinear Schrödinger equation studied by Davydov in a small amplitude approximation [Tak84, Tak85, Tak90a, Tak90b]. The same question has been studied more recently by Wang et al. [WBL89c] using a product trial wave function (as did Davydov) but with a coherent state description of the Amide-I field as well as the phonon field. Again they arrive at a nonlinear Klein–Gordon equation for which the energy of a breather soliton traveling with velocity v is given by

$$\tilde{E} = E_{\text{vib}} - E_{\text{bind}} + \frac{1}{2} m_{\text{sol}}^T v^2 + O(v^4), \quad (5.6)$$

just as in eq. (1.46). In the weak coupling limit

$$m_{\text{sol}}^T \rightarrow M_{\text{sol}} \quad (5.7)$$

given in eq. (1.66) and

$$E_{\text{bind}} \rightarrow E_{\text{BD}} \quad (5.8)$$

given in eq. (1.58). Wang et al. also present qualitative arguments that this theory implies a higher level of soliton stability (especially with respect to thermal noise) than the theory developed by Davydov.

As Lindenberg et al. have emphasized [LWB90], much work remains to be done in this area. Two aspects seem to be of particular importance:

(i) *Qualitatively different behavior.* Starting with a description of the Hamiltonian in terms of fermion (rather than boson) operators, Fedyanin et al. [FY77, FMY77, FM79, MF84] find that a correction term of the form

$$\hat{H}_1 = \alpha \sum_n (\hat{a}_n \hat{a}_{n+1} - \hat{a}_n^\dagger \hat{a}_{n+1}^\dagger) \quad (5.9)$$

should be included where α is a small parameter and $\hat{a}_n^\dagger$ ($\hat{a}_n$) are fermion creation (annihilation) operators. Defining $\phi_n = \langle \psi | \hat{a}_n | \psi \rangle$ (where $|\psi(t)\rangle$ is the wave function), making the continuum approximation and normalizing leads to the perturbed nonlinear Schrödinger equation

$$i\phi_t + \phi_{xx} + |\phi|^2 \phi = \alpha \phi_x^*. \quad (5.10)$$

These authors find a rich spectrum of solutions for $\alpha \neq 0$ that are not present when $\alpha = 0$ (the integrable case) [FM79].

(ii) *A means for soliton energy conversion.* In section 2.4 it was pointed out that Davydov's theory faces a problem in explaining how the internal energy of the soliton (approximately equal to E_0) might be transformed into mechanical energy. The reason for this difficulty is the number conservation expressed in eq. (5.2). Since the inclusion of the $(\hat{B}_n \hat{B}_{n+1} + \hat{B}_n^\dagger \hat{B}_{n+1}^\dagger)$ terms in eq. (5.3) breaks number conservation, it might provide a mechanism for energy conversion. This possibility has yet to be explored.

5.2. Globular protein

Davydov's theory (outlined in section 1.1) describes the localization and transport of Amide-I energy along the alpha-helix regions of protein. It is interesting to consider how these ideas might be extended to the less regular geometry of globular protein. To gain some insight it is appropriate to begin with eqs. (1.9) under the adiabatic approximation ($\dot{\beta}_{n\alpha} = 0$) and permit arbitrary dipole-dipole interactions between the Amide-I oscillators. With the gauge transformation $a_j = A_j \exp(-itW/\hbar)$, one obtains the system

$$[i\hbar d/dt - \text{diag}(E_1, \dots, E_N)] \mathbf{A} + \gamma \text{diag}(|A_1|^2, \dots, |A_N|^2) \mathbf{A} + \varepsilon \mathbf{M} \mathbf{A} = \mathbf{0}, \quad (5.11)$$

where $\mathbf{A} = \text{col}(A_1, \dots, A_N)$, N is the number of amino acids (Amide-I oscillators in the globular protein), $\varepsilon \mathbf{M}$ is an $N \times N$, symmetric, real matrix the elements of which specify the dipole-dipole coupling energies between each pair of Amide-I oscillators, E_j is the site energy of the j th Amide-I oscillator, and

$$\gamma \equiv \chi^2/w \quad (5.12)$$

indicates the level of anharmonicity. Equation (5.11) has been studied in some detail as a mathematical object called the "discrete self-trapping (DST) equation" [ELS85] and applied to the study of local modes in crystalline acetanilide [ELS84], and in small molecules [see eq. (4.37)].

Following the pioneering study of Lomdahl [Lom84], Feddersen has applied eq. (5.11) to the study of local modes in adenylate kinase [Fed90, Fed91], an enzyme that catalyzes the reaction



This increases the efficiency of the energetic reaction described in the introduction because only the adenosine monophosphate (AMP) must be returned for re-energizing. From the atomic coordinates in the Protein Data Bank [Dep90] the components of the dipole-dipole interaction matrix ($\varepsilon \mathbf{M}$) can be calculated using standard electromagnetic techniques [NC76, ELS84]. Since adenylate kinase is composed of 194 amino acids ($N = 194$), there are $\frac{1}{2}N(N+1) = 18915$ distinct elements in $\varepsilon \mathbf{M}$. These are easily stored in a computer, and a visual impression of the structure of the matrix is presented in fig. 8. The ten black bands along the diagonal indicate the location of ten alpha-helices while strands of a parallel beta-sheet appear as small black lines that are parallel to (but far from) the diagonal. Since it was not possible to determine the site energies $\{E_j\}$ from the X-ray data, Feddersen chose them at random from an interval of width 35 cm^{-1} [Tu82].

To the approximations involved in Davydov's theory, quantum states correspond to stationary solutions of eq. (5.11) of the form $\mathbf{A}(t) = \Phi \exp(-iEt/\hbar)$, where $\Phi = \text{col}(\phi_1, \dots, \phi_n)$ is independent of time. Thus Φ satisfies the nonlinear eigenvalue problem

$$[\tilde{M} - \gamma \text{diag}(|\phi_1|^2, \dots, |\phi_N|^2)] \Phi = E \Phi, \quad (5.14)$$

where $\tilde{M} \equiv \text{diag}(E_1, \dots, E_N) - \varepsilon \mathbf{M}$. Solutions to eq. (5.14) can be obtained numerically using path following methods [ELS84, ELS85]. The number of solutions increases dramatically from N when $\gamma = 0$ to $(3^N - 1)/2$ for γ sufficiently large [ELS85], but not all are stable. Each stable solution corresponds to an eigenstate with energy E and stability can be determined numerically using a technique developed by Carr and Eilbeck [CE85]. The degree of localization of an eigenstate can be measured by the parameter

$$\alpha = \frac{\sum_{n=1}^N |\phi_n|^4}{\sum_{n=1}^N |\phi_n|^2}. \quad (5.15)$$

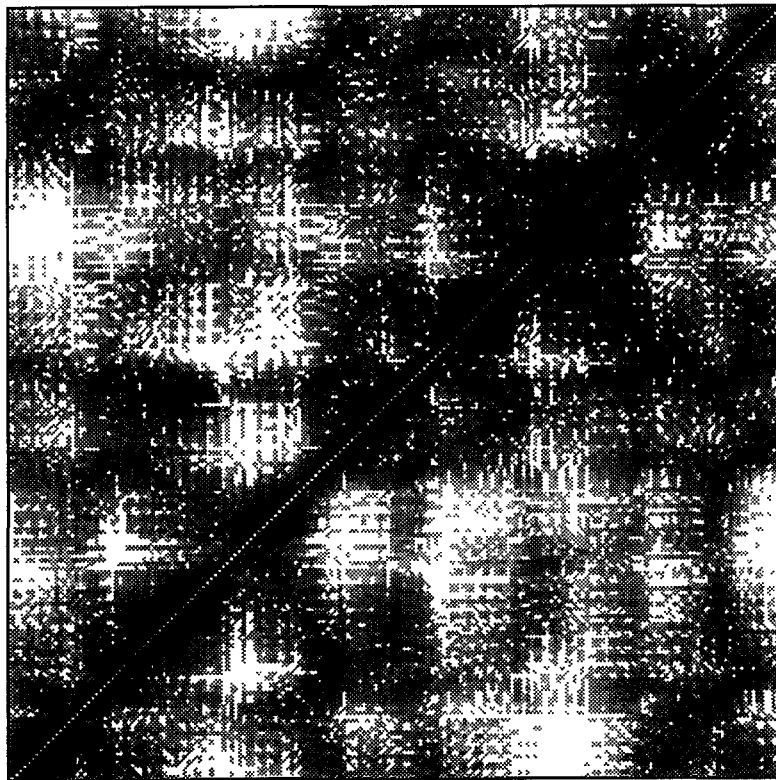


Fig. 8. A gray scale representation of the dipole-dipole interaction matrix ϵM for adenylate kinase. Black points indicate energies above 3 cm^{-1} or below -3 cm^{-1} ; white points correspond to energies between -0.009 and $+0.009 \text{ cm}^{-1}$.

For a completely localized state $\alpha = 1$, while $\alpha = 1/N$ for complete delocalization.

Some of the numerical results obtained by Feddersen for his model of adenylate kinase are presented in fig. 9. This figure shows the three solutions that are most localized for $\gamma = 0$. In addition five solutions are shown that are localized on a single site for large γ . Stable solutions (eigenstates) are shown as full lines, and unstable solutions are shown as dashed lines.

At least two conclusions can be drawn from these calculations. The first is that "Anderson localization" [And58, And78] is completely different from anharmonic localization. From fig. 9b, those solutions that are most localized (i.e. "Anderson" localized) at $\gamma = 0$ become rapidly delocalized and unstable as γ is increased to just a few wavenumbers.

The second conclusion is related to numerical estimates of γ . From measurements of the overtone spectrum of crystalline acetanilide (see section 4.1)

$$\gamma_0 \doteq 50 \text{ cm}^{-1}, \quad (5.16)$$

which is imputed to interactions with optical phonons. Comparison of this value with the numerical results in fig. 9 suggests the possibility of anharmonic localization near some (but not all) of the amino acids.

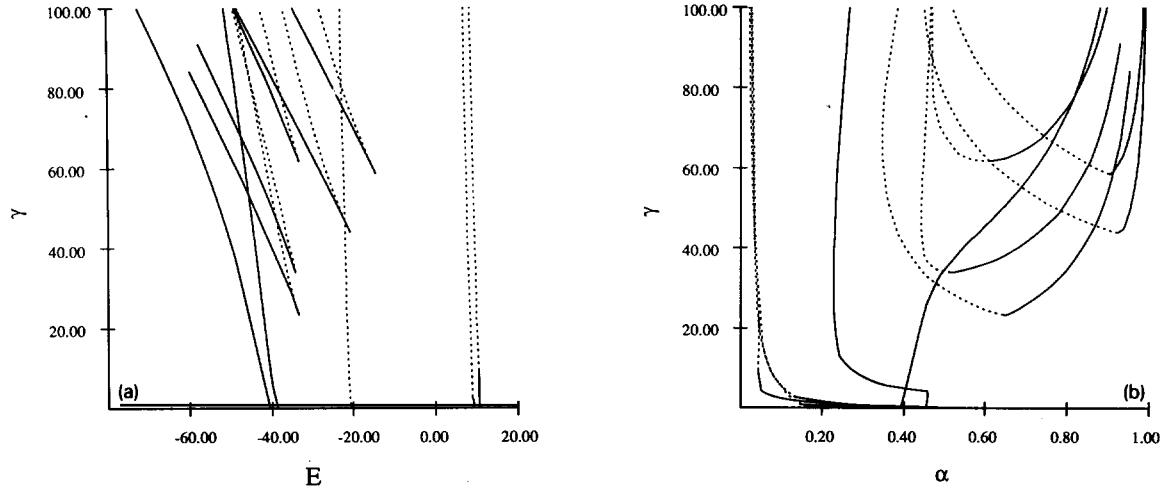


Fig. 9. (a) Anharmonicity (γ) versus energy (E) for some stationary solutions of Feddersen's adenylate kinase model. (b) The same solutions shown on the (α, γ) plane. Dotted lines indicate unstable solutions.

5.3. On-site potential well

Zolotaryuk et al. [ZS90, ZPS90] have explored a modification of Davydov's original model that includes the interactions of the amino acids with their local atomic environment. This interaction is modeled as a harmonic potential well and introduces a new parameter into the description of the problem. For a single channel (see fig. 1) each amino acid molecule rests in a quadratic potential well of spring constant K . Zolotaryuk et al. have studied the simple model of section 1.2 with this additional parameter using a novel numerical method based on a steepest descent minimization scheme. They reach the following conclusions:

- (i) For $K > 0$ the phonon field is a localized pulse rather than a kink [see eq. (1.17)] as in Davydov's theory. This difference alters the Franck–Condon factor estimated in eq. (3.16).
- (ii) In the standard Davydov theory ($K = 0$) only the soliton state is stable; excitons decay into solitons. For $K > 0$ both solitons and excitons are stable over certain ranges of the parameters.

The possibility that solitons with $K > 0$ could be more stable against thermal fluctuations needs to be explored, and it should be carried out in the context of Davydov's "three-channel" model for the alpha-helix that was presented in eqs. (1.9). The equations would then be

$$i\hbar\dot{a}_{n\alpha} = [E_0 + \chi(\beta_{n+1,\alpha} - \beta_{n\alpha})]a_{n\alpha} - J(a_{n+1,\alpha} + a_{n-1,\alpha}) + L(a_{n,\alpha+1} + a_{n,\alpha-1}), \quad (5.17a)$$

$$M\ddot{\beta}_{n\alpha} - w(\beta_{n+1,\alpha} - 2\beta_{n\alpha} + \beta_{n-1,\alpha}) + K\beta_{n\alpha} = \chi(|a_{n\alpha}|^2 - |a_{n-1,\alpha}|^2), \quad (5.17b)$$

where it is assumed that the sound energy, W , has been removed by a gauge transformation.

5.4. The electrosoliton

To this point our attention has been focused on ramifications of Davydov's original idea [Dav73, DK73]: the self-trapping and transport of Amide-I *vibrational* energy. In 1979 he indicated how this idea might be generalized to a means for self-trapping and transport of electronic charge [Dav79a].

Perhaps the best reference to this generalization is chapter VI of refs. [Dav85, Dav91c].

Davydov's concept of the electrosoliton is based upon two facts:

(i) The peptide group (CONH) in the alpha-helix has a static electric dipole moment of about 3.46 D ($= 1.55 \times 10^{-29}$ C m). This static moment is not to be confused with the transition dipole moment that appears in eq. (2.18). It arises because, with reference to fig. 4, there are the following static charges on each atom (expressed as fractions of the electronic charge) [HVB78]

$$\text{O: } -0.42, \quad \text{C: } +0.42, \quad \text{N: } -0.20, \quad \text{H: } +0.20.$$

(ii) An electron can be bound to a static electric dipole if it exceeds a minimum value of 5.42×10^{-30} C m [TAF68]. This electronic state extends away from the dipole center about 2 Å in both directions.

Thus Davydov proposed that an electron could propagate in an exciton band formed by the static dipole potential wells along a channel (see the cross-hatched path in fig. 1) of the alpha-helix. Furthermore the propagation of this electron could be influenced by interaction with acoustic mode phonons on the alpha-helix just as for the self-trapping of vibrational energy that was discussed in section 1. Taking the three channels into account, introducing a product trial wave function, and focusing attention upon the symmetric phonon mode one arrives at a modified version of eqs. (1.13),

$$i\hbar\dot{\tilde{a}}_n = [\tilde{E}_0 + 2\tilde{L} + \tilde{\chi}(\beta_{n+1} - \beta_{n-1})]\tilde{a}_n - \tilde{J}(\tilde{a}_{n+1} + \tilde{a}_{n-1}), \quad (5.18a)$$

$$3M\ddot{\beta}_n - 3w(\beta_{n+1} - 2\beta_n + \beta_{n-1}) = \tilde{\chi}(|\tilde{a}_{n+1}|^2 - |\tilde{a}_{n-1}|^2), \quad (5.18b)$$

where the sound energy, W , has been removed with the gauge transformation $\tilde{a}_n = a_n \exp(-itW/\hbar)$. The meanings of the parameters and variables in eqs. (5.18) are as in eqs. (1.13) with the following exceptions:

$\tilde{a}_n$ is the probability amplitude for an electron to be trapped at the n th turn of the helix,

$\tilde{J}$ and $\tilde{L}$ are, respectively, the nearest neighbor longitudinal and transverse electronic overlap integrals,

$\tilde{E}_0$ is the site energy of an electron, and

$\tilde{\chi}$ is a coupling constant between an electron and longitudinal phonons.

There is one other difference between eqs. (5.18) and (1.13): the interaction energy is

$$\tilde{H}_{\text{int}} = \tilde{\chi} \sum_n (\beta_{n+1} - \beta_{n-1}) |\tilde{a}_n|^2 \quad (5.19)$$

rather than

$$H_{\text{int}} = \chi \sum_n (\beta_{n+1} - \beta_n) |a_n|^2. \quad (5.20)$$

This is because an electron interacts equally with hydrogen bonds on either side of the peptide (CONH) group while an Amide-I quantum favors the hydrogen bond that is adjacent to the carbonyl (CO) group.

An "electrosoliton" theory can be developed for eqs. (5.18) using theoretical and numerical methods as in sections 1–3 [Dav85, Dav91c]. Davydov and Enol'skii [DE80, DE81, DE88] have extended this analysis to include coupling between the electron and optical phonons. They consider an extra electron in a one dimensional chain with short range interactions (i.e., the Holstein

polaron [Hol59]), and also an extra electron in a three dimensional crystal (i.e., the Pekar polaron [Pek54]).

Davydov suggests that the electrosoliton will transport charge with less loss of kinetic energy (phonon damping) because of its weak residual interaction with acoustic phonons. Furthermore a theory for a “bisoliton” (two electrosolitons with oppositely directed spins) has been developed by Brizhik and Davydov [BD84, Dav88, Dav89, Dav90] that has certain biological advantages to be discussed in section 6.7. The bisoliton concept also has led Davydov to a theory of high- T_c superconductivity, which is beyond the scope of this review [Dav88, Dav89].

Following the results of section 1.1, the exciton mass is

$$\tilde{m}^* = \hbar^2 / 2a^2 \tilde{J}, \quad (5.21)$$

the soliton mass is

$$\tilde{M}_{\text{sol}} = \tilde{m}^* \left(1 + \frac{8}{27} M \tilde{\chi}^4 / w^3 \hbar^2 \right) \quad (5.22)$$

and the soliton binding energy is

$$\tilde{E}_B = \tilde{\chi}^4 / 27 w^2 \tilde{J}. \quad (5.23)$$

The differences in the numerical factors in eqs. (1.48) and (5.22) and in eqs. (1.25) and (5.23) arise because of the different definitions of $\tilde{\chi}$ and χ in eqs. (5.19) and (5.20). In effect $\chi = 2\tilde{\chi}$.

In eqs. (5.21)–(5.23), w is the spring constant of a hydrogen bond (13–19.5 N/m) and M is the mass of an amino acid (1.9×10^{-25} kg). Davydov has estimated [Dav86]

$$\tilde{J} \approx 1 \text{ eV}, \quad \tilde{\chi}^2 / w \approx 0.5 \text{ eV}. \quad (5.24)$$

In this case the composite coupling constant ($4\pi\alpha$) defined in eq. (1.57) is equal to about 14. This is in the range of strong coupling so from eqs. (5.21)–(5.23)

$$\tilde{m}^* \approx 1.7 \times 10^{-31} \text{ kg} \quad (5.25)$$

or about one-fifth the mass of an electron, and

$$\tilde{M}_{\text{sol}} \approx 3 \times 10^{-28} \text{ kg} \quad (5.26)$$

or about 376 times the mass of an electron. Finally

$$\tilde{E}_B \approx 0.07 \text{ eV}. \quad (5.27)$$

The estimates in eqs. (5.24)–(5.27) are approximate, but they suggest that lattice interactions could significantly influence the propagation of an electron along an alpha-helix.

5.5. Proton transport in hydrogen bonded systems

In an often quoted paper Eigen and De Meyer [ED58] described ice as a “protonic semiconductor” in which the mobility of the proton is unexpectedly large. From the work of von Hippel and his colleagues, however, it appears that Eigen and De Meyer were measuring surface currents arising from multicrystallinity and not the bulk properties of pure ice [HKW71, Hip71, MHW71]. In a

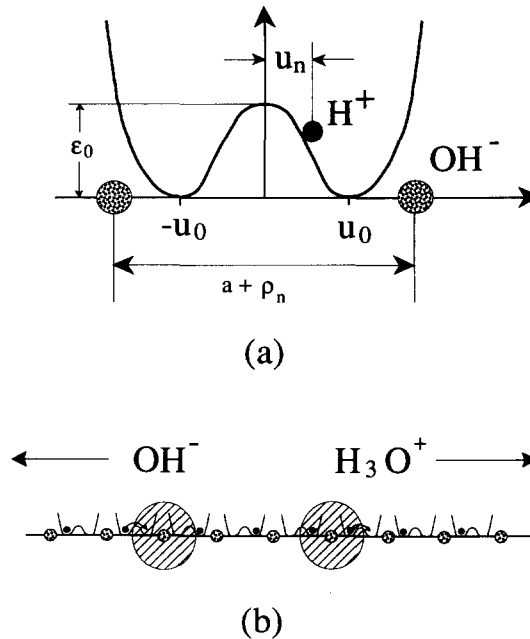


Fig. 10. Kink model for proton transport (redrawn from refs. [Dav85, Dav91c]). (a) The potential well for protons in the hydrogen bridge. (b) The model for transport of hydroxonium (H_3O^+) and hydroxyl (OH^-) ions along a water molecule chain.

biological context this suggests that strong proton currents should be expected at interfaces between hydrogen bonded phases (e.g. protein–water).

Encouraged by the successes of the soliton paradigm during the 1970s, Davydov et al. [DAZ81], Kashimori et al. [KKN82], and Yomosa [Yom82] were led independently to a solitonic theory of protonic conduction. The basic concept is to model the potential seen by a proton between two hydroxyl (OH^-) ions as the double well potential

$$U(u) = \epsilon_0 (1 - u^2/u_0^2)^2, \quad (5.28)$$

which is illustrated in fig. 10a.

Here $\pm u_0$ are the two equilibrium positions for the proton (H^+) and ϵ_0 is the height of the potential barrier between them. Since a small fraction (ca. 10^{-7}) of ice molecules are dissociated into hydroxyl and hydronium (H_3O^+) ions [ED58], the proton hopping shown in fig. 10b provides a means for ionic migration. Using Davydov's notation [Dav85, Dav91c], the Hamiltonian for the system of fig. 10 may be written

$$H_p = \sum_n \left\{ \frac{1}{2} m_p \left[\dot{u}_n^2 + \omega_1^2 (u_{n+1} - u_n)^2 \right] + U(u_n) \right\}, \quad (5.29)$$

where m_p is the proton mass, u_n is the displacement of a proton from the central position, and

$$c_0 = a\omega_1 \quad (5.30)$$

is the speed of “proton sound.” Equation (5.29) is the Hamiltonian for a nonlinear Klein–Gordon equation [Sco69], which supports “kink” and “anti-kink” solitons. A kink (anti-kink) describes the

displacement of the proton from $+u_0$ to $-u_0$ ($-u_0$ to $+u_0$) as one crosses from left to right. Thus, with reference to fig. 10b, the hydronium ion is a kink and the hydroxyl ion is an anti-kink.

The energy of a kink or anti-kink is given by [KKN82, Dav85, Dav91c]

$$E_s = E_0 + M_s v^2 + O(v^4), \quad (5.31)$$

where

$$E_0 = M_s c_0^2 \quad (5.32)$$

and

$$M_s = 4u_0 \sqrt{2m_p \epsilon_0 / a c_0} \quad (5.33)$$

is the effective mass of a kink or anti-kink. Using data given by Weiner and Asker [WA70] and by Nagle et al. [NMM80], Kashimori calculate [KKN82]

$$E_s \doteq 0.48 \text{ eV}. \quad (5.34)$$

This value is in reasonable agreement with the energy of formation of a defect in water (0.52 eV) that was obtained by Knapp et al. [KSS80], and it is comparable with the 0.42 eV of free energy released by the hydrolysis of adenosine triphosphate (see Introduction). Assuming $c_0 = 10^5$ m/s in eq. (5.32), one also obtains

$$M_s \doteq 0.005 m_p. \quad (5.35)$$

To this point in the discussion it has been assumed that the OH^- ions are “frozen” in their equilibrium positions. In 1983 Antonchenko et al. [ADZ83] and Yomosa [Yom83] independently generalized this picture to include relaxation of the OH^- sublattice. Thus the Hamiltonian was taken to be [Dav85, Dav91c]

$$H = H_p + H_{\text{OH}^-} + H_{\text{int}}, \quad (5.36)$$

where H_p is given in eq. (5.29). Also

$$H_{\text{OH}^-} = \frac{1}{2} M \sum_n \left[\dot{\rho}_n^2 + \Omega_0^2 \rho_n^2 + \Omega_1^2 (\rho_{n+1} - \rho_n)^2 \right] \quad (5.37)$$

is the sublattice Hamiltonian, ρ_n is the longitudinal displacement of an OH^- ion, M is its mass,

$$V_0 = a \Omega_1 \quad (5.38)$$

is the speed of lattice sound, and $M \Omega_0^2$ is the spring constant for an “on-site” harmonic potential similar to that introduced in section 5.3. Finally

$$H_{\text{int}} = \chi_p \sum_n \rho_n (u_n^2 - u_0^2) \quad (5.39)$$

is an interaction energy between the protons and the sublattice.

Equation (5.36) together with eqs. (5.29), (5.37) and (5.39) displays some similarity to eqs. (1.28) and (1.29), but there are important differences:

(i) Equations (1.28) and (1.29) involve quantum mechanical energy operators while eqs. (5.29), (5.36), (5.37) and (5.39) represent classical energies.

(ii) Even with $\chi_p = 0$ (no coupling between protons and the sublattice) eq. (5.29) shows soliton behavior. Thus the question is not whether this system can support solitons, but how the kink dynamics is altered by interaction with the sublattice.

Antonchenko et al. [ADZ83, Dav85, Dav91c] have shown that the kink mass in eq. (5.31) is increased by the factor

$$\left(1 + \frac{8\chi_p^2 u_0^2}{5m_p M \omega_0^4}\right) \sqrt{1 - \frac{\chi_p^2 u_0^4}{2\epsilon_0 M \omega_0^2}}, \quad (5.40)$$

and the rest energy of the kink is reduced by the factor

$$\sqrt{1 - \frac{\chi_p^2 u_0^4}{2\epsilon_0 M \omega_0^2}}. \quad (5.41)$$

During the past few years, the theory of these proton kinks has been widely studied [ZSL84, LSWZ85, KCN86, Kry86, PPF87, Pne87, PFB87, WS87, Kry88, KZ88, YK88, Kry90]. The model of fig. 10 does not permit steady (d.c.) charge transport because each proton can only move from one potential well to the other as a soliton passes. The d.c. conductivity in real hydrogen bonded systems is explained by the presence of orientational (or "Bjerrum") defects, which rotate the hydrogen bonds from upstream to downstream positions [Bje52]. Recently Zolotaryuk [Zol87] and Pnevmatikos [Pne88] have suggested that a potential with two alternating barrier heights can also represent this steady conduction. Working together and with their colleagues, they have discussed the dynamics of this model in some detail [TP89, ZP90, PSZ+91, PSS+91] and have extended it to include motion of the OH^- sublattice [ZPS91]. Good agreement with experimental data from ice has been obtained [SZ91], but additional theoretical work is needed [Zol91].

6. Biological applications

6.1. *Caveat emptor*

We now turn to biological applications of the foregoing ideas, but this task is approached with some hesitation. Ideally there should be a division of labor between the biophysicist and the biologist. It is the right (and indeed the *responsibility*) of physical science to propose and examine dynamical theories that might be important in the biological realm. Having done this, it becomes the responsibility of biological science to decide which (if any) of these theories might actually play a functional role in a living organism. When a physical scientist encroaches upon the biological realm, he or she is walking on thin ice. It is with this thought in mind that the following applications of Davydov's ideas are sketched.

6.2. *Conformons*

In a curious coincidence the term "conformon" was coined twice in 1972—once by Green and Ji [GJ72] and again by Vol'kenstein [Vol72]—and the same term was used by Kemeny and Goklany

[KG73] to express a concept that they had been independently developing. Since different authors defined the conformon in different ways, it seems useful to sort things out as follows.

(i) Green and Ji hypothesized the conformon in connection with a new model for mitochondrial function. It was defined as: "the free energy associated with a localized conformational strain" and assumed to have the following characteristic properties [GJ72]: "(a) Conformon is mobile. Conformon migration requires a relatively rigid protein framework such as α -helical structure. (b) Conformon differs from the generalized electromechanochemical free energy of protein conformational strain in the sense that conformon has the property of a 'packet of energy' associated with conformational strain localized within a relatively small volume compared with the size of the supermolecule. (c) The path of conformon migration need not be rectilinear but will be dependent on the three-dimensional arrangement of the linkage system mentioned above. (d) The properties of the conformon are believed to be intimately tied in with the vibrational coupling between adjacent bonds in polypeptide chains."

(ii) Vol'kenstein defined the conformon as [Vol72]: "The displacement of an electron or of the electronic density in a macromolecule [that] produces the deformation of the lattice, i.e. the conformational change. It can be treated as an excitation of the long wave phonons, and the system electron plus local deformation of macromolecule becomes like polaron The conformon differs from the polaron because of the lack of periodicity and homogeneity in the macromolecular globula, coil or the regular secondary structure of heteropolymer At the same time the treatment of electronic and ionic transport in the biological and artificial membranes can be based on an approximation which considers their structure as periodic. In this case the conformon obtains the properties of quasi-particles like those of polaron."

(iii) Kemeny and Goklany were motivated to explain the unexpectedly high electrical conductivity of biological substances. They state [KG73]: "Vol'kenstein's mechanism and ours are very similar from the point of view of statistical mechanics Although the applications may be different, it still seems to be worthwhile to retain Vol'kenstein's term and we will call the activated carrier plus the accompanying changes carrying energy and entropy the conformon The concept of the conformon is a generalization of the small polaron concept." In later papers [KG74, Kem74] they use Holstein's theory [Hol59] to show that "an electron spread over a molecule can form a large polaron and be bound to a number of phonons. The bound state can then propagate from molecule to molecule."

From these quotations and the foregoing discussion, it is evident the conformon of Green and Ji is an exact (albeit nonmathematical) description of the Davydov soliton presented in section 1.1. The conformon of Vol'kenstein and of Kemeny and Goklany, on the other hand, is precisely the electrosoliton described in section 5.4. Ji discussed conformon dynamics at David Green's bioenergetics meeting in 1973 [Ji74], but the response from biochemists was rather negative: they did not believe in long-lived anharmonic states. Green did not mention the conformon when, at Davydov's request, I described the soliton theory to him in the fall of 1978, but Ji (encouraged by colleagues familiar with soliton physics) has recently revived the idea in the context of a molecular model for the living cell [Ji85].

Finally it is interesting to note that Davydov was aware of Vol'kenstein's conformon at the time of his first publication on the soliton [Dav73].

6.3. Muscular contraction

In Davydov's first paper on solitons in protein [Dav73], the main application was to explain the generation of force in the "sliding filament theory" of muscular contraction. Ji also discussed

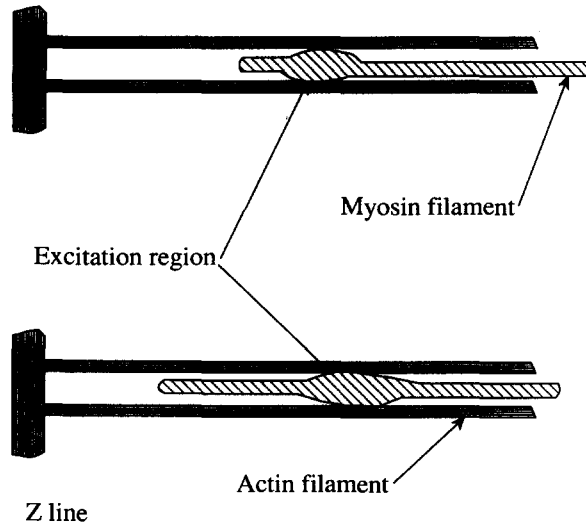


Fig. 11. Force generation by solitons in the sliding filament model of skeletal muscle (redrawn from ref. [Dav73]).

the role that the conformon might play in muscular contraction [Ji74], but the mechanical details are somewhat obscure. Davydov developed this theme in later papers [Dav74, Dav79b] and the discussion in his book [Dav85, Dav91c] is of particular interest because it includes many references to the Soviet literature.

A survey of research in the biophysics of muscular contraction is far beyond the scope of this review, but the situation in the early 1970s has been described in detail by H.E. Huxley [Hux71] and by A.F. Huxley [Hux74]. The evidence from electron microscopy and from low angle X-ray diffraction measurements in support of the sliding filament theory is very strong [Hux71]. Basically this theory asserts that skeletal muscle contracts not by changing the lengths of individual protein molecules but by sliding filaments of myosin and actin past each other. The problem arises, then, of explaining how the force between the actin and myosin filaments is generated. This question is considered in detail by A.F. Huxley and the evidence for force generation through the making and breaking of cross-bridges from the heads of myosin molecules to the actin filaments is summarized [Hux74]. In this picture, myosin heads: (i) swing out, (ii) attach to the actin, (iii) bend and therefore generate force, and finally (iv) detach from actin to begin another cycle. The force generation is rather like in a galley where the individual oars are not necessarily in step. A.F. Huxley emphasizes, however, that "None of these points should be regarded at the present day as being established beyond doubt."

Davydov [Dav73, Dav74, Dav79b, Dav85, Dav91c] raises the question: What makes the myosin heads attach to actin, pull, and then detach? His explanation can be described in relation to fig. 11, which is redrawn from ref. [Dav73]. The tails of the myosin filaments are bundles of alpha-helix, each thread being about $0.1 \mu\text{m}$ in length. (This is why the length of the alpha-helix in the numerical studies described in section 2.2 was taken to be 200 turns, and also why the mass of an amino acid, M in eq. (1.3), was taken as the average mass for a myosin tail.) Davydov suggested that a "herd" of solitons traveling along the bundle of myosin fibers would cause a swelling of the myosin region that would, in turn, provide the opportunity for the attachment, forcing and detachment cycles of the myosin heads to the actin filament.

The swelling would, in this picture, be caused by the generation of solitons without axial symmetry. To see this note from the discussion following eq. (1.11) that the alpha-helix admits solitons for which $[\beta_{n1}, \beta_{n2}, \beta_{n3}] \propto [1, \exp(\pm 2\pi i/3), \exp(\pm 4\pi i/3)]$. Since these two modes (the "E-modes") are degenerate, they can be combined to a mode of the form

$$[\beta_{n1}, \beta_{n2}, \beta_{n3}] \propto [0, 1/\sqrt{2}, -1/\sqrt{2}], \quad (6.1)$$

which introduces a local bending of the helix. Many of these bending together are assumed to produce the swelling of the excitation region shown in fig. 11.

Although attractive in its simplicity, this picture has a difficulty when viewed in the context of Davydov's soliton theory. First of all, the majority of the energy carried by a soliton is as a quantum of Amide-I vibrational energy. If one assumes conservation of Amide-I quanta, then (from the discussion of section 2.5) only a few percent of the Amide-I energy can be delivered as a mechanical force. There are two ways to circumvent this difficulty:

(i) Relax the theoretical requirement for conservation of Amide-I number as has been proposed by Takeno [Tak84, Tak85]. This eliminates a constraint from the energetic considerations of section 2.5.

(ii) Assume that the breaking of a cross-bridge attachment between a myosin head and the actin filament permits the myosin head to return to its original position without an exchange of mechanical and Amide-I vibrational energy. In terms of the discussion of section 2.5 this would mean that the mechanical coordinate (after breaking the cross-bridge) could return from x_2 to x_1 without absorbing mechanical energy. Then about 6% of the Amide-I energy could be delivered in each cycle of attachment, forcing, and detachment.

Recently Obendorf [Obe81] has presented structural evidence to suggest that the myosin fiber bundle might exert force on its neighboring actin filaments by rotating like a screw. It is not clear whether this proposal is in conflict with the suggestion of Davydov [Dav73]. It might be that solitons propagating on a helically structured myosin fiber bundle could cause an effective rotation.

6.4. Nonthermal action of electromagnetic fields on living cells

The nonthermal action of a millimeter wave electromagnetic field on living organisms was first reported independently by Webb et al. [WD68, WB69] and by Devyatkov et al. [DSV+73]. Although confirmed several times [BDRA75, GK78, Web79] this effect remained controversial for several reasons: (i) It is difficult to eliminate the possibility of "hot spots" in a microwave irradiation experiment. (ii) There is a variability in biological experiments that physical scientists are often unfamiliar with. (iii) If true, such observations threaten established economic interests (e.g. manufacturers of microwave ovens, RADAR systems, sealing equipment, antenna farms, etc.). A careful study, published by Grundler and Keilman in 1983, should have laid most doubts to rest [GK83]. In these experiments two microwave probes of different design (a "fork" and a "tube") were shown to produce the same sharp resonances in yeast growth rates at power levels well below those required to heat the organisms.

In 1983 Eremko suggested that these experimental observations might be explained as the results of electromagnetic dissociation of Davydov solitons [Ere83, Dav85, Dav91c, Ere90]. In estimating the frequency of such effects, Eremko used eq. (1.25) increased by a factor of three to account for the fact that the second term of eq. (1.20) does not change during an electromagnetic interaction. According to my calculations, this leads to a dissociation frequency in the range 0.5 to 10 GHz. This is below the experimental frequencies, which lie in the range from 40 to 140 GHz [Web79].

If, on the other hand, eq. (1.62)—from weakly coupled polaron theory—is used, the corresponding frequencies lie in the range

$$f = 30 \text{ to } 230 \text{ GHz}, \quad (6.2)$$

which is in much better agreement with the experimental observations.

With reference to fig. 2, these results support the suggestion that weakly coupled polaron theory [LLP53, Hak76, VF84, WK89] gives a better estimate of soliton binding energy than does Davydov's theory.

6.5. The turning wheel

In 1975 Wyman proposed the “turning wheel” as a general model for the steady state operation of an enzyme [Wym75]. The simplest example is illustrated in fig. 12a, which describes the conversion of the ligand (substrate) L into its product L'. In this figure it is assumed that the concentration of L is kept constant and the k 's are rate constants in the associated system of ordinary differential equations

$$[\dot{M}_1] = -(k_1[L] + k_{-4})[M_1] + k_4[M_2] + (k_L + k_{-1})[M_1L], \quad (6.3a)$$

$$[\dot{M}_2] = k_{-4}[M_1] - (k_{-3}[L] + k_4)[M_2] + k_3[M_2L], \quad (6.3b)$$

$$[\dot{M}_1L] = k_1[L][M_1] - (k_L + k_{-1} + k_2)[M_1L] + k_{-2}[M_2L], \quad (6.3c)$$

$$[\dot{M}_2L] = k_{-3}[L][M_2] + k_2[M_1L] - (k_{-2} + k_3)[M_2L], \quad (6.3d)$$

where the square brackets denote concentrations of the various species. In this simple example the enzyme is assumed to have only two conformational states, M_1 and M_2 , each of which can bind to the ligand L. Wyman assumed “one-step transitions”, so the diagonals, $M_1 \longleftrightarrow M_2L$ and $M_2 \longleftrightarrow M_1L$, are not included in his scheme.

For the steady state solution of eqs. (6.3), there is a counterclockwise circulation in the diagram of fig. 12a and the product ligand is produced at the rate

$$[L'] = k_L[M_1L]. \quad (6.4)$$

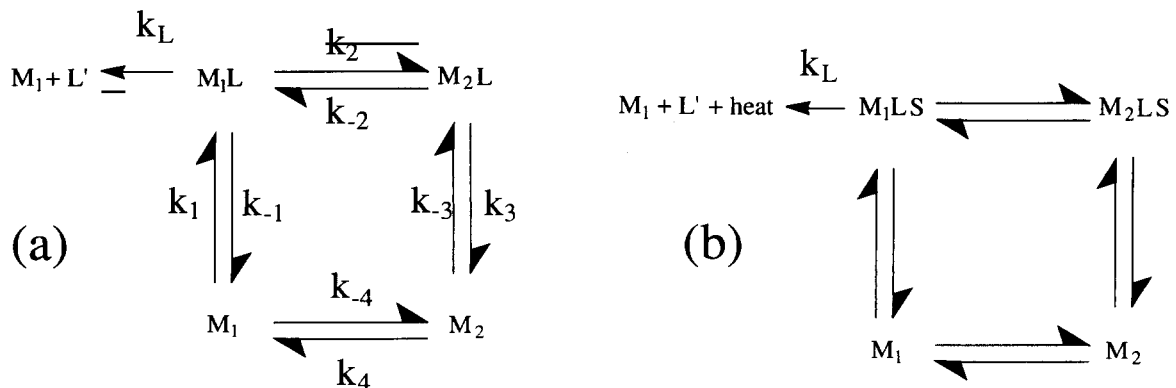


Fig. 12. Diagrams related to Wyman's “turning wheel”.

In a real enzyme, of course, the dynamics can be much more complicated. Many of these details have been discussed by Wyman [Wym75] and also by Hill and Chen [HC75] and some related experimental results have been discussed by Giacometti et al. [GFG⁺75]. Nonetheless two important questions remain:

- (i) What keeps the wheel turning?
- (ii) Why should all enzymes be proteins?

Motivated by experimental observations on crystalline acetanilide (see section 4.1), Careri and Wyman have suggested that Davydov's soliton might provide an answer to these questions [CW84]. Their basic idea is illustrated in fig. 12b, where the corresponding ordinary differential equations are again as in eqs. (6.3) and (6.4). The soliton, S , is assumed to be "an energy packet trapped ... inside a portion of a protein matrix, with the possibility of storing energy without dissipation until it decays into heat as a result of a strong perturbation from some other portion of the same protein." The soliton is assumed to be formed as the substrate binds to the ligand L and to have a long lifetime in M_2 but a short one in M_1 . This short lifetime provides an explanation for the unidirectional output reaction producing L' and the energy of the soliton pays for the energy difference between L' and L , the remainder being dissipated as heat.

The underlying assumption of this picture is that the ligand binding can, in fact, give rise to the formation of a long lived vibrational soliton. To make this assumption plausible, Careri and Wyman consider the binding of negative phosphate moieties at the NH_2 ends of alpha-helices. Because of the large peptide dipole moment (see section 5.4), binding of one negatively charged group involves an attractive energy of about 13 kcal/mole while the energy to create a soliton is only about 4.7 kcal/mole. Careri and Wyman speculate that rapid compression of the end of the helix may induce soliton formation.

It should be noted that the numerical studies on globular protein by Feddersen (see section 5.2) are consistent with the notion that M_1LS has a short lifetime while M_2LS would correspond to a local state that extends below the extant level of anharmonicity (say $\gamma = 50 \text{ cm}^{-1}$) and thus has a larger lifetime.

In a later paper [CW85] Careri and Wyman extended these ideas to a suggestion for a prebiotic mechanism to harvest solar energy that is based upon the overtone bands of crystalline acetanilide (ACN), which are presented in table 2 of section 4.1. Before the development of chlorophyll a "protoenzyme" may have developed with infrared properties similar to ACN. The fundamental band is ruled out of consideration because of the high absorption by water and the first overtone is similarly ruled out because of high protein absorption. But the second overtone ($Q = 3$) at 4803 cm^{-1} falls in a frequency region where water displays a narrow window for infrared transmission.

6.6. Protonics

In section 5.5 some theoretical descriptions for proton transport in ice were discussed and it was argued that the dynamics should be "solitonic" even without lattice interactions. Recent work by Careri and his colleagues has shown that proton conductivity (or "protonics") is an important feature of the life process. From measurements of the conductivities of nearly dry biological samples as the moisture content of normal and heavy water was increased, he was able to demonstrate three important facts:

- (i) The conductivity was primarily protonic because the conductivity ratio for normal to heavy water was the square root of two [CG84].
- (ii) The onset of protonic conductivity coincides with the onset of biological activity.

Table 4
Experimental values for critical exponents of conductivity on
biological systems.

System	t	Reference
Lysozyme powder	1.29 ± 0.05	[CGR88]
Purple membrane fragments	1.23 ± 0.05	[RSCB88]
Maize seeds	1.23 ± 0.05	[BCL89]
<i>Artemia</i> cysts	1.65 ± 0.05	[BCC89]

(iii) The onset of protonic conductivity (σ_p) obeys a "percolation" law of the form

$$\sigma_p \propto (p - p_c)^t, \quad (6\text{r}5)$$

where p is the probability of site occupancy and t is a critical exponent. Experimental values of t are summarized in table 4.

For *two dimensional* systems theoretical values for the critical exponent are

$$t = 1.26 \pm 0.05 \quad [\text{SHSD83}],$$

$$t = 1.28 \pm 0.03 \quad [\text{DV82}],$$

while for *three dimensional* systems

$$t = 1.87 \pm 0.04 \quad [\text{SHSD83}],$$

$$t = 1.95 \pm 0.03 \quad [\text{FH78}].$$

Thus the evidence indicates two dimensional protonic conductivity in lysozyme powder, purple membrane and maize seeds. In the *artemia* cysts the mechanism for protonic conductivity appears to be three dimensional.

From this work it appears that protonic currents become functional only if there is long range connectivity. In this case the solitonic mechanisms discussed in section 5.5 might increase proton mobility. Finally we note that Krimm and Divivedi have suggested a protonic pathway that lies along a distorted alpha-helix in purple membrane [KD82].

6.7. Oxidative phosphorylation

In this last item on our list of biological applications, we turn to a question that may have arisen while reading the introduction: How does adenosine diphosphate (ADP) get converted back into the high energy form of adenosine triphosphate (ATP)? In cells with nuclei this process takes place in the mitochondria and is called *oxidative phosphorylation*. The mitochondria are subcellular organelles that may have evolved from separate organisms, and one of them is sketched in fig. 13 [HM78, Mit79, Fox82].

Within the mitochondria a series of chemical transformations, called the citric acid (or Krebs) cycle, breaks down the carbon chain of glucose to carbon dioxide and produces nicotamide adenine dinucleotide (NAD^+) plus a proton (H^+) and two electrons ($2e^-$). These two electrons make three trips across a region of intrinsic membrane protein, carrying two protons from inside the mitochondrion to outside on each pass. The result of this process is twofold:

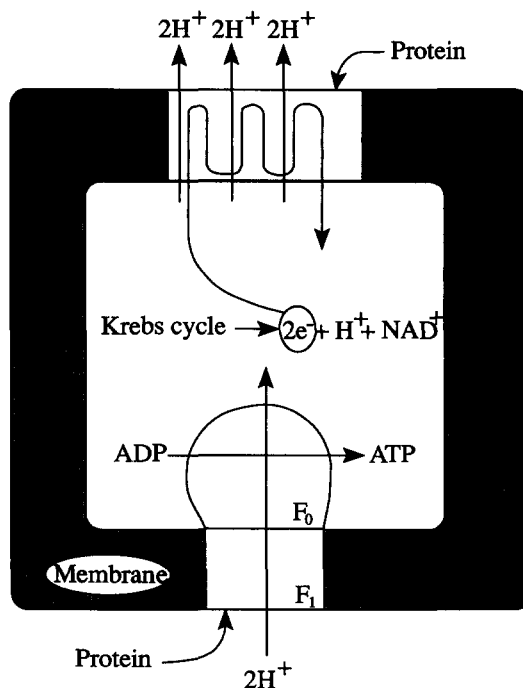


Fig. 13. Simplified sketch of a mitochondrion. Only a single membrane protein of each type is indicated.

(i) The pH inside is increased by about 1.4 with respect to the outside, and (ii) the membrane potential is increased on the outside by about 140 mV. Thus the electrochemical gradient between the outside and inside is about 5.3 kilocal per mole of protons [HM78]. Acting through the " F_1 - F_0 complex" the flow of two protons provides sufficient energy to drive the reaction



which is the reverse of the reaction discussed in the second paragraph of the Introduction.

In the context of this review, the dynamics of the two electrons ($2e^-$) through the region of intrinsic membrane protein is of primary interest. In section 5.4, Davydov's concept of the electrosoliton was introduced and an expression for the binding energy was given in eq. (5.23). Brizhik and Davydov have also described a *bisoliton* that carries two electrons and has a binding energy that is four times as large [BD84, Dav88, Dav89, Dav90, Dav91c], and they have suggested that the electron pair of fig. 13 may be a bisoliton. The key question again is whether or not local distortion of the protein structure facilitates the motion of the electron pair.

Considering the discussion of proton transport in section 5.5, one is led to surmise that the relevant dynamical object on an outward pass might be a complex of two protons and two electrons plus lattice binding energy ($2H^+ + 2e^- +$ lattice distortion). The theory of biological solitons does not yet encompass such an object.

7. Conclusions

Perhaps the most important impression to carry away from this review is that a functional protein might well be a "little universe" in which energy (and therefore information) can be stored and transported. The storage and transport mechanisms would involve local distortions of the backbone structure that can be described as "solitonic". The energy may be in three forms: (i) intramolecular vibrations, (ii) electrons, and (iii) protons. Such mechanisms could play functional roles in muscular contraction, enzymatic activity and oxidative phosphorylation to name but a few possibilities.

The so-called "Davydov soliton" (DS) or "vibron", which involves the coupling of Amide-I (CO stretching) vibrations of the peptide groups to acoustic modes on an alpha-helix, has received the majority of theoretical attention over the past decade and is the main topic of the foregoing discussion. It is an unusual area of biophysics because there are no adjustable parameters in the fundamental energy operator that defines the problem.

Many useful theoretical tools for dealing with this energy operator have been developed in the area of polaron physics over the past half century. For parameters that represent the propagation of a single quantum of Amide-I energy on an alpha-helix, the quantum mechanical description of the DS should be based on weak coupling polaron theory. This implies a binding energy that is about two orders of magnitude larger than is predicted by Davydov's theory.

Although there is not yet clear experimental evidence for the DS, the following points should be considered:

(i) The DS is similar to a "local mode", which is a well established phenomenon in physical chemistry.

(ii) Spectral studies in the Amide-I region of crystalline acetanilide provide evidence that CO vibrational energy can be localized through interactions with lattice phonon modes.

(iii) Theoretical and numerical studies that have concluded that the DS is not a viable dynamical entity at physiological temperature are unconvincing. This point is discussed in detail in section 3.

In the future more experimental effort is needed. The pump-probe experiment described in section 4.4 is, in my view, particularly important. Also studies of "model proteins", such as crystalline acetanilide and its chemical cousins, will continue to provide interesting data, and may establish a basis for extending the techniques of modern electronics (lasers, heterodyning, etc.) into the far infrared region of the electromagnetic spectrum. A number of specific suggestions for further work are made in section 4.1. Turning to biology, studies of the relationship between the onset of protonic conduction and the emergence of metabolic activity—introduced by Careri and described in section 6.6—are opening a new chapter in biochemistry, which will contribute to our understanding of the mechanisms of charge transport in protein.

On the theoretical side, more effort is needed to develop the quantum theory for a soliton that carries two (or more) quanta. For vibrational solitons this is important because two quanta of Amide-I energy are about equal to the free energy released by the hydrolysis of adenosine triphosphate (ATP). For electrosolitons it is important because electrons seem to move in pairs as they pump protons across the mitochondrial membrane during oxidative phosphorylation. Also the processes by which the energy from ATP hydrolysis might be converted into a soliton and back into useful work has received scant attention from biophysicists. It deserves serious consideration in the future.

Theoretical and experimental studies of Davydov's soliton have stimulated the imaginations and engaged the efforts of many scientists throughout the world, and this activity has sometimes led to disagreements and conflicting points of view. Several of us have taken positions that later turned out to be incorrect. We should not feel overly uncomfortable about this; it is the way science makes

progress. As C.G. Jung has pointed out in a somewhat similar context [Jun55]:

Der Irrtum ist eine ebenso wichtige Lebensbedingung wie die Wahrheit.

Error is just as important a condition of life's progress as truth.

Our aim is to know whether or not solitonic mechanisms play a functional role in enzymatic activity.

Acknowledgements

It is a pleasure to thank all who have contributed to this review. The list is long—see the bibliography—but special thanks are extended to those who have provided comments on a draft version including: M. Barthes, L.J. Bernstein, L. Brizhik, D.W. Brown, G. Careri, P.L. Christiansen, J. Cottingham, L. Cruzeiro-Hansson, A.S. Davydov, V. Enol'skii, A.A. Eremko, W. Förner, H. Haken, Z. Ivić, D. Kapor, R.S. Knox, D. Kratochvilová (née Krausová), V.A. Kuprievich, P.S. Lomdahl, L. MacNeil, L. Mašković, O. Faurskov Nielsen, B.M. Pierce, J.W. Schweitzer, M. Škrinjar, S. Stojanović, S. Takeno, J.A. Tuszyński, M. Wagner, and A.V. Zolotaryuk. Support from the National Science Foundation under Grant No. DMS-8902579 and the Air Force Office of Scientific Research is also gratefully acknowledged.

References

- [ABL⁺83] B. Alberts, D. Bray, J. Lewis, M. Raff, K. Roberts and J.D. Watson, *Molecular Biology* (Garland, New York, 1983) pp. 550–609.
- [ABV⁺88] V.V. Artamonov, T.L. Botte, M.Ya. Valakh, M.P. Lisitsa, G.S. Litvinov, A.P. Litvinchuk, M.A. Roshavin and V.I. Struk, Raman scattering on *E. coli* in different metabolic states, *Dokl. Akad. Nauk Ukr. SSR Ser. A* 9 (1988) 42–45 (in Ukrainian and Russian).
- [ADZ83] V.Ya. Antonchenko, A.S. Davydov and A.V. Zolotaryuk, Solitons and proton motion in ice-like structures, *Phys. Stat. Sol. (b)* 115 (1983) 631–640.
- [AK86] D.M. Alexander and J.A. Krumhansl, Localized excitations in hydrogen-bonded molecular crystals, *Phys. Rev. B* 33 (1986) 7172–7185.
- [Ale85] D.M. Alexander, Analog of small Holstein polaron in hydrogen-bonded amide systems, *Phys. Rev. Lett.* 54 (1985) 138–141.
- [And58] P.W. Anderson, Absence of diffusion in certain random lattices, *Phys. Rev.* 109 (1958) 1492–1505.
- [And78] P.W. Anderson, Local moments and localized states, *Rev. Mod. Phys.* 50 (1978) 191–201.
- [ASNI91] G. Araki, K. Suzuki, H. Nakayama and K. Ishii, Polaron-like vibrational band of molecular crystal with one-dimensional hydrogen-bond chains: N-methylacetamide, *Phys. Rev. B* 43 (1991) 12662–12664.
- [Bar90] M. Barthes, Incoherent neutron scattering and infra-red measurements in acetanilide and derivatives, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Bar91] M. Barthes, private communication (1991).
- [BAS⁺88] M. Barthes, R. Almairac, J.L. Sauvajol, R. Currat, J. Moret and J.L. Ribet, Neutron scattering investigation of deuterated crystalline acetanilide, *Europhys. Lett.* 7 (1988) 55–60.
- [BAS⁺91] M. Barthes, R. Almairac, J.L. Sauvajol, J. Moret, R. Currat and J. Dianoux, Incoherent neutron scattering in acetanilide and three deuterated derivatives, *Phys. Rev. B* 43 (1991) 5223–5227.
- [BBG⁺80] V.S. Bannikov, S.M. Bezruchko, E.V. Grishankova, S.V. Kuz'min, Yu.A. Mityagin, R.Yu. Orlov, S.B. Rozhkov and V.A. Sokolina, Investigation of *Bacillus megaterium* cells by Raman scattering, *Dokl. Akad. Nauk SSSR* 253 (1980) 479–480 [*Dokl. Biophys.* 253 (1980) 119–120].
- [BC89] I. Barvik and V. Čápek, Towards memory effects for electrons and phonons on nonrigid lattices, *Phys. Rev. B* 40 (1989) 9973–9976.

- [BCC89] F. Bruni, G. Careri and J.S. Clegg, Dielectric properties of *artemia* cysts at low water contents, *Biophys. J.* 55 (1989) 331–338.
- [BCL89] B. Bruni, G. Careri and A.C. Leopold, Critical exponents of protonic percolation in maize seeds, *Phys. Rev. A* 40 (1989) 2803–2805.
- [BD83] L.S. Brizhik and A.S. Davydov, Soliton excitations in one-dimensional molecular systems, *Phys. Stat. Sol. (b)* 115 (1983) 615–630.
- [BD84] L.S. Brizhik and A.S. Davydov, The electrosoliton pairing in soft molecular chains, *Fiz. Nizk. Temp.* 10 (1984) 748–753.
- [BDRA75] M.A.J. Bertrand, M. Dardalhon, N. Rebeyrotte and M.D. Averbach, Action d'un rayonnement électromagnétique a longueur d'onde millimétrique sur la croissance bactérienne, *C.R. Acad. Sci. Paris D* 281 (1975) 843–846.
- [BE84] L.S. Brizhik and V.Z. Enol'skii, Extra electron interacting with optical and acoustic phonons in 1-D molecular chain, *Ukr. Fiz. Zh.* 29 (1984) 340–345.
- [BE91] L.S. Brizhik and A.A. Eremko, Soliton states in a chain with two atoms per elementary cell, *Phys. Stat. Sol. (b)* 164 (1991) 525–536.
- [Ber91a] L.J. Bernstein, Energy localization in a nonlinear dimer, *Nanobiology* (1991), to appear.
- [Ber91b] L.J. Bernstein, Nonlinear self-trapping in a quantum dimer, *Physica D* 53 (1991) 240–248.
- [BES90] L.J. Bernstein, J.C. Eilbeck and A.C. Scott, The quantum theory of local modes in a coupled system of nonlinear oscillators, *Nonlinearity* 3 (1990) 293–323.
- [BGVV88] L.S. Brizhik, Yu.B. Gaididei, A.A. Vakhnenko and V.A. Vakhnenko, Soliton generation in semi-infinite molecular chains, *Phys. Stat. Sol. (b)* 146 (1988) 605–612.
- [BI89] D.W. Brown and Z. Ivić, Unification of polaron and soliton theories of electron transport, *Phys. Rev. B* 40 (1989) 9876–9887.
- [Big89] I.J. Bigio, private communication (1989).
- [Bje52] N. Bjerrum, Structure and properties of ice, *Science* 115 (1952) 385–390.
- [BLW86] D.W. Brown, K. Lindenberg and B.J. West, On the applicability of Hamilton's equation in the quantum soliton problem, *Phys. Rev. A* 33 (1986) 4104–4109.
- [BLW87] D.W. Brown, K. Lindenberg and B.J. West, Nonlinear density-matrix equation for the study of finite temperature soliton dynamics, *Phys. Rev. B* 35 (1987) 6169–6181.
- [BLW90] D.W. Brown, K. Lindenberg and X. Wang, When is a soliton?, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [BO91] H. Bolterauer and M. Opper, The quantum lifetime of the Davydov soliton, *Z. Phys. B* 82 (1991) 95–103.
- [Bol90] H. Bolterauer, Quantum effects on the Davydov soliton, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Bri90] L. Brizhik, Soliton generation in infinite and half-infinite molecular chains, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Bro88] D.W. Brown, Balancing the Schrödinger equation with Davydov Ansätze, *Phys. Rev. A* 37 (1988) 5010–5011.
- [Bro91] D.W. Brown, private communication (1991).
- [BWL86] D.W. Brown, B.J. West and K. Lindenberg, Davydov solitons: new results at variance with standard derivations, *Phys. Rev. A* 33 (1986) 4110–4115.
- [CA83] M.S. Cooper and N.M. Amer, The absence of coherent vibrations in the Raman spectra of living cells, *Phys. Lett. A* 98 (1983) 138–142.
- [Car73] G. Careri, Search for cooperative phenomena in hydrogen-bonded amide structures, in: *Cooperative Phenomena*, eds. H. Haken and M. Wagner (Springer, Berlin, 1973).
- [Car90] G. Careri, The Amide-I band in acetanilide: physical properties and biochemical suggestions, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [CBG+84] G. Careri, U. Buontempo, F. Galluzzi, A.C. Scott, E. Gratton and E. Shyamsunder, Spectroscopic evidence for Davydov-like solitons in acetanilide, *Phys. Rev. B* 30 (1984) 4689–4702.
- [CE85] J. Carr and J.C. Eilbeck, Stability of stationary solutions of the discrete self-trapping equation, *Phys. Lett. A* 109 (1985) 201–204.
- [CG84] G. Careri and A. Giansanti, Deuterium effect in the dielectric losses of wheat seeds, *Lett. Nuovo Cimento* 40 (1984) 193–196.
- [CGR88] G. Careri, A. Giansanti and J.A. Rupley, Critical exponents of protonic percolation in hydrated lysozyme powders, *Phys. Rev. A* 37 (1988) 2703–2705.
- [CGS88] G. Careri, E. Gratton and E. Shyamsunder, Fine structure of the amide-I band in acetanilide, *Phys. Rev. A* 37 (1988) 4048–4051.
- [CHC+88] L. Cruzeiro, J. Halding, P.L. Christiansen, O. Skovgaard and A.C. Scott, Temperature effects on the Davydov soliton, *Phys. Rev. A* 37 (1988) 880–887.

- [CK87] V. Čápek and D. Krausová, To the relation between energies of molecular soliton and extended states in chains, Czech. J. Phys. B 37 (1987) 1201–1202.
- [CN73] Yu.N. Chirgadze and N.A. Nevskaya, Resonance coupling between amide modes in ordered peptide crystals, Dokl. Akad. Nauk SSSR 208 (1973) 447–450 (in Russian).
- [CS89] J.P. Cottingham and J.W. Schweitzer, Calculation of the lifetime of a Davydov soliton at finite temperature, Phys. Rev. Lett. 62 (1989) 1792–1795.
- [CS90] P.L. Christiansen and A.C. Scott, eds., Davydov's Soliton Revisited (Plenum, New York, 1990).
- [CW84] G. Careri and J. Wyman, Soliton-assisted unidirectional circulation in a biochemical cycle, Proc. Natl. Acad. Sci. USA 81 (1984) 4386–4388.
- [CW85] G. Careri and J. Wyman, Unidirectional circulation in a prebiotic photochemical cycle, Proc. Natl. Acad. Sci. USA 82 (1985) 4115–4116.
- [Dav73] A.S. Davydov, The theory of contraction of proteins under their excitation, J. Theor. Biol. 38 (1973) 559–569.
- [Dav74] A.S. Davydov, Quantum theory of muscular contraction, Biofiz. 19 (1974) 670–676 [Biophys. 19 (1974) 684–691].
- [Dav77] A.S. Davydov, Solitons and energy transfer along protein molecules, J. Theor. Biol. 66 (1977) 379–387.
- [Dav79a] A.S. Davydov, The effect of electron–phonon interaction on the electron motion in a one-dimensional molecular system, Teor. Mat. Fiz. 40 (1979) 408–421 (in Russian).
- [Dav79b] A.S. Davydov, Solitons, bioenergetics and the mechanism of muscle contraction, Int. J. Quantum Chem. 16 (1979) 5–17.
- [Dav79c] A.S. Davydov, Solitons in molecular systems, Phys. Scr. 20 (1979) 387–394.
- [Dav80] A.S. Davydov, Soliton motion in a one-dimensional molecular lattice with account taken of thermal oscillations, Zh. Eksp. Teor. Fiz. 78 (1980) 789–796 [Sov. Phys. – JETP 51 (1980) 397–400].
- [Dav82a] A.S. Davydov, Biology and Quantum Mechanics (Pergamon, New York, 1982).
- [Dav82b] A.S. Davydov, Solitons in quasi-one-dimensional molecular structures, Usp. Fiz. Nauk 138 (1982) 603–643 [Sov. Phys. Usp. 25 (1982) 898–918].
- [Dav85] A.S. Davydov, Solitons in Molecular Systems (Reidel, Dordrecht, 1985).
- [Dav86] A.S. Davydov, Nonlinear phenomena in electron transfer molecular systems, internal report ITP-86-129E, Institute for Theoretical Physics, Kiev (1986).
- [Dav88] A.S. Davydov, Bisoliton mechanism of high-temperature superconductivity, Phys. Stat. Sol. (b) 146 (1988) 619.
- [Dav89] A.S. Davydov, Nonlinear bisoliton model of high-temperature superconductivity of ceramic compounds, Non-linearity 2 (1989) 383–390.
- [Dav90] A.S. Davydov, Solitons in biology and possible role of bisolitons in high- T_c superconductivity, in: Davydov's Soliton Revisited, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Dav91a] A.S. Davydov, The lifetime of molecular solitons, J. Biol. Phys. 18 (1991) 111–125.
- [Dav91b] A.S. Davydov, private communication (1991).
- [Dav91c] A.S. Davydov, Solitons in Molecular Systems, 2nd Ed. (Reidel, Dordrecht, 1991).
- [DAZ81] A.S. Davydov, V.Ya. Antonchenko and A.V. Zolotariuk, Solitons and proton motion in ice, preprint ITP-81-60R, Institute for Theoretical Physics, Kiev (1981) (in Russian).
- [DE77] A.S. Davydov and A.A. Eremko, Radiative lifetime of solitons in molecular chains, Ukr. Fiz. Zh. 22 (1977) 881–892 (in Russian).
- [DE80] A.S. Davydov and V.Z. Enol'skii, Motion of an extra electron in a molecular chain including interaction with optical phonons, Zh. Eksp. Teor. Fiz. 79 (1980) 1888–1897 (in Russian).
- [DE81] A.S. Davydov and V.Z. Enol'skii, A three dimensional soliton in an ionic crystal, Zh. Eksp. Teor. Fiz. 81 (1981) 1088–1098 (in Russian).
- [DE88] A.S. Davydov and V.Z. Enol'skii, On the question of the effective mass of Pekar's polaron, Zh. Eksp. Teor. Fiz. 94 (1988) 177–181 (in Russian).
- [DES78] A.S. Davydov, A.A. Eremko and A.I. Sergienko, Solitons in α -helical protein molecules, Ukr. Fiz. Zh. 23 (1978) 983–993 (in Russian).
- [DK73] A.S. Davydov and N.I. Kislukha, Solitary excitations in one-dimensional molecular chains, Phys. Stat. Sol. (b) 59 (1973) 465–470.
- [Dep90] Department of Chemistry, Brookhaven National Laboratory, Upton, L.I., NY 11973, Protein Data Bank (1990).
- [DSV+73] N.D. Devyatkov, A.Z. Smolyanskaya, R.L. Vilenskaya, S.E. Maniulov, E.N. Chistyakova, V.F. Kondrativa and M.A. Strelkova, Influence of millimeter-band electromagnetic radiation on biological objects, Usp. Fiz. Nauk 110 (1973) 452–469 [Sov. Phys. Usp. 16 (1974) 568–579].
- [DV82] B. Derrida and J. Vannimenus, A transfer matrix approach to random resistor networks, J. Phys. A 15 (1982) L557–L564.

- [ED58] M. Eigen and L. De Maeyer, Self-dissociation and protonic charge transport in water and ice, *Proc. R. Soc. London A* 247 (1958) 505–533.
- [EGV85] A.A. Eremko, Yu.B. Gaididei and A.A. Vakhnenko, Dissociation-accompanied Raman scattering of Davydov solitons, *Phys. Stat. Sol. (b)* 127 (1985) 703–713.
- [Ell29] J.W. Ellis, Molecular absorption spectra of liquids below 3μ , *Trans. Faraday Soc.* 25 (1929) 888–897.
- [ELS84] J.C. Eilbeck, P.S. Lomdahl and A.C. Scott, Soliton structure in crystalline acetanilide, *Phys. Rev. B* 30 (1984) 4703–4712.
- [ELS85] J.C. Eilbeck, P.S. Lomdahl and A.C. Scott, The discrete self-trapping equation, *Physica D* 16 (1985) 318–338.
- [Eno90] V.Z. Eno'skii, Interaction of an extra electron with optical phonons in long molecular chains and ionic crystals, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Ere83] A.A. Eremko, Photodissociation of Davydov solitons, *Dokl. Akad. Nauk Ukr. SSR* 3 (1983) (in Ukrainian and Russian).
- [Ere90] A.A. Eremko, Dissociation of Davydov solitons by electromagnetic waves, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Fed90] H. Feddersen, Chaos and localization in biomolecular dynamics, PhD thesis, Lab. Appl. Math. Phys., Techn. Univ. of Denmark (1990).
- [Fed91] H. Feddersen, Localization of vibrational energy in globular protein, *Phys. Lett. A* 154 (1991) 391–395.
- [FG61] R.L. Fulton and M. Gouterman, Vibronic coupling. I. Mathematical treatment for two electronic states, *J. Chem. Phys.* 35 (1961) 1059–1071.
- [FH78] R. Fisch and A.B. Harris, Critical behavior of random resistor networks near the percolation threshold, *Phys. Rev. B* 18 (1978) 416–420.
- [Fit68] D.B. Fitchen, Zero phonon transitions, in: *Physics of Color Centers*, ed. W.B. Fowler (Academic Press, New York, 1968).
- [FL90] W. Förner and J. Ladik, Influence of heat bath and disorder on Davydov solitons, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [FM79] V.K. Fedyanin and V.G. Makhankov, Soliton-like solutions in one-dimensional systems with resonance interaction, *Phys. Scr.* 20 (1979) 552–557.
- [FMY77] V.K. Fedyanin, V.G. Makhankov and L.V. Yakushevich, Exciton–phonon interaction in long-wave approximation, *Phys. Lett. A* 61 (1977) 256–258.
- [For91a] W. Förner, Davydov soliton dynamics: temperature effects, *J. Phys. C* (1991), to appear.
- [For91b] W. Förner, Quantum and disorder effects in Davydov soliton theory, *Phys. Rev. A* 44 (1991) 2694–2708.
- [For91c] W. Förner, Thermal stability of protein solitons, *Nanobiology* (1991), to appear.
- [For91d] W. Förner, Quantum and temperature effects on Davydov soliton dynamics, *Phys. Rev. Lett.* (1991), to appear.
- [For91e] W. Förner, Davydov soliton dynamics: two quantum states and diagonal disorder, *J. Phys. Cond. Matter* (1991), to appear.
- [Fox82] R.F. Fox, *Biological Energy Transduction* (Wiley, New York, 1982).
- [Fro52] H. Fröhlich, Interaction of electrons with lattice vibrations, *Proc. R. Soc. London A* 215 (1952) 291–298.
- [FRR+90] W. Fann, L. Rothberg, M. Roberson, S. Benson, J. Madey, S. Etemad and R. Austin, Dynamical test of Davydov-type solitons in acetanilide using a picosecond free-electron laser, *Phys. Rev. Lett.* 64 (1990) 607–610.
- [FY77] V.K. Fedyanin and L.V. Yakushevich, Exciton–phonon interaction in one-dimensional molecular crystals, *Teor. Mat. Fiz.* 30 (1977) 133–137 [*Theor. Math. Phys.* 30 (1977) 85–87].
- [GFG+75] G.M. Giacometti, A. Focesi, B. Giardina, M. Brunori and J. Wyman, Kinetics of binding of carbon monoxide to *Lumbricus erythrocytorin*: a possible model, *Proc. Natl. Acad. Sci. USA* 72 (1975) 4313–4316.
- [GJ72] D.E. Green and S. Ji, The electromechanochemical model of mitochondrial structure and function, in: *The Molecular Basis of Electron Transport*, eds. J. Schultz and B.F. Cameron (Academic Press, New York, 1972).
- [GK78] W. Grundler and F. Keilmann, Nonthermal effects of millimeter microwaves on yeast growth, *Z. Naturforsch. C* 33 (1978) 15–22.
- [GK83] W. Grundler and F. Keilmann, Sharp resonances in yeast growth prove nonthermal sensitivity to microwaves, *Phys. Rev. Lett.* 51 (1983) 1214–1216.
- [GL91] B. Gerlach and H. Löwen, Analytical properties of polaron systems: do polaronic phase transitions exist or not?, *Rev. Mod. Phys.* 63 (1991) 63–90.
- [Gla63a] R.J. Glauber, The quantum theory of optical coherence, *Phys. Rev.* 130 (1963) 2529–2539.
- [Gla63b] R.J. Glauber, Coherent and incoherent states of the radiation field, *Phys. Rev.* 131 (1963) 2766–2788.
- [GR80] I.S. Gradshteyn and I.M. Ryzhik, *Table of Integrals, Series, and Products* (Academic Press, New York, 1980).
- [Hak76] H. Haken, *Quantum Field Theory of Solids* (North-Holland, Amsterdam, 1976).
- [HC75] T.L. Hill and Y.-D. Chen, Stochastics of cycle completions (fluxes) in biochemical kinetic diagrams, *Proc.*

- Natl. Acad. Sci. USA 72 (1975) 1291–1295.
- [HCSN86] J. Halding, P.L. Christiansen, O. Skovgaard and O. Faurskov Nielsen, Experimental and computational study of N-methylacetamide, *Phys. Lett. A* 117 (1986) 123–126.
- [Hen76] B.R. Henry, Local modes and their application to the analysis of polyatomic overtone spectra, *J. Phys. Chem.* 80 (1976) 2160–2164.
- [Hen87] B.R. Henry, private communication (1987).
- [Hip71] A. von Hippel, Transfer of protons through “pure” ice I_h single crystals. II. Molecular models for polarization and conduction, *J. Chem. Phys.* 54 (1971) 145–149.
- [HKW71] A. von Hippel, D.B. Knoll and W.B. Westphal, Transfer of protons through “pure” ice I_h single crystals. I. Polarization spectra of ice I_h , *J. Chem. Phys.* 54 (1971) 104–123.
- [HM78] P.G. Hinkle and R.E. McCarty, How cells make ATP, *Sci. Am.* 288(3) (1978) 134–144.
- [HMS81] J.M. Hyman, D.W. McLaughlin and A.C. Scott, On Davydov's alpha-helix solitons, *Physica D* 3 (1981) 23–44.
- [Hoh61] G. Höhler, The polaron model, in: *Lectures on Field Theory and the Many-Body Problem*, ed. E.R. Caianiello (Academic Press, New York, 1961) pp. 285–304.
- [Hol59] T. Holstein, Studies of polaron motion. Part I. The molecular crystal model. Part II. The “small” polaron, *Ann. Phys.* 8 (1959) 325–389.
- [Hol81] T. Holstein, Dynamics of self-localized charge-carriers in quasi 1-D solids, *Mol. Cryst. Liq. Cryst.* 77 (1981) 235–252.
- [Hux71] H.E. Huxley, The structural basis of muscular contraction, *Proc. R. Soc. London B* 178 (1971) 131–149.
- [Hux74] A.F. Huxley, Muscular contraction, *J. Physiol.* 243 (1974) 1–43.
- [Huy77] W.J. Huybrechts, Internal excited state of the optical polaron, *J. Phys. C* 10 (1977) 3761–3768.
- [HVB78] W.G.J. Hol, P.T. van Duijnen and H.J. Berendsen, The α -helix dipole and the properties of proteins, *Nature* 273 (1978) 443–446.
- [IS72] K. Itoh and T. Shimanouchi, Vibrational spectra of crystalline formamide, *J. Mol. Spectrosc.* 42 (1972) 86–99.
- [Ji74] S. Ji, A general theory of ATP synthesis and utilization, *Ann. NY Acad. Sci.* 227 (1974) 211–226.
- [Ji85] S. Ji, The Bhopalator: a molecular model of the living cell based on the concepts of conformons and dissipative structures, *J. Theor. Biol.* 116 (1985) 339–426.
- [Joh85] C.T. Johnston, private communication (1985).
- [JS85] C.T. Johnston and B.I. Swanson, Temperature dependence of the vibrational spectra of acetanilide: Davydov solitons or Fermi coupling?, *Chem. Phys. Lett.* 114 (1985) 547–552.
- [Jun55] C.G. Jung, Versuch einer Darstellung der psychoanalytischen Theorie (Rascher, Zürich, 1955) p. 160 [Freud and Psychoanalysis (Pantheon Books, New York, 1961) paragraph 451].
- [KCD84] T.J. Koscic, R.E. Cline Jr. and D.D. Dlott, Picosecond coherent Raman investigation of the relaxation of low frequency vibrational modes in amino acids and peptides, *J. Chem. Phys.* 81 (1984) 4932–4949.
- [KCN86] Y. Kasimori, F. Chien and K. Nishimoto, Theoretical study of soliton dynamics of a finite one-dimensional hydrogen-bonded system, *Chem. Phys.* 107 (1986) 389–396.
- [KD82] S. Krimm and A.M. Divivede, Infrared spectrum of the purple membrane: clue to a conduction mechanism?, *Science* 216 (1982) 407–408.
- [Kem74] G. Kemeny, Collective aspects of conformons and the electron transfer chain, *J. Theor. Biol.* 48 (1974) 231–241.
- [KG73] G. Kemeny and I.M. Goklany, Polarons and conformons, *J. Theor. Biol.* 40 (1973) 107–123.
- [KG74] G. Kemeny and I.M. Goklany, Quantum mechanical model for conformons, *J. Theor. Biol.* 48 (1974) 23–28.
- [KK82a] V.A. Kuprievich and Z.G. Kudritskaya, Exciton description of interacting vibrations in a molecular chain, technical report ITP-82-62E, Institute for Theoretical Physics, Kiev (1982).
- [KK82b] V.A. Kuprievich and Z.G. Kudritskaya, On the interaction of vibrational excitons with acoustic phonons in a molecular chain, technical report ITP-82-63E, Institute for Theoretical Physics, Kiev (1982).
- [KK82c] V.A. Kuprievich and Z.G. Kudritskaya, Numerical evaluation of the exciton–phonon interaction parameters in the theory of Davydov solitons in polypeptide chains, technical report ITP-82-64E, Institute for Theoretical Physics, Kiev (1982).
- [KKN82] Y. Kashimori, T. Kikuchi and K. Nishimoto, The solitonic mechanism for proton transport in a hydrogen bonded chain, *J. Chem. Phys.* 77 (1982) 1904–1907.
- [KL87] W.C. Kerr and P.S. Lomdahl, Quantum-mechanical derivation of the equations of motion for Davydov solitons, *Phys. Rev. B* 35 (1987) 3629–3632.
- [KL90] W.C. Kerr and P.S. Lomdahl, Quantum-mechanical derivation of the Davydov equations for multi-quanta states, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [KMW90] R.S. Knox, S. Maiti and P. Wu, Search for remote transfer of vibrational energy in proteins, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).

- [KP84] V.M. Kenkre and S.M. Phatak, Exact probability propagators for motion with arbitrary degree of transfer coherence, *Phys. Lett. A* 100 (1984) 101–104.
- [Kru87] J.A. Krumhansl, Comparisons of optical absorption by impurity centers and by polarons in crystalline acetanilide, in: *Energy Transfer Dynamics*, eds. T.W. Barrett and H.A. Pohl (Springer, Berlin, 1987).
- [Kry86] E.S. Kryachko, On the red shift of OH stretching region vibrations in ice and water, *Int. J. Quant. Chem.* 30 (1986) 495–508.
- [Kry88] E.S. Kryachko, Model of ionic defect in symmetrical hydrogen-bonded ice: continuum approach, *Solid State Commun.* 65 (1988) 1609–1612.
- [Kry90] E.S. Kryachko, Collective proton transfer in the $(A-H\cdots)_\infty$ system with double-Morse symmetric potential. I. Model of proton kink defect, *Chem. Phys.* 143 (1990) 359–370.
- [KSS80] E.W. Knapp, K. Schulten and Z. Schulten, Proton conduction in linear hydrogen-bonded systems, *Chem. Phys.* 46 (1980) 215–229.
- [Kup90] V.A. Kuprievich, On the calculations of the exciton–phonon coupling parameters in the theory of Davydov solitons, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Kup91] V.A. Kuprievich, private communication (1991).
- [KZ88] L.N. Kristoforov and A.V. Zolotaryuk, Dynamics of ionic and bonding defects in quasi-one-dimensional hydrogen-bonded chains, *Phys. Stat. Sol. (b)* 146 (1988) 487–501.
- [Lan33] L.D. Landau, Über die Bewegung der Elektronen in Kristallgitter, *Phys. Z. Sowjetunion* 3 (1933) 664–665.
- [LB86] S.P. Layne and I.J. Bigio, Raman spectroscopy of *Bacillus megaterium* using an optical multi-channel analyzer, *Phys. Scr.* 33 (1986) 91–96.
- [LBSL85] S.P. Layne, I.J. Bigio, A.C. Scott and P.S. Lomdahl, Transient fluorescence in synchronously dividing *Escherichia coli*, *Proc. Natl. Acad. Sci. (USA)* 82 (1985) 7599–7603.
- [LEK72] P.W. Langhoff, S.T. Epstein and M. Karplus, Aspects of time dependent perturbation theory, *Rev. Mod. Phys.* 44 (1972) 602–644.
- [LH89] Y. Lai and H.A. Haus, Quantum theory of solitons in optical fibers, *Phys. Rev. A* 40 (1989) 844–866.
- [LK85] P.S. Lomdahl and W.C. Kerr, Do Davydov solitons exist at 300 K?, *Phys. Rev. Lett.* 55 (1985) 1235–1238.
- [LK90] P.S. Lomdahl and W.C. Kerr, Davydov solitons at 300 kelvin: the final search, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [LLP53] T.D. Lee, F.E. Low and D. Pines, The motion of slow electrons in a polar crystal, *Phys. Rev.* 90 (1953) 297–302.
- [LMC+ 86] A.F. Lawrence, J.C. McDaniel, D.B. Chang, B.M. Pierce and R.R. Birge, Dynamics of the Davydov model in α -helical proteins: effects of the coupling parameter and temperature, *Phys. Rev. A* 33 (1986) 1188–1201.
- [LMS+ 82] P.S. Lomdahl, L. MacNeil, A.C. Scott, M.E. Stoneham and S.J. Webb, An assignment to internal soliton vibrations of laser-Raman lines from living cells, *Phys. Lett. A* 92 (1982) 207–210.
- [Lom84] P.S. Lomdahl, Nonlinear dynamics of globular proteins, in: *Nonlinear Electrodynamics in Biological Systems*, eds. W.R. Adey and A.F. Lawrence (Plenum, New York, 1984).
- [LSWZ85] E.W. Laedke, K.H. Spatschek, M.W. Wilkens Jr. and A.V. Zolotaryuk, Two-component solitons and their stability in hydrogen-bonded chains, *Phys. Rev. A* 32 (1985) 1161–1179.
- [LWB90] K. Lindenberg, X. Wang and D.W. Brown, Vibron solitons: a semiclassical approach, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [McC72] C.W.F. McClare, A “molecular energy” muscle model, *J. Theor. Biol.* 35 (1972) 569–595.
- [MF53] P.M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw-Hill, New York, 1953).
- [MF84] V.G. Makhankov and V.K. Fedyanin, Nonlinear effects in quasi-one-dimensional models of condensed matter theory, *Phys. Rep.* 104 (1984) 1–86.
- [MFL89] H. Motschmann, W. Förner and J. Ladik, *J. Phys. Condens. Matter* 1 (1989) 5083.
- [MHW71] M.A. Maidique, A. von Hippel, D.B. Knoll and W.B. Westphal, Transfer of protons through “pure” ice I_h single crystals. III. Extrinsic versus intrinsic polarization; surface versus volume conduction, *J. Chem. Phys.* 54 (1971) 150–160.
- [Mit79] P. Mitchell, Compartmentation and communication in living systems, *Eur. J. Biochem.* 95 (1979) 1–20.
- [MS84] L. MacNeil and A.C. Scott, Launching a Davydov soliton: II. Numerical analysis, *Phys. Scr.* 29 (1984) 284–287.
- [MS88] B. Mechtly and P.B. Shaw, Evolution of a molecular exciton on a Davydov lattice at $T = 0$, *Phys. Rev. B* 38 (1988) 3075–3087.
- [Nag87] P. Nagy, The ground state of a one-dimensional exciton–phonon system, *J. Phys. C* 20 (1987) 5527–5535.
- [NC76] N.A. Nevskaya and Yu.N. Chirgadze, Infrared spectra and resonance interactions of Amide-I and II vibrations of α -helix, *Biopolymers* 15 (1976) 637–648.
- [NMM80] J.F. Nagle, M. Mille and H.J. Morowitz, Theory of hydrogen bonded chains in bioenergetics, *J. Chem. Phys.* 72 (1980) 3959–3971.

- [Obe81] P. Obendorf, A rotating myosin filament theory of muscular contraction, *J. Theor. Biol.* 93 (1981) 667–680.
- [OS81] R.A. O'Sullivan and L. Santo, Experimental aspects in Raman spectroscopy of micro-organisms, *Can. J. Spectrosc.* 26 (1981) 143–148.
- [OSPN88] O.H. Olsen, M.R. Samuelsen, S.B. Petersen and L. Nørskov, Excitations in three-dimensional models of α -helix in protein, *Phys. Rev. A* 38 (1988) 5856–5866.
- [Ost91] N. Østergård, Ab initio calculations in biomolecular dynamics, PhD thesis, Lab. Appl. Math. Phys., Techn. Univ. of Denmark (1991).
- [Pek54] I. Pekar, Untersuchungen über die Electronentheorie der Kristalle (Akademie Verlag, Berlin, 1954).
- [PFB87] St. Pnevmatikos, N. Flytzanis and A.R. Bishop, Soliton dynamics of an extended ϕ^4 model with dissipation and an external field, *J. Phys. C* 20 (1987) 2829–2851.
- [PFV91] J. Pokorný, J. Fiala and K. Vacek, Fröhlich coherent vibrations and Raman scattering, *Czech. J. Phys.* 41 (1991) 484–491.
- [Pie90] B.M. Pierce, Quantum chemical calculations of molecular parameters defining Davydov soliton dynamics in polypeptides, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Pie91] B.M. Pierce, private communication (1991).
- [Pne87] St. Pnevmatikos, Collective nonlinear dynamics of hydrogen-bonded like chains, *Phys. Lett. A* 122 (1987) 249–252.
- [Pne88] St. Pnevmatikos, Soliton dynamics of hydrogen-bonded networks: a mechanism for proton conductivity, *Phys. Rev. Lett.* 60 (1988) 1534–1537.
- [PPF87] M. Peyrard, St. Pnevmatikos and N. Flytzanis, Dynamics of two-component solitary waves in hydrogen-bonded chains, *Phys. Rev. A* 36 (1987) 903–914.
- [PSS+91] St. Pnevmatikos, A.V. Savin, I. Stylianou, M.J. Velgakis and A.V. Zolotaryuk, Proton transport by solitons, *Physica D* 51 (1991) 316–332.
- [PSZ+91] St. Pnevmatikos, A.V. Savin, A.V. Zolotaryuk, Yu.S. Kivshar and M.J. Velgakis, Nonlinear transport in hydrogen-bonded chains: free solitonic excitations, *Phys. Rev. A* 43 (1991) 5518–5536.
- [Ras68] E.I. Rashba, Dynamic theory of vibronic spectra of molecular crystals, *Zh. Eksp. Teor. Fiz.* 54 (1968) 542–588 [*Sov. Phys. – JETP* 27 (1968) 292–300].
- [Ret91] S. Rettrup, private communication (1991).
- [RHB82] K.V. Reddy, D.F. Heller and M.J. Berry, Highly vibrationally excited benzene: overtone spectroscopy and intramolecular dynamics of C_6H_6 , C_6D_6 and partially deuterated or substituted benzenes, *J. Chem. Phys.* 76 (1982) 2814–2837.
- [RN90] W. Rhodes and A. Nicholls, Localization versus delocalization of the Davydov soliton: dispersive collapse under the Schrödinger equation, *Phys. Rev. Lett.* 64 (1990) 1174–1177.
- [RSCB88] J.A. Rupley, L. Siemankowski, G. Careri and F. Bruni, Two-dimensional protonic percolation on lightly hydrated purple membrane, *Proc. Natl. Acad. Sci. USA* 85 (1988) 9022–9025.
- [SAMB89] J.L. Sauvajol, R. Almairac, J. Moret, M. Barthes and J.L. Ribet, Temperature dependence of the Raman spectrum of fully deuterated acetanilide, *J. Raman Spectrosc.* 20 (1989) 12883–12887.
- [SBJ89] A.C. Scott, I.J. Bigio and C.T. Johnston, Polarons in acetanilide, *Phys. Rev. B* 39 (1989) 517–521.
- [SC90] J.W. Schweitzer and J.P. Cottingham, Perturbation estimate of the lifetime of the Davydov soliton at 300 K, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Sch91] J.W. Schweitzer, private communication (1991).
- [Sco69] A.C. Scott, A nonlinear Klein–Gordon equation, *Am. J. Phys.* 37 (1969) 52–61.
- [Sco81] A.C. Scott, The laser-Raman spectrum of a Davydov soliton, *Phys. Lett. A* 86 (1981) 60–62.
- [Sco82a] A.C. Scott, Dynamics of Davydov solitons, *Phys. Rev. A* 26 (1982) 578–595; errata 27 (1983) 2767.
- [Sco82b] A.C. Scott, The vibrational structure of Davydov solitons, *Phys. Scr.* 25 (1982) 651–658.
- [Sco84] A.C. Scott, Launching a Davydov soliton I. Soliton analysis, *Phys. Scr.* 29 (1984) 279–283.
- [Sco87] A.C. Scott, On Davydov solitons at 310 K, in: *Energy Transfer Dynamics*, eds. T.W. Barrett and H.A. Pohl (Springer, Berlin, 1987).
- [Sco90] A.C. Scott, A nonresonant discrete self-trapping equation, *Phys. Scr.* 42 (1990) 14–18.
- [SDA+91] J.L. Sauvajol, G. De Nunzio, R. Almairac, J. Moret and M. Barthes, "Anomalous" excitation in hydrogen-bonded molecular crystals—a Raman scattering study of specifically deuterated acetanilide L4842 ($C_6D_5CONHCD_3$), *Solid State Commun.* 77 (1991) 199–205.
- [SE86] A.C. Scott and J.C. Eilbeck, On the CH stretch overtones of benzene, *Chem. Phys. Lett.* 132 (1986) 23–28.
- [SGSC85] A.C. Scott, E. Gratton, E. Shyamsunder and G. Careri, IR overtone spectrum of the vibrational soliton in crystalline acetanilide, *Phys. Rev. B* 32 (1985) 5551–5553.
- [SHSD83] M. Sahimi, B.D. Hughes, L.E. Scriven and H.T. Davis, Critical exponent of percolation conductivity by finite

- size scaling, *J. Phys. C* 16 (1983) L521–L527.
- [SIZ90] M. Satrić, Z. Ivić and R. Žakula, The temperature dependence of exciton–phonon coupling in the context of Davydov's model: the dynamic damping of soliton, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [SKS88] M.J. Škrinjar, D.V. Kapor and S. Stojanović, Classical and quantum approach to Davydov's soliton theory, *Phys. Rev. A* 38 (1988) 6402–6408.
- [SLE85] A.C. Scott, P.S. Lomdahl and J.C. Eilbeck, Between the local-mode and normal-mode limits, *Chem. Phys. Lett.* 113 (1985) 29–36.
- [SZ91] A.V. Savin and A.V. Zolotaryuk, Dynamics of ionic defects and lattice solitons in a thermalized hydrogen-bonded chain, *Phys. Rev. A* (1991), in press.
- [TAF68] J.E. Turner, V.E. Anderson and K. Fox, Ground-state energy eigenvalues and eigenfunctions for an electron in an electric-dipole field, *Phys. Rev.* 174 (1968) 81–89.
- [Tak84] S. Takeno, Vibron solitons in one-dimensional molecular crystals, *Prog. Theor. Phys.* 71 (1984) 395–398.
- [Tak85] S. Takeno, Vibron solitons and coherent polarization in an exactly tractable oscillator–lattice system, *Prog. Theor. Phys.* 73 (1985) 853–873.
- [Tak90a] S. Takeno, A classical and quantum theory of dynamical self-trapping in nonlinear systems and its implication to energy transfer in biological systems, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [Tak90b] S. Takeno, Quantum theory of vibron solitons—coherent states of a vibron–phonon system and self-localized modes, *J. Phys. Soc. Jpn.* 59 (1990) 3127–3141.
- [TGC87] A. Tenenbaum, A. Campa and A. Giansanti, On the unconventional Amide I band in acetanilide, *Phys. Lett. A* 121 (1987) 126–130.
- [TP89] G.P. Tsironis and St. Pnevmatikos, Proton conductivity in quasi-one-dimensional hydrogen-bonded systems: nonlinear approach, *Phys. Rev. B* 39 (1989) 7161–7173.
- [Tu82] A.T. Tu, *Raman Spectroscopy in Biology* (Wiley, New York, 1982) ch. 3.
- [VF84] G. Venzl and S.F. Fischer, Excitonic and solitonic states in one-dimensional exciton–phonon systems, *J. Phys. Chem.* 81 (1984) 6090–6095.
- [Vol72] M.V. Vol'kenstein, The conformon, *J. Theor. Biol.* 34 (1972) 193–195.
- [WA70] J.H. Weiner and A. Asker, Proton migration in hydrogen-bonded chains, *Nature* 226 (1970) 842–844.
- [WB69] S.J. Webb and A.D. Booth, Absorption of microwaves by microorganisms, *Nature* 222 (1969) 1199–1200.
- [WBL89a] X. Wang, D.W. Brown and K. Lindenberg, Quantum Monte Carlo simulation of the Davydov model, *Phys. Rev. Lett.* 62 (1989) 1796–1799.
- [WBL89b] X. Wang, D.W. Brown and K. Lindenberg, A study of vibron solitons, *J. Mol. Liq.* 41 (1989) 123–142.
- [WBL89c] X. Wang, D.W. Brown and K. Lindenberg, Vibron solitons, *Phys. Rev. B* 39 (1989) 5366–5385.
- [WBL90] X. Wang, D.W. Brown and K. Lindenberg, Quantum Monte Carlo simulations of the Davydov model, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [WD68] S.J. Webb and D.D. Dodds, Inhibition of bacterial cell growth by 136gc microwaves, *Nature* 218 (1968) 374–375.
- [Web79] S.J. Webb, Factors affecting the induction of lambda prophages by millimeter waves, *Phys. Lett. A* 73 (1979) 145–148.
- [Web80] S.J. Webb, Laser-Raman spectroscopy of living cells, *Phys. Rep.* 60 (1980) 201–224.
- [Web85] S.J. Webb, private communication (1985).
- [WK89] M. Wagner and A. Köngeter, A Fulton–Gouterman approach to exciton localization and excitonic solitons, *J. Chem. Phys.* 91 (1989) 3036–3044.
- [Wri90] E.M. Wright, Quantum theory of self-phase modulation, *J. Opt. Soc. Am. B* 7 (1990) 1142–1146.
- [WS77] S.J. Webb and M.E. Stoneham, Resonances between 10^{11} and 10^{12} Hz in active bacterial cells as seen by laser Raman spectroscopy, *Phys. Lett. A* 60 (1977) 267–268.
- [WS87] H. Weberpals and K.H. Spatscheck, Dynamics of solitary waves in hydrogen bonded chains, *Phys. Rev. A* 36 (1987) 2946–2952.
- [Wym75] J. Wyman, The turning wheel: a study in steady states, *Proc. Natl. Acad. Sci. USA* 72 (1975) 3983–3987.
- [YK88] O.E. Yanovitskii and E.S. Kryachko, Model for orientational defects in quasi-one-dimensional ice crystals, *Phys. Stat. Sol. (b)* 147 (1988) 69–81.
- [Yom82] S. Yomosa, Dynamics of the protons in one-dimensional hydrogen-bonded systems, *J. Phys. Soc. Jpn.* 51 (1982) 3318–3324.
- [Yom83] S. Yomosa, Solitary waves in one-dimensional hydrogen-bonded systems, *J. Phys. Soc. Jpn.* 52 (1983) 1866–1873.

- [YSW79] E. Young, P.B. Shaw and G. Whitfield, Asymptotic spectrum of momentum eigenstates of one-dimensional polarons, *Phys. Rev. B* 19 (1979) 1225–1229.
- [Zol87] A.V. Zolotaryuk, Kink models in biophysics, in: *Biophysical Aspects of Cancer*, eds. J. Fiala and J. Pokorný, (Charles Univ., Prague, 1987) pp. 179–188.
- [Zol88] A.V. Zolotaryuk, Many-particle Davydov solitons, *Fiz. Mnogochast. Sistem* 13 (1988) 40–52 (in Russian).
- [Zol91] A.V. Zolotaryuk, private communication (1991).
- [ZP90] A.V. Zolotaryuk and St. Pnevmatikos, One-component model for proton transport in hydrogen-bonded chains, *Phys. Lett. A* 143 (1990) 233–238.
- [ZPS90] A.V. Zolotaryuk, St. Pnevmatikos and A.V. Savin, Self-trapping in a molecular chain with substrate potential, in: *Davydov's Soliton Revisited*, eds. P.L. Christiansen and A.C. Scott (Plenum, New York, 1990).
- [ZPS91] A.V. Zolotaryuk, St. Pnevmatikos and A.V. Savin, Two-sublattice kink dynamics and ionic defects in hydrogen-bonded chains, *Physica D* 51 (1991) 407–417.
- [ZRRS88] Q. Zhang, V. Romero-Rochin and R. Silbey, Variational approach to the Davydov soliton, *Phys. Rev. A* 38 (1988) 6409–6415.
- [ZS71] V.E. Zakharov and A.B. Shabat, Exact theory of two-dimensional self-focusing and one-dimensional self-modulation of waves in nonlinear media, *Zh. Eksp. Theor. Fiz.* 61 (1971) 118–134 [*Sov. Phys. – JETP* 34 (1972) 62–69].
- [ZS90] A.V. Zolotaryuk and A.V. Savin, Solitons in molecular chains with intramolecular nonlinear interactions, *Physica D* 46 (1990) 295–314.
- [ZSL84] A.V. Zolotaryuk, K.H. Spatschek and E.W. Laedke, Stability of activation-barrier-lowering solitons, *Phys. Lett. A* 101 (1984) 517–520.